

# A THURSTON COMPACTIFICATION OF THE SPACE OF STABILITY CONDITIONS

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**ABSTRACT.** We propose compactifications of the moduli space of Bridgeland stability conditions of a triangulated category. Our construction arises from viewing a stability condition as a metric on the underlying category and is inspired by the Thurston compactification of the Teichmüller space of hyperbolic metrics on a surface. The key ingredient in the construction are maps from the stability manifold to an infinite projective space. We prove that, under suitable hypotheses, these maps are injective and their image has a compact closure. We identify a family of points in the boundary that are categorical analogues to the intersection functionals in Teichmüller theory.

We study in detail the geometry of the resulting compactification for the 2-Calabi–Yau categories of graphs, and fully work out the cases of the  $A_2$  and  $\widehat{A}_1$  graphs. To do so, we carefully examine the dynamics of Harder–Narasimhan multiplicities under auto-equivalences of the category. We introduce finite automata to study this dynamics and employ it in our analysis of the  $A_2$  and  $\widehat{A}_1$  categories.

## 1. INTRODUCTION

The space of Bridgeland stability conditions has emerged as an important invariant of triangulated categories, evidenced by its prominent role in homological algebra, algebraic geometry, mirror symmetry, representation theory, and mathematical physics. This paper takes the point of view of studying stability conditions through the metrics they define on the underlying triangulated category [24]. This perspective infuses the study of stability conditions with ideas from metric geometry, opening up new possibilities inspired by the study of moduli spaces of metrics elsewhere in mathematics.

In recent years, a significant body of work has established connections between metric geometry and homological algebra, particularly by relating Teichmüller theory and Bridgeland stability [9, 10, 12, 14, 17, 24, 37, 39]. A particularly promising prospect of this connection is a theory of dynamics in triangulated categories parallel to the theory of dynamics in surface geometry. To further develop this subject, we must consider not only spaces of Bridgeland stability conditions but also their modular compactifications.

In this paper, we propose a family of such compactifications, inspired by Thurston’s compactification of Teichmüller space. The first part of the paper establishes general results about this construction by studying a stability condition via its associated family of metrics. The highlights of this part of the paper are Theorems 3.9 and 4.3. Theorem 3.9 shows that the stable objects of a stability condition (though not necessarily the central charge) are determined by any one of its associated metrics. Theorem 4.3 shows that in fact the entire stability condition (including the central charge) is determined by any pair of its associated metrics.

The second part of the paper focuses on a family of well-studied categories, namely the 2-Calabi–Yau categories associated to graphs. In this case, we prove that a stability condition is completely determined by any one of its associated metrics. In Sections 7

and 8 respectively, we work out our proposed compactification explicitly for two small but highly non-trivial cases, namely the  $A_2$  and  $\widehat{A}_1$  graphs. To do so, we introduce a number of general ideas, including the use of automata to describe the evolution of Harder–Narasimhan filtrations (Section 1.4 and Section 5) and the notion of rectifiable filtrations (Section A).

**1.1. The construction.** Thurston’s compactification of Teichmüller space proceeds by embedding it in an infinite projective space, and then taking its closure. To do this, one first fixes a suitable collection of curves  $\mathbf{S}$ . The embedding sends a point of Teichmüller space, represented by a metric  $\mu$ , to the real-valued function on  $\mathbf{S}$  defined by the length with respect to  $\mu$ .

In our categorical analogue of Thurston’s construction, Bridgeland stability conditions provide the metrics. The analogue of the set of curves tends to be flexible—the best choice may vary for different categories. In the examples studied in the second part of the paper, we take  $\mathbf{S}$  to be the set of spherical objects. Indeed, as described by Khovanov and Seidel [29], in a 2-Calabi–Yau category of type A there is a precise correspondence between spherical objects and curves on the punctured disc. In greater generality, a reasonable choice for “curves” may be the objects that are stable for some stability condition. The categorical analogue of the length of a curve with respect to a metric is the mass of such an object with respect to a stability condition, whose definition we now recall.

Let  $\mathcal{C}$  be a triangulated category. Given a stability condition  $\tau$  on  $\mathcal{C}$  with central charge  $Z$ , the *mass* of an object  $x$  is defined as

$$(1) \quad m_\tau(x) = \sum_i |Z(a_i)|,$$

where the  $a_i$  are the semistable Harder–Narasimhan factors of  $x$ . The notion of mass admits a  $q$ -analogue for any positive real number  $q > 0$ . The  $q$ -*mass* of an object  $x$  is defined as

$$m_{q,\tau}(x) = \sum_i q^{\phi(a_i)} |Z(a_i)|,$$

where  $\phi(a_i)$  is the phase of the semistable object  $a_i$ .

The  $q$ -mass satisfies the triangle inequality [24]. That is, in a distinguished triangle  $x \rightarrow y \rightarrow z \xrightarrow{+1}$ , we have  $m_{q,\tau}(y) \leq m_{q,\tau}(x) + m_{q,\tau}(z)$ . It easily follows that the function that sends a morphism  $f$  to the non-negative real number  $m_{q,\tau}(\text{Cone}(f))$  defines a translation invariant metric on the category in the sense of [34]. In our paper, all the metrics on categories are of this kind.

Let  $\mathbf{S}$  be a subset of the set of objects of  $\mathcal{C}$ . Denote by  $\mathbf{R}^{\mathbf{S}}$  the space of functions from  $\mathbf{S}$  to  $\mathbf{R}$ , endowed with the product topology. Let  $\mathbf{P}^{\mathbf{S}}$  denote the quotient  $(\mathbf{R}^{\mathbf{S}} \setminus \{0\})/\mathbf{R}^\times$ . Let  $\text{Stab}(\mathcal{C})$  be the space of stability conditions on  $\mathcal{C}$ , and recall that it admits a standard action of  $\mathbf{C}$ . For each  $q$ , the association  $\tau \mapsto m_{q,\tau}$  defines a continuous map

$$m_q: \text{Stab}(\mathcal{C})/\mathbf{C} \rightarrow \mathbf{P}^{\mathbf{S}}.$$

Our proposed “Thurston compactification” of  $\text{Stab}(\mathcal{C})/\mathbf{C}$  is the closure of the image.

**1.2. The results.** Thurston’s compactification of Teichmüller space has the following salient features.

- (1) The length function up to scaling determines the metric up to isotopy. That is, the map taking a metric to its length function is injective. In fact, it is a homeomorphism onto its image.
- (2) The closure of the image is a compact real manifold with boundary.

- (3) The boundary contains a distinguished collection of functionals, namely the intersection functionals. This collection is in fact dense in the boundary. Finally, all points of the boundary have a modular interpretation as projective measured foliations.

We now address each of these points in our categorical context. In the remarks below, we have chosen  $\mathbf{S}$  to be the set of objects  $x$  such that there is a stability condition in which  $x$  is semistable. We remind the reader that the best choice for  $\mathbf{S}$  may vary in applications.

**1.2.1. Injectivity and homeomorphic embedding.** The family of functions  $m_{q,\tau}$  determines the stability condition  $\tau \in \text{Stab}(\mathcal{C})/\mathbf{C}$ . In fact, we prove in Theorem 4.3 that for any two distinct positive real numbers  $q_1, q_2$ , the product map  $m_{q_1} \times m_{q_2}$  is injective. We conjecture that a product of mass maps gives a homeomorphic embedding of  $\text{Stab}(\mathcal{C})/\mathbf{C}$ .

We prove in Theorem 3.9 that for any single  $q$ , the function  $m_{q,\tau}$  determines the stable objects of  $\tau$ . On the other hand, we note that a single  $m_{q,\tau}$  may fail to determine the central charge of  $\tau$ . As a result, we cannot expect in general that a single  $m_q$  gives an embedding of the stability manifold (a phenomenon also observed in [30]).

Nevertheless, for 2-CY categories of graphs, we prove in Proposition 6.14 that for any  $q$ , the map  $m_q$  is injective. Moreover, in types  $A_2$  and  $\widehat{A}_1$ , we prove in Propositions 8.15 and 7.22 that  $q = 1$  already gives a homeomorphic embedding. (The case of  $q \neq 1$  is handled in the follow-up work [1] using similar techniques). Our key technical tool is *Harder–Narasimhan automata*, which we briefly discuss in Section 1.4 and develop in Section 5.

**1.2.2. Properties of the closure.** We prove that the closure of the image of  $m_q$  is compact under mild hypotheses. Precisely, as long as the set  $\mathbf{S}$  contains a classical generator of the category, the closure of the image of  $m_q$  is compact (Proposition 4.1). Categories that have a classical generator include derived categories of coherent sheaves on a quasi-projective variety, and the category of perfect complexes over a dg algebra.

For applications to dynamics, it is particularly important to understand when the compactification is homeomorphic to a closed Euclidean ball, since in that case every autoequivalence will act with a fixed point on the compactification. For the  $A_2$  2-Calabi–Yau category, we prove that the closure of  $m_1$  is homeomorphic to a closed Euclidean ball (Proposition 7.31).

It is an important question to understand analogous results for other categories, for example, 2-CY graph categories and derived categories of K3 surfaces.

**1.2.3. Functionals on the boundary.** Assume that  $\mathcal{C}$  is  $\mathbf{k}$ -linear and of finite type. Let  $a \in \mathcal{C}$  be a spherical object. Associated to  $a$ , we have a function on  $\mathbf{S}$  defined by

$$x \mapsto \dim \text{Hom}^\bullet(a, x) = \sum_i \dim \text{Hom}^i(a, x).$$

In Definition 4.9, we define a close analogue of this function which also incorporates the  $q$ , denoted by  $\overline{\text{hom}}_q(a)$ . This is a categorical analogue of the intersection functional of a curve. We prove in Corollary 4.14 that, in direct analogy with geometry, the function  $\overline{\text{hom}}_q(a)$  represents a boundary point in our compactification.

Whether or not the  $\overline{\text{hom}}$  functionals are dense in the boundary is a subtle question, whose answer depends on  $q$ . In the case of the 2-CY category of type  $A_2$  and  $q = 1$ , the answer is “yes”—they are essentially the rational numbers  $\mathbf{Q} \cup \{\infty\}$  on the boundary circle  $\mathbf{R} \cup \{\infty\}$ . For the same category and  $q \neq 1$ , however, they are not dense in the boundary, but a fractal subset, which may be identified with the left  $q$ -rational numbers of [1].

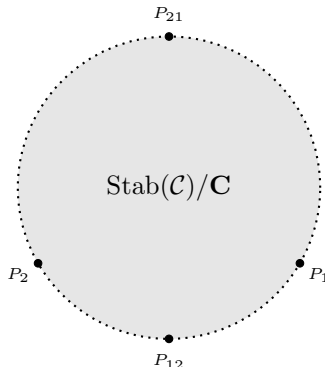


FIGURE 1. The compactification of  $\text{Stab}(\mathcal{C})/\mathbf{C}$ . The spherical objects appear as a dense subset of the boundary.

Giving a moduli interpretation to the boundary points, as a kind of categorical analogue of measured foliations, remains an excellent outstanding problem for future work.

**1.3. Departures from Teichmüller theory.** The categorical picture, although motivated by Thurston’s geometric picture, also differs from it interesting ways.

While a stability condition defines a family of metrics on the underlying triangulated category, those metrics are not “hyperbolic” in a meaningful sense. They behave more like flat metrics, which is consonant with [10], where the authors establish a precise relationship between quadratic differentials and stability condition in certain 3-CY settings.

In the 2-CY category of type  $A_n$ , the stability manifold and the Teichmüller space of a disk with  $(n + 1)$  punctures are both universal covers of the complement of the hyperplane arrangement of type  $A_n$ . For  $n > 2$ , however, we believe that no homeomorphism between them extends to a homeomorphism between the Thurston compactification and any of our compactifications. In Thurston’s construction, the Teichmüller space maps homeomorphically onto the interior of the Thurston compactification. In our case, on the other hand, we see that for  $n > 2$ , the interior of our compactification contains additional points which are not themselves stability conditions. (Some of them can be identified with degenerate stability conditions constructed by Bolognese [4].) It seems likely that a closer Teichmüller-theoretic analogue of our construction is a compactification of the moduli space of flat metrics on a surface, constructed, for example, by Duchin, Leininger, and Rafi [13]. It would be interesting to relate our construction and theirs.

Our theory has a natural  $q$ -analogue, so we obtain a family of compactifications of the space of stability conditions, depending on a positive real parameter  $q$ . This should not be confused with existing literature on quantizations of Teichmüller space (e.g. [16]), which pertain to quantizing algebras of functions.

**1.4. Harder–Narasimhan automata.** After establishing the general results, we focus on the 2-CY categories  $\mathcal{C}_\Gamma$  associated to finite connected graphs  $\Gamma$  without self-loops or multiple edges. The Artin–Tits braid group  $B_\Gamma$  acts (conjecturally faithfully) on the 2-CY category  $\mathcal{C}_\Gamma$ . To understand the compactification of the stability manifold, we heavily use this action. Along the way, we develop techniques that may be broadly applicable to the study of stability conditions in the presence of a group action.

The key question we address is how the HN filtration of an object transforms under an auto-equivalence. To make this precise, fix a stability condition and let  $\Sigma$  be the set of indecomposable semistable objects up to shift. For an object  $x$ , let  $\text{HN}(x) \in \mathbf{Z}^\Sigma$  be its HN multiplicity vector—the function that assigns to  $s \in \Sigma$  the number of times it appears (up to shift) among the HN factors of  $x$ . Suppose a group  $G$  acts on our category by auto-equivalences. This action may not induce a compatible linear action on  $\mathbf{Z}^\Sigma$ . Nevertheless, in many situations—including the  $A_2$  and  $\widehat{A}_1$  cases studied in the latter part of the paper—it induces a “piecewise linear” action. Roughly speaking, this means the following. We can partition the objects of the category into subsets, called *states*. For each ordered pair of states, there are certain allowable elements of  $G$  that send all objects of the first state into objects of the second state and induce linear transformations on their HN multiplicities. We formalise this structure as a decorated finite automaton called an *HN automaton* (see Section 5).

An HN automaton allows us to express the evolution of HN multiplicities as a linear map. This can lead to a direct method to compute the categorical entropy of an autoequivalence (see [18]).

We construct HN automata in type  $A_2$  and  $\widehat{A}_1$ . These automata recognize all braids, meaning that every braid can be represented by a sequence of allowable arrows in the decorated graph. We use these automata as a tool to prove all of the basic properties of our compactifications. As an additional application, we recover a theorem of Rouquier–Zimmermann [35, Proposition 4.8] in the type  $A_2$  case and prove a new analogue of their theorem in the type  $\widehat{A}_1$  case. We anticipate automata to play an important role in the study of group actions on triangulated categories.

**1.5. Organisation.** In Section 2, we recall some background on Bridgeland stability conditions, spherical objects and spherical twists. In Section 3, we prove foundational results about the mass functions associated to stability conditions, including the fact that the mass function determines the stable objects (Theorem 3.9). In Section 4, we use the mass to define a map from the stability manifold to a projective space, and establish its basic properties. These include the compactness of the closure of the image (Proposition 4.1), injectivity for a pair of masses (Theorem 4.3), and the presence of hom functionals in the closure (Corollary 4.14). In Section 5, we introduce the notion of Harder–Narasimhan automata.

So far, all the discussion applies generally. From Section 6 onwards, we focus on specific categories, namely the 2-Calabi–Yau categories of graphs. Section 6 begins with the definition and basic properties of these categories, and proceeds to prove the injectivity of a single mass map (Proposition 6.14). Section 7 contains the  $A_2$  case in detail. Using a suitable HN automaton, we prove that the mass map is a homeomorphic embedding (Proposition 7.22; the boundary is a circle (Proposition 7.16 and Proposition 7.29); and construct an explicit homeomorphism from the closure to a 2-disk (Proposition 7.31). Along the way, we reprove a result of Rouquier–Zimmermann [35, Proposition 4.8]. Section 8 contains analogous results for the  $\widehat{A}_1$  case, including a new analogue of Rouquier–Zimmermann’s result.

The paper has two appendices. They contain technical results, which may have broader use. Appendix A addresses the question of when a filtration in a triangulated category can be re-arranged to obtain the Harder–Narasimhan filtration. Appendix B studies homological properties of self-extensions of a spherical object.

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## 2. BACKGROUND

**2.1. Stability conditions.** We assume familiarity with the theory of Bridgeland stability conditions (see [7]), but we recall the basic definitions. Let  $\mathcal{C}$  be a triangulated category. A stability condition  $\tau$  on  $\mathcal{C}$  is specified by two compatible structures  $(\mathcal{P}, Z)$ :

- (1) the *slicing*  $\mathcal{P}$  consists of additive subcategories  $\mathcal{P}(\phi)$  for each real number  $\phi$ ;
- (2) the *central charge*  $Z$  is a homomorphism of additive groups from the Grothendieck group  $K(\mathcal{C})$  to the complex numbers  $\mathbf{C}$ .

The slicing and the central charge satisfy the following conditions:

- (1)  $\mathcal{P}(\phi + 1) = \mathcal{P}(\phi)[1]$ ;
- (2) if  $a \in \mathcal{P}(\phi)$  and  $b \in \mathcal{P}(\psi)$  with  $\phi > \psi$ , then  $\text{Hom}(a, b) = 0$ ;
- (3) for every nonzero object  $x \in \mathcal{C}$ , there exists a sequence of morphisms

$$0 = x_0 \rightarrow x_1 \rightarrow \cdots \rightarrow x_n = x$$

such that if we define  $a_i$  so that each triangle in

$$\begin{array}{ccccccc} 0 = x_0 & \longrightarrow & x_1 & \longrightarrow & x_2 & \cdots & x_{n-1} \longrightarrow x_n = x, \\ & \nwarrow & \nearrow & & \nwarrow & \nearrow & \\ & +1 & & +1 & & & +1 \\ & a_1 & & a_2 & & \cdots & a_n \end{array}$$

is exact, then  $a_i \in \mathcal{P}(\phi_i)$  and  $\phi_1 > \cdots > \phi_n$ ;

- (4) for every nonzero  $a \in \mathcal{P}(\phi)$ , there is some positive real number  $r$  such that

$$Z([a]) = r \cdot e^{i\pi\phi}.$$

- (5) The slicing  $\mathcal{P}$  is locally finite as defined in [7, Definition 5.7].

The additive category  $\mathcal{P}(\phi)$  turns out to be abelian and of finite length. Its objects are called *semistable of phase  $\phi$* .

We call a sequence of maps  $x_0 \rightarrow \cdots \rightarrow x_n = x$  a *filtration* of  $x$ . For each  $i$ , let  $a_i$  be such that  $x_{i-1} \rightarrow x_i \rightarrow a_i \xrightarrow{+1}$  is a distinguished triangle. We then say that the objects  $a_i$  are *factors* of the filtration.

The specific filtration described in (3) is called the *Harder–Narasimhan (HN) filtration* of  $x$ . Its factors are called *HN factors* and their phases are called *HN phases*. The HN filtration is unique up to isomorphisms (see, e.g. [7, §3]).

Fix a stability condition  $\tau = (\mathcal{P}, Z)$  on  $\mathcal{C}$ . We recall from [7, §3] the induced truncation structure on  $\mathcal{C}$ . This structure depends on the slicing, but the notation hides this dependence. Let  $I \subset \mathbf{R}$  be an interval. Denote by  $\mathcal{C}_I \subset \mathcal{C}$  the full subcategory consisting of objects whose HN phases lie in  $I$ . The notation  $\mathcal{C}_{\geq \alpha}$  means  $\mathcal{C}_{[\alpha, \infty)}$ , and likewise for  $\mathcal{C}_{< \alpha}$  and  $\mathcal{C}_{\leq \alpha}$  and  $\mathcal{C}_{> \alpha}$ . The pair of categories  $(\mathcal{C}_{> \alpha}, \mathcal{C}_{\leq \alpha+1})$  defines a *t-structure* on  $\mathcal{C}$  with the heart  $\mathcal{C}_{(\alpha, \alpha+1]}$ . As a result, for every  $\alpha \in \mathbf{R}$  we have truncation functors

$$\text{tr}_{> \alpha}: \mathcal{C} \rightarrow \mathcal{C}_{> \alpha} \text{ and } \text{tr}_{\leq \alpha}: \mathcal{C} \rightarrow \mathcal{C}_{\leq \alpha}$$

that are right and left adjoints, respectively, to the inclusions  $\mathcal{C}_{>\alpha} \rightarrow \mathcal{C}$  and  $\mathcal{C}_{\leq\alpha} \rightarrow \mathcal{C}$  (see [3, 1.3]; however their convention for denoting the two subcategories is opposite to ours). The pair  $(\mathcal{C}_{\geq\alpha}, \mathcal{C}_{<\alpha+1})$  also defines a  $t$ -structure with heart  $\mathcal{C}_{[\alpha, \alpha+1)}$ , and thus we have similar functors  $\mathrm{tr}_{\geq\alpha}$  and  $\mathrm{tr}_{<\alpha}$ . We abbreviate  $\mathrm{tr}_{>\alpha} x$  to  $x_{>\alpha}$ , and similarly for the other functors. For an interval  $I = [\alpha, \beta]$ , we define  $\mathrm{tr}_I$  as

$$\mathrm{tr}_I = \mathrm{tr}_{\geq\alpha} \circ \mathrm{tr}_{\leq\beta} = \mathrm{tr}_{\leq\beta} \circ \mathrm{tr}_{\geq\alpha},$$

and set  $x_I = \mathrm{tr}_I x$ . We have analogous functors and notation for other kinds of intervals, namely  $(\alpha, \beta)$ ,  $(\alpha, \beta]$ , and  $[\alpha, \beta)$ .

Let  $\mathrm{Stab}(\mathcal{C})$  be the set of all stability conditions. It is proved in [7] that  $\mathrm{Stab}(\mathcal{C})$  has a natural topology. Furthermore, the main result of [7] states that each connected component  $\Sigma$  of  $\mathrm{Stab}(\mathcal{C})$  is a complex manifold, and the forgetful map

$$\mathrm{Stab}(\mathcal{C}) \ni (\mathcal{P}, Z) \mapsto Z \in \mathrm{Hom}(K(\mathcal{C}), \mathbf{C})$$

is a local homeomorphism to a linear subspace  $V(\Sigma)$ .

The complex numbers  $\mathbf{C}$  act on  $\mathrm{Stab}(\mathcal{C})$  as follows. Say  $\tau = (\mathcal{P}, Z)$  and  $\omega = a + i\pi b$ , then

$$(\omega \cdot \mathcal{P})(\phi) = \mathcal{P}(\phi - b) \text{ and } \omega \cdot Z = e^\omega Z.$$

The action of  $\mathbf{C}$  on  $\mathrm{Stab}(\mathcal{C})$  lifts the action of  $\mathbf{C}^*$  on  $\mathrm{Hom}(K(\mathcal{C}), \mathbf{C})$  by scaling. It follows that the induced forgetful map

$$\mathrm{Stab}(\mathcal{C})/\mathbf{C} \rightarrow \mathrm{Hom}(K(\mathcal{C}), \mathbf{C})/\mathbf{C}^*$$

maps each connected component  $\Sigma/\mathbf{C}$  locally homeomorphically to  $V(\Sigma)/\mathbf{C}^*$ .

Observe that the purely imaginary numbers  $i\mathbf{R} \subset \mathbf{C}$  simply shift the slicing and rotate the central charge. We call such transformations *translations*.

**2.2. Spherical objects and twists.** Let  $\mathbf{k}$  be a field and  $\mathcal{C}$  a  $\mathbf{k}$ -linear triangulated category. Assume that  $\mathcal{C}$  is of finite type, that is, for all objects  $x, y$  of  $\mathcal{C}$ , the vector space

$$\mathrm{Hom}^\bullet(x, y) = \bigoplus_n \mathrm{Hom}(x, y[n])$$

is a finite-dimensional  $\mathbf{k}$ -vector space.

We recall the notion of a spherical object introduced in [36, Definition 2.9] (see also [20, Chapter 8]). An object  $x$  of  $\mathcal{C}$  is said to have a Serre dual if the functor  $\mathrm{Hom}^\bullet(x, -)$  is representable. If  $x$  has a Serre dual, the representing object  $Sx$  is unique up to a unique isomorphism, and there are functorial isomorphisms

$$\mathrm{Hom}^\bullet(x, y) \cong \mathrm{Hom}^\bullet(y, Sx)^\vee.$$

Fix  $d \geq 2$ . A  $d$ -Calabi–Yau object is an object  $x$  such that its shift  $x[d]$  is its Serre dual. A  $d$ -spherical object is a  $d$ -Calabi–Yau object  $x$  whose endomorphism algebra is isomorphic to the cohomology algebra of the  $d$ -sphere:

$$\mathrm{Hom}^\bullet(x, x) \cong H^\bullet(S^d, \mathbf{k}).$$

Note that this algebra is simply  $k[t]/t^2$ , where  $t$  has degree  $d$ . We denote by  $\mathrm{loop}_x$  an unspecified homogeneous generator of degree  $d$  of  $\mathrm{Hom}^\bullet(x, x)$ .

If  $x$  is a spherical object and  $y$  is any object, then there is a perfect pairing

$$(2) \quad \mathrm{Hom}^i(x, y) \times \mathrm{Hom}^{d-i}(y, x) \rightarrow \mathrm{Hom}^d(x, x) \cong \mathbf{k},$$

induced by composition [28, Proposition 2.2].

If every object in the category  $\mathcal{C}$  is  $d$ -Calabi–Yau (for the same  $d$ ), then the category  $\mathcal{C}$  is said to be a  $d$ -Calabi–Yau category. Thus, if  $\mathcal{C}$  is a  $d$ -Calabi–Yau category, then for any  $x, y \in \mathcal{C}$ , there exist functorial isomorphisms

$$\mathrm{Hom}^i(x, y) \cong \mathrm{Hom}^{d-i}(y, x)^\vee.$$

Moreover, it follows that if  $x$  is any object in a  $d$ -Calabi–Yau category such that the  $\mathbf{k}$ -vector space  $\mathrm{Hom}^\bullet(x, x)$  is two-dimensional, then  $x$  is automatically a  $d$ -spherical object of  $\mathcal{C}$ .

There is also a notion of a *strongly*  $d$ -Calabi–Yau category, in which the pairing corresponding to the functorial isomorphism (2) is required to be anti-symmetric (see [28]). The categories we study in the later part of the paper are strongly  $d$ -Calabi–Yau.

Assume that  $\mathcal{C}$  admits a dg-enhancement. This is true, for instance, if  $\mathcal{C}$  is algebraic in the sense of [27] or enhanced in the sense of [5], and will hold in all the examples we consider. Fix a dg-enhancement on  $\mathcal{C}$ , which guarantees that  $\mathcal{C}$  has functorial cones. Then any spherical object  $x$  gives rise to an autoequivalence  $\sigma_x: \mathcal{C} \rightarrow \mathcal{C}$  called the *spherical twist* in  $x$  (see [36, §2.2]). For  $y \in \mathcal{C}$ , the object  $\sigma_x(y)$  is defined to be the cone of the evaluation map

$$x \otimes \mathrm{Hom}^\bullet(x, y) \xrightarrow{\mathrm{ev}} y.$$

The twist  $\sigma_x$  and its inverse  $\sigma_x^{-1}$  give rise to distinguished triangles

$$\mathrm{Hom}^\bullet(x, y) \otimes x \xrightarrow{\mathrm{ev}} y \rightarrow \sigma_x(y) \xrightarrow{+1} \text{ and } \sigma_x^{-1}y \rightarrow y \xrightarrow{\mathrm{coev}} \mathrm{Hom}^\bullet(y, x)^\vee \otimes x \xrightarrow{+1}.$$

The evaluation map  $\mathrm{ev}$  is self-explanatory. The co-evaluation map  $\mathrm{coev}$  is the adjoint to the evaluation map

$$\mathrm{Hom}^\bullet(y, x) \otimes y \rightarrow x.$$

### 3. THE MASS ASSOCIATED TO A STABILITY CONDITION

Let  $\mathcal{C}$  be a triangulated category and let  $\tau$  be a stability condition on  $\mathcal{C}$ . Fix a positive real number  $q$ ; for us the most important case is  $q = 1$ .

Let  $x$  be an object of  $\mathcal{C}$ . Let

$$\begin{array}{ccccccc} 0 = x_0 & \longrightarrow & x_1 & \longrightarrow & x_2 & \cdots & x_{n-1} \longrightarrow x_n = x, \\ & \nwarrow \text{dashed} & \swarrow & \nwarrow \text{dashed} & \swarrow & & \nwarrow \text{dashed} & \swarrow \\ & +1 & & +1 & & & +1 & \\ & a_1 & & a_2 & & \cdots & & a_n \end{array}$$

be the HN filtration of  $x$ , where  $a_i$  is semistable of phase  $\phi_i$ . We define the  $(q, \tau)$ -mass of  $x$  by the formula

$$m_{q, \tau}(x) = \sum_i q^{\phi_i} |Z(a_i)|.$$

We take the mass of the zero object to be zero.

Let  $H^i(-)$  be the  $i$ th cohomology with respect to any fixed heart associated to  $\tau$ . Then it is easy to verify that

$$(3) \quad m_{q, \tau}(x) = \sum_i q^{-i} m_{q, \tau}(H^i(x)).$$

A key property of the mass is that it satisfies a triangle inequality.

**Proposition 3.1** (Triangle inequality). *Let  $x \rightarrow y \rightarrow z \xrightarrow{+1}$  be an exact triangle. Then*

$$m_{q, \tau}(y) \leq m_{q, \tau}(x) + m_{q, \tau}(z).$$

*Proof.* See [24, Theorem 3.8]. □



We devote Appendix A to understanding more about when equality holds in Proposition 3.1. The following is a particularly simple instance.

**Proposition 3.2.** *Let  $x \rightarrow y \rightarrow z \xrightarrow{+1}$  be an exact triangle. Assume that the lowest HN phase of  $x$  is greater than or equal to the highest HN phase of  $z$ . Then*

$$m_{q,\tau}(y) = m_{q,\tau}(x) + m_{q,\tau}(z).$$

*Proof.* This proposition is immediate from Example A.3 (4) and Proposition A.8. However, we also give a direct argument here.

Using the exact triangle, we can splice together filtrations of  $x$  and  $z$  to get a filtration of  $y$ , as follows. Suppose we are given  $0 = x_0 \rightarrow \cdots \rightarrow x_n = x$  with factors  $a_1, \dots, a_n$  and  $0 = z_0 \rightarrow \cdots \rightarrow z_m = z$  with factors  $b_1, \dots, b_m$ .

Set  $y_{m+n} = y$ , and recall that  $z_m = z$ . For each  $i = m, \dots, 1$ , define  $y_{n+i-1}$  by descending induction so that it fits into a map of distinguished triangles as follows:

$$\begin{array}{ccccc} y_{n+i-1} & \longrightarrow & z_{i-1} & \longrightarrow & x[1] \xrightarrow{+1} \\ \downarrow & & \downarrow & & \downarrow = \\ y_{n+i} & \longrightarrow & z_i & \longrightarrow & x[1] \xrightarrow{+1} . \end{array}$$

By the octahedral axiom, we see that for each  $i = m, \dots, 1$ , there is a distinguished triangle

$$y_{n+i-1} \rightarrow y_{n+i} \rightarrow b_i \xrightarrow{+1} .$$

Note that  $y_n = x$ . For each  $i = n, \dots, 1$ , set  $y_{i-1} = x_{i-1}$ . Then for  $i = n, \dots, 1$ , there is a distinguished triangle

$$y_{i-1} \rightarrow y_i \rightarrow a_i \xrightarrow{+1} .$$

Therefore we have constructed a filtration

$$(4) \quad 0 = y_0 \rightarrow y_1 \rightarrow \cdots \rightarrow y_{m+n} = y$$

with factors  $a_1, \dots, a_n, b_1, \dots, b_m$ .

Suppose the filtrations  $(x_i)$  and  $(y_j)$  are the HN filtrations of  $x$  and  $y$ . By hypothesis, we have  $\phi(a_n) \geq \phi(b_1)$ . If  $\phi(a_n) > \phi(b_1)$ , then (4) is the HN filtration of  $y$ , and hence

$$(5) \quad m_{q,\tau}(y) = \sum m_{q,\tau}(a_i) + \sum m_{q,\tau}(b_j) = m_{q,\tau}(x) + m_{q,\tau}(z).$$

If  $\phi(a_n) = \phi(b_1)$ , then shorten the filtration (4) by dropping the  $y_n$  term so that in the middle it looks like

$$\cdots \rightarrow y_{n-1} \rightarrow y_{n+1} \rightarrow \cdots .$$

Let  $c$  complete the triangle  $y_{n-1} \rightarrow y_{n+1} \rightarrow c \xrightarrow{+1}$ . By the octahedral axiom, we have an exact triangle  $a_n \rightarrow c \rightarrow b_1 \xrightarrow{+1}$ . Since  $a_n$  and  $b_1$  are semistable of the same phase, so is  $c$ . As a result, the shortened (4) is the HN filtration of  $y$ . Furthermore, we have

$$m_{q,\tau}(c) = m_{q,\tau}(a_n) + m_{q,\tau}(b_1),$$

and hence (5) holds.  $\square$

The following key proposition shows that there are no unexpected mass-preserving decompositions of semistable objects.

**Proposition 3.3.** *Let  $y$  be a  $\tau$ -semistable object of phase  $\phi$ . Let*

$$x \rightarrow y \rightarrow z \xrightarrow{+1}$$

*be an exact triangle such that*

$$m_{q,\tau}(y) = m_{q,\tau}(x) + m_{q,\tau}(z).$$

*Then  $x$  and  $z$  are also semistable of the same phase  $\phi$ .*

The proof follows the same ideas as the proof of the triangle inequality in [24]. It uses the following  $q$ -analogue of the triangle inequality for vectors, which is [24, Lemma 3.6].

**Lemma 3.4.** *Let  $a, b, c$  be complex numbers of the form*

$$a = r_a e^{i\pi\phi_a}, \quad b = r_b e^{i\pi\phi_b}, \quad c = r_c e^{i\pi\phi_c},$$

*where  $r_a, r_b, r_c$  are positive real numbers and  $\phi_a, \phi_b, \phi_c$  are real numbers lying in an interval of length 1. Suppose  $b = a + c$ . Then*

$$r_a q^{\phi_a} + r_c q^{\phi_c} \geq r_b q^{\phi_b}$$

*with equality if and only if  $\phi_a = \phi_b = \phi_c$ .*

**Lemma 3.5.** *Let  $I$  be a half-open interval of length 1, and let  $x$  be an object in the  $I$ -heart associated to  $\tau$ . Write  $Z(x) = re^{i\pi\phi}$  for  $\phi \in I$ . Then*

$$m_{q,\tau}(x) \geq r q^{\phi},$$

*with equality if and only if  $x$  is semistable of phase  $\phi$ .*

*Proof.* Apply Lemma 3.4 inductively to the central charges of the HN factors of  $x$ .  $\square$

*Proof of Proposition 3.3.* Abbreviate  $m_{q,\tau}$  by  $m$  and assume that both  $x$  and  $z$  are nonzero. The statement is unaffected if we translate the slicing of  $\tau$ . Translate so that  $y$  has phase 0.

We first show that  $z$  lies in the  $[0, 1)$ -heart associated to  $\tau$ . Taking cohomology with respect to the corresponding  $t$ -structure gives

$$0 \rightarrow H^{-1}(z) \rightarrow H^0(x) \rightarrow H^0(y) = y \rightarrow H^0(z) \rightarrow H^1(x) \rightarrow 0.$$

Let  $s = \text{coker}(H^{-1}(z) \rightarrow H^0(x))$  and  $t = \ker(H^0(z) \rightarrow H^1(x))$ . Then we have the short exact sequence

$$0 \rightarrow s \rightarrow y \rightarrow t \rightarrow 0.$$

By the triangle inequality (Proposition 3.1), we have

$$(6) \quad m(y) \leq m(s) + m(t).$$

The triangle inequality applied to (suitable rotations of) the defining triangles of  $s$  and  $t$  gives

$$(7) \quad m(s) \leq m(H^0(x)) + qm(H^{-1}(z))$$

and

$$(8) \quad m(t) \leq m(H^0(z)) + q^{-1}m(H^1(x)).$$

By adding (7) and (8), we get

$$m(s) + m(t) \leq m(H^0(x)) + q^{-1}m(H^1(x)) + qm(H^{-1}(z)) + m(H^0(z)).$$

By (6), the left hand side dominates  $m(y)$ . For the right hand side, we use

$$(9) \quad qm(H^{-1}(z)) + m(H^0(z)) \leq \sum_i q^{-i}m(H^i(z)) = m(z),$$

where the last step uses (3). A similar inequality holds for  $x$ . Thus, the right hand side is dominated by  $m(x) + m(z)$ , which by the hypothesis is also  $m(y)$ . Hence, equalities must hold in (6), (7), (8), and (9). From (9), we conclude that  $H^i(z) = 0$  for all  $i \notin \{-1, 0\}$ . In the exact sequence

$$0 \rightarrow s \rightarrow y \rightarrow t \rightarrow 0,$$

the object  $y$  is semistable of the lowest possible phase in the heart, namely 0. It follows that  $s$  is also semistable of phase 0.

Now consider the exact sequence

$$0 \rightarrow H^{-1}(z) \rightarrow H^0(x) \rightarrow s \rightarrow 0.$$

Since  $s$  is semistable of the lowest phase in the heart, Proposition 3.2 implies that

$$m(H^0(x)) = m(s) + m(H^{-1}(z)).$$

On the other hand, we know that (7) is an equality, so

$$m(s) = m(H^0(x)) + qm(H^{-1}(z)).$$

The last two equations imply that  $H^{-1}(z) = 0$ . That is,  $z$  lies in the  $[0, 1)$ -heart.

A parallel argument shows that  $x$  lies in the  $(-1, 0]$ -heart.

Write

$$Z(x) = r_x e^{i\pi\phi_x}, \quad Z(y) = r_y, \quad Z(z) = r_z e^{i\pi\phi_z},$$

where  $r_x, r_y, r_z$  are positive real numbers,  $\phi_x \in (-1, 0]$  and  $\phi_z \in [0, 1)$ . Lemma 3.5 says

$$(10) \quad m(x) \geq r_x q^{\phi_x} \text{ and } m(z) \geq r_z q^{\phi_z}$$

with equalities if and only if  $x$  and  $z$  are semistable. Since  $Z(y) = Z(x) + Z(z)$  lies on the non-negative real axis, we must have

$$\phi_z \leq \phi_x + 1.$$

Therefore, Lemma 3.4 gives

$$(11) \quad r_x q^{\phi_x} + r_z q^{\phi_z} \geq r_y$$

with equality if and only if  $\phi_x = \phi_z = 0$ . Inequalities (10) and (11), together with the hypothesis  $r_y = m(y) = m(x) + m(z)$  imply that equalities must hold in (10) and (11). Thus,  $x$  and  $z$  are semistable of phase 0.  $\square$

**Corollary 3.6.** *Let  $y$  be a semistable object of phase  $\phi$ . Let*

$$0 = y_0 \rightarrow y_1 \rightarrow \cdots \rightarrow y_n = y$$

*be a filtration of  $y$  with factors  $a_i$ . Suppose that*

$$m_{q,\tau}(y) = \sum_{i=1}^n m_{q,\tau}(a_i).$$

*Then for all  $i$ , the object  $a_i$  is semistable of phase  $\phi$ .*

*Proof.* By definition, for every  $j$ , we have a distinguished triangle  $y_{j-1} \rightarrow y_j \rightarrow a_j \xrightarrow{+1}$ . By the triangle inequality (Proposition 3.1),

$$m_{q,\tau}(y_j) \leq m_{q,\tau}(y_{j-1}) + m_{q,\tau}(a_j).$$

Applying this result repeatedly, we see that

$$m_{q,\tau}(y_j) \leq m_{q,\tau}(y_{j-1}) + m_{q,\tau}(a_j) \leq \sum_{i=1}^j m_{q,\tau}(a_i).$$

On the other hand, we know that equality holds everywhere for  $j = n$ , and thus by induction for all  $j$ . Now the result follows by applying Proposition 3.3 inductively to each triangle  $y_{j-1} \rightarrow y_j \rightarrow a_j \xrightarrow{+1}$ .  $\square$

**Corollary 3.7.** *Let  $y$  be a  $\tau$ -stable object of phase  $\phi$ . Let*

$$0 = y_0 \rightarrow y_1 \rightarrow \cdots \rightarrow y_n = y$$

*be a filtration of  $y$  with factors  $a_i$ . Suppose that*

$$m_{q,\tau}(y) = \sum_{i=1}^n m_{q,\tau}(a_i).$$

*Then for all  $i$  except possibly one, we have  $a_i = 0$ .*

*Proof.* By Corollary 3.6, we conclude that all  $a_i$  are  $\tau$  semistable of the same phase  $\phi$ . Since  $y$  is stable, it is a simple object of abelian category of semistable objects of phase  $\phi$ . Thus, its Jordan–Hölder filtration is of length 1.  $\square$

**Remark 3.8.** The Harder–Narasimhan factors of an object  $x$  are semistable, but not necessarily stable. However, the abelian category of semistable objects of a given phase is of finite length [7, Definition 5.7]. As a result, every semistable object has a finite Jordan–Hölder filtration whose factors are stable of the same phase. By splicing together these Jordan–Hölder filtrations, we obtain a filtration of  $x$  with stable factors. We call such a filtration a *stable HN filtration*. The filtration is not unique in general, but its factors are unique up to reordering. Moreover, the mass of  $x$  is the sum of the masses of the (stable) factors in any such filtration.

**Theorem 3.9.** *Let  $\tau$  and  $\tau'$  be two stability conditions such that for each object  $x$  of  $\mathcal{C}$ , we have  $m_{q,\tau}(x) = m_{q,\tau'}(x)$ . Then an object of  $\mathcal{C}$  is  $\tau$ -stable if and only if it is  $\tau'$ -stable.*

*Proof.* Let  $x$  be a  $\tau$ -stable object. As described in Remark 3.8, consider a filtration of  $x$  with  $\tau'$ -stable factors  $y_1, \dots, y_k$  such that

$$m_{q,\tau'}(x) = \sum_{i=1}^k m_{q,\tau'}(y_i).$$

Since  $m_{q,\tau}(x) = m_{q,\tau'}(x)$  and  $m_{q,\tau}(y_i) = m_{q,\tau'}(y_i)$  for every  $i$ , we have:

$$m_{q,\tau}(x) = \sum_{i=1}^k m_{q,\tau}(y_i).$$

Since  $x$  is  $\tau$ -stable, we conclude by Corollary 3.7 that all but one of the objects  $y_i$  are zero. It follows that  $x$  is  $\tau'$ -stable.  $\square$

**Proposition 3.10.** *Let  $\tau$  and  $\tau'$  be stability conditions. Suppose that  $q_1$  and  $q_2$  are two distinct positive real numbers such that for each object  $x$  we have*

$$\begin{aligned} m_{q_1,\tau}(x) &= m_{q_1,\tau'}(x) \text{ and} \\ m_{q_2,\tau}(x) &= m_{q_2,\tau'}(x). \end{aligned}$$

*Then  $\tau = \tau'$ .*

*Proof.* Let  $x$  be any stable object of  $\tau$ . By Theorem 3.9,  $x$  is also a stable object of  $\tau'$ . Let  $\phi$  be the  $\tau$ -phase of  $x$  and  $\phi'$  be the  $\tau'$ -phase of  $x$ . We have

$$\begin{aligned} q_1^\phi |Z_\tau(x)| &= q_1^{\phi'} |Z_{\tau'}(x)| \text{ and} \\ q_2^\phi |Z_\tau(x)| &= q_2^{\phi'} |Z_{\tau'}(x)|. \end{aligned}$$

Noting that both  $|Z_\tau(x)|$  and  $|Z_{\tau'}(x)|$  are nonzero, we have  $(q_1/q_2)^\phi = (q_1/q_2)^{\phi'}$ , which gives  $\phi = \phi'$  and hence  $|Z_\tau(x)| = |Z_{\tau'}(x)|$ . We conclude that the phases and central charges of  $x$  in both  $\tau$  and  $\tau'$  are equal. Since this holds for all stable objects  $x$ , we must have  $\tau = \tau'$ .  $\square$

It is useful to record generalisations of Theorem 3.9 and Proposition 3.10 that allow us to restrict the objects  $x$ . Let  $U$  be any subset of  $\text{Stab}(\mathcal{C})$ . Let  $\mathbf{S}$  be a set of objects of  $\mathcal{C}$ . Suppose that  $U$  and  $\mathbf{S}$  satisfy the following.

**Conditions 3.11.**

- (1) If  $x \in \mathbf{S}$  and  $x \cong y$ , then  $y \in \mathbf{S}$ .
- (2) If  $x \in \mathbf{S}$  and  $\tau \in U$ , then the factors of any  $\tau$ -stable HN filtration of  $x$  (as in Remark 3.8) also lie in  $\mathbf{S}$ .

**Example 3.12.** In applications, we usually take  $U = \text{Stab}(\mathcal{C})$  or a connected component. Examples of  $U$  and  $\mathbf{S}$  that satisfy Conditions 3.11 include the following.

- (a)  $U$  is any subset of  $\text{Stab}(\mathcal{C})$ , and  $\mathbf{S}$  is the union over all  $\tau \in U$  of all  $\tau$ -stable objects.
- (b) Suppose that  $\mathcal{C}$  is 2-Calabi–Yau. Take  $U$  to be any subset of  $\text{Stab}(\mathcal{C})$ , and  $\mathbf{S}$  to be the set of all spherical objects in  $\mathcal{C}$ . The second condition holds thanks to the Mukai lemma (see, e.g., [8, Lemma 12.2] or [21, Corollary 2.3]).
- (c) Similarly, suppose that  $\mathcal{C}$  is 2-Calabi–Yau. Take  $U$  to be any subset of  $\text{Stab}(\mathcal{C})$ , and  $\mathbf{S}$  to be the set of all spherical objects in  $\mathcal{C}$  that are stable for some  $\tau \in U$ .

**Corollary 3.13.** *Let  $U \subset \text{Stab}(\mathcal{C})$  and  $\mathbf{S}$  be a set of objects of  $\mathcal{C}$  satisfying Conditions 3.11. Let  $\tau, \tau' \in U$  such that for all  $x \in \mathbf{S}$ , we have  $m_{q,\tau}(x) = m_{q,\tau'}(x)$ . Then an object  $x \in \mathbf{S}$  is  $\tau$ -stable if and only if it is  $\tau'$ -stable.*

*Proof.* The proof of Theorem 3.9 goes through verbatim.  $\square$

**Corollary 3.14.** *Let  $U \subset \text{Stab}(\mathcal{C})$  and  $\mathbf{S}$  be a set of objects of  $\mathcal{C}$  satisfying Conditions 3.11. Let  $\tau, \tau' \in U$ . Suppose that  $q_1$  and  $q_2$  are two distinct positive real numbers such that for each object  $x \in \mathbf{S}$  we have*

$$\begin{aligned} m_{q_1,\tau}(x) &= m_{q_1,\tau'}(x) \text{ and} \\ m_{q_2,\tau}(x) &= m_{q_2,\tau'}(x). \end{aligned}$$

*Then  $\tau = \tau'$ .*

*Proof.* Identical to the proof of Proposition 3.10, using Corollary 3.13 in place of Theorem 3.9.  $\square$

#### 4. THE PROJECTIVE EMBEDDING AND ITS BOUNDARY

Let  $\mathcal{C}$  be a triangulated category and  $\text{Stab}(\mathcal{C})$  the space of stability conditions on  $\mathcal{C}$ . Let  $\mathbf{S}$  be a non-empty set of nonzero objects of  $\mathcal{C}$ . Let  $\mathbf{R}^{\mathbf{S}}$  be the space of functions from  $\mathbf{S} \rightarrow \mathbf{R}$  with the product topology, and set

$$\mathbf{P}^{\mathbf{S}} = (\mathbf{R}^{\mathbf{S}} \setminus \{0\}) / \mathbf{R}^\times,$$

where  $\mathbf{R}^\times$  acts by simultaneous scaling. Fix a positive real number  $q$ .

Every stability condition  $\tau$  gives a nonzero function

$$m_{q,\tau}: \mathbf{S} \rightarrow \mathbf{R}$$

that sends  $x \in \mathbf{S}$  to its  $\tau$ -mass  $m_{q,\tau}(x)$  defined in Section 3. Changing  $\tau$  to a translate by the action of  $\mathbf{C}$  only rescales the function above. As a result, we have a well-defined map

$$(12) \quad m_q: \text{Stab}(\mathcal{C})/\mathbf{C} \rightarrow \mathbf{P}^{\mathbf{S}},$$

which we call the  $q$ -mass map. If  $q = 1$ , we drop it from the notation. The choice of  $\mathbf{S}$  is flexible, but will often satisfy Conditions 3.11 for a suitable subset of stability conditions. The aim of this section is to prove some general properties of the  $q$ -mass map.

**4.1. Pre-compactness.** Recall that an object  $p \in \mathcal{C}$  is called a (classical) generator if every object  $y \in \mathcal{C}$  is a direct summand of some object  $x \in \mathcal{C}$  that admits a filtration

$$0 \rightarrow x_0 \rightarrow \cdots \rightarrow x_m = x,$$

whose factors are shifts of  $p$ .

**Proposition 4.1** (Pre-compactness). *Suppose the set  $\mathbf{S}$  contains objects  $p_1, \dots, p_n$  such that  $p = \oplus p_i$  is a generator of  $\mathcal{C}$ . Then the closure of the image of  $m_q$  in  $\mathbf{P}^{\mathbf{S}}$  is compact.*

*Proof.* We first show that for every  $y \in \mathcal{C}$ , there is some positive real number  $N(y)$  such that for every stability condition  $\tau$ , we have

$$(13) \quad m_{q,\tau}(y) \leq N(y)m_{q,\tau}(p).$$

To see this, take  $y \in \mathcal{C}$ . Consider an object  $x \in \mathcal{C}$  such that  $y$  is a direct summand of  $x$ , and  $x$  has a filtration

$$0 = x_0 \rightarrow x_1 \rightarrow \cdots \rightarrow x_m = x$$

whose factors are  $p[a_0], \dots, p[a_m]$ . By the triangle inequality (Proposition 3.1), we have for every stability condition  $\tau$  that

$$m_{q,\tau}(x) \leq (q^{a_0} + \cdots + q^{a_m})m_{q,\tau}(p).$$

Since  $y$  is a summand of  $x$ , we also have  $m_{q,\tau}(y) \leq m_{q,\tau}(x)$ . Taking,  $N(y) = (q^{a_0} + \cdots + q^{a_m})$ , we have the desired inequality (13).

Note in particular by (13) that for every  $\tau \in \text{Stab}(\mathcal{C})/\mathbf{C}$ , we have

$$m_{q,\tau}(p) = \sum_i m_{q,\tau}(p_i) > 0.$$

Let  $\tilde{m}_q: \text{Stab}(\mathcal{C})/\mathbf{C} \rightarrow \mathbf{R}^{\mathbf{S}}$  be the lift of  $m_q$  characterised by

$$\sum_i \tilde{m}_{q,\tau}(p_i) = 1.$$

Let  $\tilde{B}$  be the closure in  $\mathbf{R}^{\mathbf{S}}$  of the image of  $\tilde{m}_q$ . By (13), we see that the image of  $\tilde{m}_q$  lies in the compact set

$$\prod_{y \in \mathbf{S}} [0, N(y)].$$

Thus,  $\tilde{B}$  is a closed subset of a compact set, and hence compact. Note that by continuity, we have for all  $\mu \in \tilde{B}$  that  $\sum \mu(p_i) = 1$ , and therefore  $\tilde{B} \subset \mathbf{R}^{\mathbf{S}} - 0$ .

Let  $\pi: (\mathbf{R}^{\mathbf{S}} - 0) \rightarrow \mathbf{P}^{\mathbf{S}}$  be the projection. Since  $\tilde{B}$  is compact, so is  $\pi(\tilde{B})$ . By construction,  $m_q(\text{Stab}(\mathcal{C})/\mathbf{C})$  is contained in the compact set  $\pi(\tilde{B}) \subset \mathbf{P}^{\mathbf{S}}$ . Therefore, its closure is a closed subset of a compact set, and hence compact.  $\square$

**Corollary 4.2.** *Under the same hypotheses as Proposition 4.1, fix positive real numbers  $q_1, \dots, q_n$ . Consider the map*

$$m_{q_1} \times \dots \times m_{q_n} : \text{Stab}(\mathcal{C})/\mathbf{C} \rightarrow \mathbf{P}^{\mathbf{S}} \times \dots \times \mathbf{P}^{\mathbf{S}}.$$

*Then the closure of its image is compact.*

*Proof.* Let  $m = m_{q_1} \times \dots \times m_{q_n}$ . Then the image of  $m$  is contained in the product of the images of  $m_{q_i}$ . Hence, the closure of the image of  $m$  is contained in the product of their closures, which is compact by Proposition 4.1. The result follows.  $\square$

**4.2. Injectivity.** The following proposition says that if  $\mathbf{S}$  is sufficiently large, then two mass maps together determine the stability condition up to translation.

**Theorem 4.3.** *Assume that  $\mathbf{S}$  contains all the stable objects in every stability condition on  $\mathcal{C}$ . Then for  $n \geq 2$  and distinct positive real numbers  $q_1, \dots, q_n$ , the map*

$$m = m_{q_1} \times \dots \times m_{q_n} : \text{Stab}(\mathcal{C})/\mathbf{C} \rightarrow \mathbf{P}^{\mathbf{S}} \times \dots \times \mathbf{P}^{\mathbf{S}}$$

*is injective.*

*Proof.* It is sufficient to show that  $m$  is injective for  $n = 2$ ; it then follows for any  $n \geq 2$ . Note also that  $\mathbf{S}$  satisfies Conditions 3.11 for  $U = \text{Stab}(\mathcal{C})$ .

Let  $\tau, \tau' \in \text{Stab}(\mathcal{C})$ , and suppose  $m(\tau) = m(\tau')$ . Then, for  $i = 1, 2$ , there exist positive real numbers  $\lambda_i$  such that

$$m_{q_i}(\tau) = \lambda_i m_{q_i}(\tau') \in \mathbf{R}^{\mathbf{S}}.$$

Let  $\tau''$  be obtained from  $\tau'$  by acting by  $u + i\pi v \in \mathbf{C}$ . Then

$$m_{q_i}(\tau'') = e^u q_i^v \in \mathbf{R}^{\mathbf{S}}$$

We choose  $u$  and  $v$  such that

$$m_{q_i}(\tau) = m_{q_i}(\tau'') \in \mathbf{R}^{\mathbf{S}}.$$

Let  $x \in \mathbf{S}$  be stable for  $\tau$ . By Corollary 3.13,  $x$  is also stable for  $\tau''$ . Now by Corollary 3.14, we have  $\tau = \tau''$ . Thus  $\tau$  and  $\tau'$  agree up to the action of  $\mathbf{C}$ .  $\square$

*Remark 4.4.* Suppose  $\mathcal{C}$  is 2-Calabi–Yau and the spherical objects in  $\mathcal{C}$  span its Grothendieck group. Then Theorem 4.3 holds even if we replace  $\mathbf{S}$  by its subset consisting only of *spherical* objects. See Corollary 3.13.

*Remark 4.5.* In general, the mass map for a single  $q$ , namely

$$m_q : \text{Stab}(\mathcal{C})/\mathbf{C} \rightarrow \mathbf{P}^{\mathbf{S}},$$

need not be injective. See [30, § 6.1] for an example, or consider the following example. For simplicity, take  $q = 1$ . Consider the quiver  $1 \rightarrow 2$  of type  $A_2$ , and let  $\mathcal{C}$  be the bounded derived category of finite-dimensional representations of this quiver. Let  $S_1$  and  $S_2$  be the irreducible representations  $k \rightarrow 0$  and  $0 \rightarrow k$  respectively. Let  $E$  be the representation  $k \xrightarrow{1} k$ , which fits into the exact sequence

$$0 \rightarrow S_2 \rightarrow E \rightarrow S_1 \rightarrow 0.$$

Recall that any object of  $\mathcal{C}$  is a direct sum of triangulated shifts of copies of  $S_1$ ,  $S_2$ , and  $E$ . Also recall that the mass of a direct sum is the sum of the masses of the summands. Consider a stability condition in which  $S_1$  and  $S_2$  are stable objects of the  $[0, 1)$  heart, such that  $\phi(S_1) < \phi(S_2)$ . Due to this inequality,  $E$  is not semistable, and its HN factors are  $S_2$  and  $S_1$ . So we have

$$m(E) = m(S_1) + m(S_2).$$

Consider two stability conditions  $\tau$  and  $\tau'$  in which  $S_1$  and  $S_2$  are stable objects of the  $[0, 1)$  heart, satisfying the following property:

$$Z_\tau(S_1) = Z_{\tau'}(S_1) = 1, \text{ and}$$

both  $Z_\tau(S_2)$  and  $Z_{\tau'}(S_2)$  are complex numbers in the upper half plane such that

$$Z_{\tau'}(S_2) = e^{ic} Z_\tau(S_2)$$

for some small  $c > 0$ . That is,  $Z_{\tau'}(S_2)$  is a slight rotation of  $Z_\tau(S_2)$ . Then it is clear that  $\tau$  and  $\tau'$  are not equal points of  $\text{Stab}(\mathcal{C})/\mathbf{C}$ , whereas  $m_\tau = m_{\tau'}$ .

*Remark 4.6.* For specific categories  $\mathcal{C}$ , we can prove stronger injectivity results. For example, for the 2-CY categories  $\mathcal{C}_\Gamma$  associated to a connected graph  $\Gamma$ , we prove in Proposition 6.14 that the mass map for a single  $q$  is injective.

*Remark 4.7.* It is a natural question to ask whether the map  $m$  from Theorem 4.3 is a homeomorphism onto its image. Since  $m$  is continuous and injective, the question is whether it is an open map. In the case of 2-CY graph categories, we prove a sufficient criterion for  $m$  to be an open map (Proposition 6.19). We check this criterion for the  $A_2$  and  $\widehat{A}_1$  cases in Proposition 7.22 and Proposition 8.15 respectively.

**4.3. Points in the boundary of the mass map.** In this section, we obtain a family of elements in  $\mathbf{P}^{\mathbf{S}}$  that lie in the closure of the image of the stability manifold. These are analogous to the intersection functionals in Teichmüller theory, and arise as limits under successive applications of a spherical twist.

We assume that the category  $\mathcal{C}$  is  $\mathbf{k}$ -linear and hom-finite with a fixed dg-enhancement. We also assume that  $\mathcal{C}$  is a Krull–Schmidt category — namely, every object decomposes into a finite direct sum of objects with local endomorphism rings. Then every object has a unique decomposition into indecomposable objects up to reordering and isomorphisms.

*Remark 4.8.* We are most interested in the case when  $\mathcal{C}$  admits a Bridgeland stability condition, and thus a bounded  $t$ -structure. In this case, it turns out that  $\mathcal{C}$  is automatically Krull–Schmidt. Indeed, an additive category is Krull–Schmidt if and only if it is Karoubian (i.e. idempotent-complete) and the endomorphism ring of every object is semiperfect (see, e.g. [11, Theorem A.1]). Any triangulated category that admits a bounded  $t$ -structure is Karoubian [32]. Finally, every finitely generated algebra over a field is semiperfect (see, e.g. [31, Example (23.3)]). By the hom-finiteness of  $\mathcal{C}$ , the endomorphism ring of any object is semiperfect.

**Definition 4.9.** Let  $a$  be a  $d$ -spherical object in  $\mathcal{C}$  and let  $q > 0$  be a fixed real number. Let  $x$  be any object of  $\mathcal{C}$  such that

$$\text{Hom}^i(x, x) = 0 \text{ if } i < 0.$$

(1) If  $x$  is indecomposable, set

$$\overline{\text{hom}}_q(a, x) = \begin{cases} \sum_i q^{-i} \dim(\text{Hom}^i(a, x)) & x \not\cong a[j] \text{ for any } j \in \mathbf{Z}, \\ q^j \cdot |q^{-d} - q^{-1}| & x \cong a[j]. \end{cases}$$

(2) If  $x = \oplus_i y_i$  for indecomposable objects  $y_i$ , set

$$\overline{\text{hom}}_q(a, x) = \sum_i \overline{\text{hom}}_q(a, y_i).$$



We should regard  $\overline{\text{hom}}_q(x, y)$  as a categorical version of the minimal intersection number between two curves. Then, in analogy with [15, §3.3]), we can regard the function  $\overline{\text{hom}}_q(a, -): \mathbf{S} \rightarrow \mathbf{R}$  as the intersection functional.

The following theorem shows that the quantity  $\overline{\text{hom}}_q(a, x)$  governs the growth of the  $\tau$ -mass of an object under repeated applications of  $\sigma_a$ .

**Theorem 4.10.** *Fix  $d \geq 2$ . Let  $\tau$  be a stability condition on  $\mathcal{C}$ . Let  $a$  be a  $\tau$ -semistable  $d$ -spherical object, and let  $x$  be an object that has no endomorphisms of negative degree. Let  $q > 0$  and set  $e = d - 1$ .*

(1) *If  $0 < q \leq 1$ , then*

$$\lim_{n \rightarrow \infty} \frac{m_{q,\tau}(\sigma_a^n x)}{q + q^{1-e} + \dots + q^{1-(n-1)e}} = m_{q,\tau}(a) \cdot \overline{\text{hom}}_q(a, x).$$

(2) *If  $q \geq 1$ , then*

$$\lim_{n \rightarrow \infty} \frac{m_{q,\tau}(\sigma_a^{-n} x)}{q^{1+e} + q^{1+2e} + \dots + q^{1+ne}} = m_{q,\tau}(a) \cdot \overline{\text{hom}}_q(a, x).$$

In particular, if  $q = 1$ , then

$$(14) \quad \lim_{n \rightarrow \pm\infty} \frac{m_\tau(\sigma_a^n x)}{n} = m_\tau(a) \cdot \overline{\text{hom}}(a, x).$$

The rest of Section 4.3 is devoted to the proof of Theorem 4.10.

Recall that for a  $d$ -spherical object  $a$  and an arbitrary object  $x$ , we have a perfect pairing

$$\text{Hom}^i(a, x) \otimes \text{Hom}^{d-i}(x, a) \rightarrow \mathbf{k},$$

which is obtained by the composition followed by a map

$$\text{Tr}: \text{Hom}^d(a, a) \rightarrow \mathbf{k}.$$

The graded vector space  $\text{Hom}^\bullet(a, a)$  is two-dimensional, generated by the identity map  $\text{id}_a$  of degree zero, and a map  $\text{loop}_a$  of degree  $d$ , which we choose so that  $\text{Tr}(\text{loop}_a) = 1$ .

Let  $x$  be any object. Then we have the distinguished triangles

$$(15) \quad \text{Hom}^\bullet(a, x) \otimes a \xrightarrow{\text{ev}} x \rightarrow \sigma_a(x) \xrightarrow{+1}$$

and

$$\sigma_a^{-1}x \rightarrow x \xrightarrow{\text{coev}} \text{Hom}^\bullet(x, a)^\vee \otimes a \xrightarrow{+1}.$$

The two triangles together give the filtration

$$(16) \quad \sigma_a^{-1}x \rightarrow x \rightarrow \sigma_a x$$

with factors  $\text{Hom}^\bullet(x, a)^\vee \otimes a$  and  $\text{Hom}^\bullet(a, x) \otimes a[1]$ . Using that  $a$  is  $d$ -spherical, we have the isomorphism

$$(17) \quad \gamma: \text{Hom}^\bullet(x, a)^\vee \xrightarrow{\sim} \text{Hom}^\bullet(a, x)[d].$$

**Lemma 4.11.** *In the setup above, suppose  $x$  does not contain any shift of  $a$  as a direct summand. Consider the filtration*

$$\sigma_a^{-1}x \rightarrow x \rightarrow \sigma_a x.$$

*Then, with the identification given by  $\gamma$  from (17), the connecting map*

$$(18) \quad \text{Hom}^\bullet(a, x) \otimes a[1] \rightarrow \text{Hom}^\bullet(a, x) \otimes a[d+1]$$

*is a shift of  $\text{id} \otimes \text{loop}_a$ .*

*Proof.* Shifted by  $-1$ , the connecting map is the composite

$$(19) \quad \mathrm{Hom}^\bullet(a, x) \otimes a \xrightarrow{\mathrm{ev}} x \xrightarrow{\mathrm{coev}} \mathrm{Hom}^\bullet(x, a)^\vee \otimes a \xrightarrow{\gamma \otimes \mathrm{id}} \mathrm{Hom}^\bullet(a, x)[d] \otimes a.$$

Extend  $\mathrm{Tr}$  on  $\mathrm{Hom}^d(a, a)$  to  $\mathrm{Hom}^\bullet(a, a)$  by setting it to be 0 on  $\mathrm{Hom}^0(a, a)$ . Then we have a perfect pairing

$$\mathrm{Hom}^\bullet(a, x) \otimes \mathrm{Hom}^\bullet(x, a) \rightarrow \mathbf{k} \text{ given by } (f, g) \mapsto \mathrm{Tr}(g \circ f).$$

Choose a homogeneous basis  $(f_i)$  of  $\mathrm{Hom}^\bullet(a, x)$ . Thanks to the perfect pairing, there exists a homogeneous basis  $(g_j)$  of  $\mathrm{Hom}^\bullet(x, a)$  such that  $\mathrm{Tr}(g_j \circ f_i) = \delta_{ij}$ . Let  $(g_j^\vee)$  be the dual basis of  $\mathrm{Hom}^\bullet(x, a)^\vee$ . In the bases  $(f_i)$  of  $\mathrm{Hom}^\bullet(a, x)$  and  $(g_j^\vee)$  of  $\mathrm{Hom}^\bullet(x, a)^\vee$ , we can consider the composite  $\mathrm{Hom}^\bullet(a, x) \otimes a \xrightarrow{\mathrm{coev} \circ \mathrm{ev}} \mathrm{Hom}^\bullet(x, a)^\vee \otimes a$  as a matrix whose entries are elements in  $\mathrm{Hom}^\bullet(a, a)$ . By definition of evaluation and coevaluation, the  $(i, j)$ th entry of this matrix is  $g_j \circ f_i$ . Since  $x$  does not have any shift of  $a$  as a direct summand, a composition  $a[m] \rightarrow x \rightarrow a[n]$  cannot be a nonzero multiple of the identity map. Therefore,  $g_j \circ f_i$  is either zero or a multiple of  $\mathrm{loop}_a$ . Since  $\mathrm{Tr}(g_j \circ f_i) = \delta_{ij}$ , we conclude that  $g_j \circ f_i = \delta_{ij} \cdot \mathrm{loop}_a$ . In other words, the matrix of  $\mathrm{coev} \circ \mathrm{ev}$  is the identity matrix times  $\mathrm{loop}_a$ .

In the bases  $(g_j^\vee)$  of  $\mathrm{Hom}^\bullet(x, a)^\vee$  and  $(f_i[d])$  of  $\mathrm{Hom}^\bullet(a, x)[d]$ , the  $(i, j)$ -entry of the matrix of  $\gamma$  is  $\mathrm{Tr}(g_j \circ f_i)$ . So this matrix is the identity matrix. It follows that the composite  $\gamma \circ \mathrm{coev} \circ \mathrm{ev}$  is  $\mathrm{id} \otimes \mathrm{loop}_a$ .  $\square$

We recall the objects  $a_n$  studied in Appendix B. By Proposition B.2, the object  $a_n$  is characterised uniquely by the existence of a filtration with factors

$$a[0], a[-(d-1)], \dots, a[-n(d-1)]$$

such that the connecting map  $a[-i(d-1)] \rightarrow a[-(i-1)(d-1)+1]$  is a shift of  $\mathrm{loop}_a$ . By Proposition B.3,  $\mathrm{Hom}^\bullet(a, a_n)$  is of dimension 2, generated by

$$i_n: a \rightarrow a_n \text{ and } l_n: a[-n(d-1)-d] \rightarrow a_n.$$

Similarly,  $\mathrm{Hom}^\bullet(a_n, a)$  is also of dimension 2, generated by

$$t_n: a_n \rightarrow a[d] \text{ and } q_n: a_n \rightarrow a[-n(d-1)].$$

We have

$$(20) \quad t_n \circ i_n = \mathrm{loop}_a, \quad q_n \circ l_n = \mathrm{loop}_a[-n(d-1)-d], \quad t_n \circ l_n = 0, \quad q_n \circ i_n = 0.$$

Recall from (15) that we have a natural map  $x \rightarrow \sigma_a x$ . We can repeatedly apply  $\sigma_a$  to this map and compose the resulting maps to obtain a map  $x \rightarrow \sigma_a^{n+1} x$  for all  $n \geq 0$ .

**Lemma 4.12.** *Let  $x \in \mathcal{C}$  be any object that does not contain a shift of  $a$  as a direct summand and which does not have endomorphisms of negative degree. For every  $n \geq 0$ , the natural map  $x \rightarrow \sigma_a^{n+1} x$  completes to a distinguished triangle*

$$\mathrm{Hom}^\bullet(a, x) \otimes a_n \rightarrow x \rightarrow \sigma_a^{n+1} x.$$

*Proof.* Recall that we have the triangle

$$(21) \quad \mathrm{Hom}^\bullet(a, x) \otimes a \rightarrow x \rightarrow \sigma_a x \xrightarrow{+1}.$$

Set  $e = d - 1$ . By repeatedly applying  $\sigma_a$  to the triangle above, we get the triangles

$$(22) \quad \mathrm{Hom}^\bullet(a, x) \otimes a[-ie] \rightarrow \sigma_a^i x \rightarrow \sigma_a^{i+1} x \xrightarrow{+1}.$$

Set  $H = \text{Hom}^\bullet(a, x)$  for brevity. Then the triangles above assemble into the diagram

$$(23) \quad \begin{array}{ccccccc} x & \xrightarrow{\quad} & \sigma_a x & \longrightarrow & \cdots & \longrightarrow & \sigma_a^n x & \xrightarrow{\quad} & \sigma_a^{n+1} x \\ & \nwarrow \text{dashed} & \swarrow & & & & \nwarrow \text{dashed} & \swarrow & \\ & H \otimes a[1] & & \cdots & \cdots & & H \otimes a[1 - ne] & & \end{array}$$

Suppose  $y_n$  completes  $x \rightarrow \sigma_a^{n+1} x$  to a distinguished triangle

$$x \rightarrow \sigma_a^{n+1} x \rightarrow y_n \xrightarrow{+1} .$$

Then (23) gives the filtration

$$(24) \quad \begin{array}{ccccccc} 0 = y_0 & \xrightarrow{\quad} & y_1 & \longrightarrow & \cdots & \longrightarrow & y_{n-1} & \xrightarrow{\quad} & y_n \\ & \nwarrow \text{dashed} & \swarrow & & & & \nwarrow \text{dashed} & \swarrow & \\ & H \otimes a[1] & & \cdots & H \otimes a[1 - (n-1)e] & & H \otimes a[1 - ne] & & \end{array}$$

$\alpha_{n-1}$        $\beta_n$        $\alpha_n$

We want to prove that  $y_n \cong H \otimes a_n[1]$ . We prove a slightly stronger statement by induction on  $n$ : there is an isomorphism  $y_n \cong H \otimes a_n[1]$  under which the map  $\alpha_n$  is a shift of  $\text{id} \otimes q_n$ .

The base case  $n = 1$  is a tautology. Assume the statement for  $n - 1$ , and identify  $y_{n-1}$  with  $H \otimes a_{n-1}[1]$  so that  $\alpha_{n-1}$  is  $\text{id} \otimes q_{n-1}$ . We claim that  $\beta_n$  is a shift of  $\text{id} \otimes l_{n-1}$ . Granting this claim, by the second distinguished triangle in Proposition B.3, it follows that we have an isomorphism  $y_n \cong H \otimes a_n[1]$  such that the map  $\alpha_n$  is  $\text{id} \otimes q_n$ . The induction step is then complete.

It remains to prove the claim about  $\beta_n$ . By choosing a homogeneous basis of  $H$ , we view  $\beta_n: H \otimes a[1 - ne] \rightarrow H \otimes a_{n-1}[1]$  as a matrix with entries in  $\text{Hom}^\bullet(a, a_{n-1})$ . Then the entries are multiples of  $l_{n-1}$  and  $i_{n-1}$ , up to shift. Observe that  $\beta_n$  is the composite  $H \otimes a[1 - ne] \rightarrow \sigma_a^n x \rightarrow y_{n-1}$ . By Lemma B.6, a shift of  $i_{n-1}$  cannot be a matrix entry of  $\beta_n$ . So all matrix entries are multiples of shifts of  $l_{n-1}$ . By the induction hypothesis, we know that  $\alpha_{n-1}$  is  $\text{id} \otimes q_{n-1}$ . From Lemma 4.11, we know that  $\alpha_{n-1} \circ \beta_n$  is a shift of  $\text{id} \otimes \text{loop}_a$ . Since  $q_{n-1} \circ l_{n-1}$  is a shift of  $\text{loop}_a$ , it follows that the matrix of  $\beta_n$  has shifts of  $l_{n-1}$  on the diagonal and no other non-zero entries.  $\square$

We now employ the machinery of *rectifiable filtrations*, as discussed in Section A, specifically Definition A.10.

**Lemma 4.13.** *Let  $d \geq 2$ . Fix a stability condition  $\tau$ . Let  $a$  be a  $\tau$ -semistable  $d$ -spherical object of  $\mathcal{C}$ . Let  $x \in \mathcal{C}$  be any object that does not have endomorphisms of negative degree and does not have any shift of  $a$  as a direct summand. There exists  $N$  (depending on  $x$  and  $a$ ) such that for every  $n \geq N$ , the filtration*

$$0 \rightarrow \sigma_a^n x \rightarrow \sigma_a^{n+1} x$$

*is rectifiable (with respect to  $\tau$ ).*

*Proof.* As before, set  $e = d - 1$ .

The factors of the filtration are  $\sigma_a^n x$  and  $\text{Hom}^\bullet(a, x) \otimes a[1 - ne]$ . We need to show that the connecting map

$$\text{Hom}^\bullet(a, x) \otimes a[1 - ne] \rightarrow \sigma_a^n x[1]$$

is rectifiable. Let us show, equivalently, that its shift

$$\chi: \text{Hom}^\bullet(a, x) \otimes a[-ne] \rightarrow \sigma_a^n x$$

is rectifiable.

Let  $\alpha$  be a real number. We need to show that the map

$$(25) \quad (\mathrm{Hom}^\bullet(a, x) \otimes a[-ne])_{\geq \alpha} \rightarrow (\sigma_a^n x)_{< \alpha+1}$$

induced by  $\chi$  is zero. Without loss of generality, assume that  $x_{\leq 0} = 0$ , which can be achieved by replacing  $x$  by a sufficiently positive shift.

First assume that  $\alpha \geq -1$ . Then, if  $n$  is large enough, we have

$$(\mathrm{Hom}^\bullet(a, x) \otimes a[-ne])_{\geq \alpha} = 0.$$

So the map (25) is also zero.

Now assume  $\alpha < -1$ . By Lemma 4.12, we have the exact triangle

$$(26) \quad \mathrm{Hom}^\bullet(a, x) \otimes a_{n-1} \rightarrow x \rightarrow \sigma_a^n x \xrightarrow{+1}.$$

It is easy to see that,  $\alpha < -1$  and  $x_{\leq 0} = 0$  imply that the  $+1$  map in (26) induces an isomorphism

$$(\sigma_a^n x)_{< \alpha+1} \rightarrow (\mathrm{Hom}^\bullet(a, x) \otimes a_{n-1}[1])_{< \alpha+1}.$$

Under this isomorphism, the map in (25) is equal to the map

$$(27) \quad (\mathrm{Hom}^\bullet(a, x) \otimes a[-ne])_{\geq \alpha} \rightarrow (\mathrm{Hom}^\bullet(a, x) \otimes a_{n-1}[1])_{< \alpha+1}.$$

The map above is induced by the connecting map in the filtration

$$x \rightarrow \sigma_a^n x \rightarrow \sigma_a^{n+1} x.$$

We have seen in the proof of Lemma 4.12 that this map (called  $\beta_n$  in (24)) is a shift of  $\mathrm{id} \otimes l_{n-1}$ . The source of  $l_{n-1}$  is  $a[-(n-1)e - d]$ , which is semistable. The target of  $l_{n-1}$  is  $a_{n-1}$ , whose HN factors are  $a[0], \dots, a[-(n-1)e]$ . Since  $d \geq 2$ , we have  $-(n-1)e - 1 \geq -(n-1)e - d$ , and hence

$$\phi([a_{n-1}[-1]]) \geq \phi(a[-(n-1)e - d]).$$

By Example A.3 (4), the map  $l_{n-1}$  is rectifiable. The map

$$\mathrm{id} \otimes l_{n-1}: \mathrm{Hom}^\bullet(x, a) \otimes a[-(n-1)e - d] \rightarrow \mathrm{Hom}^\bullet(x, a) \otimes a_{n-1}$$

is a direct sum of shifts of  $l_{n-1}$ . By Example A.3 (2) and (3), the direct sum is also rectifiable. Therefore, the map (27) is zero.  $\square$

*Proof of Theorem 4.10.* We prove the result for  $0 < q \leq 1$ ; the case of  $q \geq 1$  is analogous. We abbreviate  $m_{q,\tau}$  by  $m$  and  $\mathrm{hom}_q$  by  $\mathrm{hom}$ .

First, let  $x \cong a[j]$ . Then

$$\sigma_a^n x = a[j - ne],$$

and hence

$$m(\sigma_a^n x) = m(a) \cdot q^{j-en}.$$

Therefore,

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{m(\sigma_a^n x)}{q + q^{1-e} + \dots + q^{1-(n-1)e}} &= \lim_{n \rightarrow \infty} \frac{m(a) \cdot q^{j-en}}{q + q^{1-e} + \dots + q^{1-(n-1)e}} \\ &= m(a) \cdot q^j (q^{-d} - q^{-1}) \\ &= m(a) \cdot \overline{\mathrm{hom}}(a, x). \end{aligned}$$

Next, suppose  $x$  is indecomposable and not isomorphic to a shift of  $a$ . Let  $N$  be as guaranteed by Lemma 4.13. Then, for every  $i \geq N$ , we apply Theorem A.19 to get

$$\begin{aligned} m(\sigma_a^{i+1}x) &= m(\sigma_a^i x) + m(\operatorname{Hom}^\bullet(a, x) \otimes a[1 - ie]) \\ &= m(\sigma_a^i x) + q^{1-ie} \cdot m(a) \cdot \overline{\operatorname{hom}}(a, x). \end{aligned}$$

By taking the sum as  $i$  ranges from  $N$  to  $n-1$ , we get

$$m(\sigma_a^n x) = m(\sigma_a^N x) + \left( q^{1-(n-1)e} + \dots + q^{1-Ne} \right) \cdot m(a) \cdot \overline{\operatorname{hom}}(a, x).$$

Dividing by  $q^{1-(n-1)e} + \dots + q^1$  and letting  $n \rightarrow \infty$  yields

$$\lim_{n \rightarrow \infty} \frac{m(\sigma_a^n x)}{q + q^{1-e} + \dots + q^{1-(n-1)e}} = m(a) \cdot \overline{\operatorname{hom}}(a, x),$$

as desired.

Finally, if  $x$  is decomposable, then the desired equality follows by adding up the equalities for each indecomposable summand.  $\square$

Let  $\mathbf{S}$  be a set of objects of  $\mathcal{C}$  such that no object of  $\mathbf{S}$  has endomorphisms of negative degree. Recall from (12) the mass map

$$m_q: \operatorname{Stab}(\mathcal{C})/\mathbf{C} \rightarrow \mathbf{P}^{\mathbf{S}}.$$

We also have a map  $h_q: \mathbf{S} \rightarrow \mathbf{P}^{\mathbf{S}}$  defined as

$$(28) \quad h_q: x \mapsto [\overline{\operatorname{hom}}_q(x, -)]$$

For  $q = 1$ , we write  $m$  and  $h$  to mean  $m_q$  and  $h_q$  respectively.

**Corollary 4.14.** *Let  $d \geq 2$ . Let  $a$  be a  $d$ -spherical object that is semistable in a stability condition  $\tau$ . Then, in  $\mathbf{P}^{\mathbf{S}}$  we have the following.*

(1) *If  $0 < q \leq 1$ , then*

$$\lim_{n \rightarrow \infty} m_q(\sigma_a^n x) = h_q(a).$$

(2) *If  $q \geq 1$ , then*

$$\lim_{n \rightarrow \infty} m_q(\sigma_a^{-n} x) = h_q(a).$$

*In particular, if  $q = 1$ , then*

$$\lim_{n \rightarrow \pm \infty} m(\sigma_a^n x) = h(a).$$

*In all cases,  $h_q(a)$  is in the closure of the image of the mass map.*

*Proof.* Follows immediately from Theorem 4.10.  $\square$

**Remark 4.15** (Degenerate stability conditions). Theorem 4.10 describes one way to approach the boundary of  $m_q(\operatorname{Stab}(\mathcal{C})/\mathbf{C}) \subset \mathbf{P}^{\mathbf{S}}$ , namely by applying powers of a spherical twist. There are more direct ways to approach the boundary. Start with a stability condition  $\tau$  with heart  $\heartsuit$  and central charge  $Z: K(\mathcal{C}) \rightarrow \mathbf{C}$ . Recall that  $Z$  must map nonzero objects of  $\heartsuit$  to nonzero complex numbers. Take a deformation  $Z_t$  of  $Z$  such that  $Z_0$  sends some objects of  $\heartsuit$  to 0. Let  $\tau_t$  be the stability condition associated with  $(\heartsuit, Z_t)$ . Then the limit of  $m_q(\tau_t)$  as  $t \rightarrow 0$  exists in  $\mathbf{P}^{\mathbf{S}}$ . However, it may not be in the image of the stability manifold. We can consider the limit of  $\tau_t$  as a degenerate stability condition in the sense of [4], and the limit of  $m_q(\tau_t)$  as the mass function associated to the limiting degenerate stability condition. Note that these two kinds of limits can yield distinct boundary points (see, e.g., the results of [1]).

## 5. HARDER–NARASIMHAN (HN) AUTOMATA

The aim of this section is to introduce some general machinery for a piecewise generalisation of a  $G$ -set or a  $G$ -representation. In Sections 7.1 and 8.2.2, we apply this to the action of the auto-equivalence group on the Harder–Narasimhan filtration of an object. The reader may choose to defer reading the remainder of this section until required for the applications.

Let  $\Theta$  be a directed graph with vertex set  $\Theta_0$  and edge set  $\Theta_1$ . For an edge  $\alpha \in \Theta_1$ , let  $s(\alpha)$  and  $t(\alpha)$  denote its source and target, respectively. A *path* in  $\Theta$  is a finite sequence of edges  $(\alpha_1, \dots, \alpha_n)$ , such that for each  $i$ , we have

$$s(\alpha_{i+1}) = t(\alpha_i).$$

A  $\Theta$ -*representation*  $X$  with values in a fixed category consists of objects  $\{X_v \mid v \in \Theta_0\}$  indexed by vertices, and morphisms  $\{\phi(X)_\alpha: X_{s(\alpha)} \rightarrow X_{t(\alpha)} \mid \alpha \in \Theta_1\}$  indexed by edges. A *morphism* of  $\Theta$ -representations from  $X$  to  $Y$  consists of morphisms  $\{f_v: X_v \rightarrow Y_v \mid v \in \Theta_0\}$  such that all squares of the form

$$\begin{array}{ccc} X_{s(\alpha)} & \xrightarrow{f_{s(\alpha)}} & Y_{s(\alpha)} \\ \downarrow \phi(X)_\alpha & & \downarrow \phi(Y)_\alpha \\ X_{t(\alpha)} & \xrightarrow{f_{t(\alpha)}} & Y_{t(\alpha)} \end{array}$$

commute. A  $\Theta$ -*set* is a representation of  $\Theta$  in the category of sets. Similarly, if  $A$  is any ring, then a  $\Theta$ -*representation of  $A$ -modules* is a representation of  $\Theta$  in the category of  $A$ -modules. Clearly, any  $\Theta$ -representation of  $A$ -modules is also a  $\Theta$ -set. A  $\Theta$ -*subset*  $T \subset S$  is a morphism of  $\Theta$ -sets in which for each  $v \in \Theta_0$ , the corresponding map  $T_v \rightarrow S_v$  is an inclusion.

Fix a group  $G$ . We say that a function  $\ell: \Theta_1 \rightarrow G$  is a  $G$ -*labelling* of  $\Theta$ . We can extend this function to paths in  $\Theta$  in a natural way, as follows. If  $(\alpha_1, \dots, \alpha_n)$  is a path, then

$$\ell((\alpha_1, \dots, \alpha_n)) = \ell(\alpha_n) \cdots \ell(\alpha_1).$$

Consider a directed graph  $\Theta$  with a  $G$ -labelling  $\ell: \Theta_1 \rightarrow G$ . Let  $S$  be a set with a  $G$ -action. These data naturally give rise to a  $\Theta$ -set  $S$ , in which the object associated to each vertex  $v \in \Theta_0$  is simply  $S_v = S$ , and the map associated to any edge  $\alpha$  is the action of  $\ell(\alpha)$  on  $S$ . Similarly, a  $G$ -representation  $M$  naturally gives rise to a  $\Theta$ -representation. In particular, if  $\mathcal{C}$  is a (triangulated) category with a  $G$ -action, then  $\text{ob } \mathcal{C}$  is a  $\Theta$ -set.

We now take up the question of evolutions of HN filtrations under group actions. Let  $\mathcal{C}$  be a  $k$ -linear, hom-finite triangulated category and  $\tau$  a stability condition on  $\mathcal{C}$ . Recall from Remark 4.8 that  $\mathcal{C}$  is Krull–Schmidt. Let  $\Sigma$  be the set of all indecomposable  $\tau$ -semistable objects in the  $[0, 1)$ -heart.

**Definition 5.1.** Let  $x$  be any object of  $\mathcal{C}$  with  $\tau$ -semi-stable HN factors  $x_1, \dots, x_\ell$ . The  $\tau$ -Harder–Narasimhan multiplicity vector  $\text{HN}_\tau(x) \in \mathbf{Z}_{\geq 0}^\Sigma$  is the function that sends  $s \in \Sigma$  to

$$\text{HN}_\tau(x)(s) = \sum_{n \in \mathbf{Z}} (\text{multiplicity of } s[n] \text{ in the direct sum decomposition of } x_1 \oplus \cdots \oplus x_\ell).$$

*Remark 5.2.* The definition has a natural  $q$ -analog  $\text{HN}_{q,\tau}(x) \in \mathbf{Z}_{\geq 0}[q^\pm]^\Sigma$ , where

$$\text{HN}_{q,\tau}(x)(s) = \sum_{n \in \mathbf{Z}} (\text{multiplicity of } s[n] \text{ in the direct sum decomposition of } x_1 \oplus \cdots \oplus x_\ell) \cdot q^n.$$

We now come to the key definition that will capture the piecewise linear nature of the evolution of HN multiplicities. Recall that  $\tau$  is a stability condition on a triangulated category  $\mathcal{C}$ . The set of indecomposable  $\tau$  semistable objects in the  $[0, 1)$ -heart is denoted  $\Sigma$ , and  $G$  is a group acting on  $\mathcal{C}$ .

**Definition 5.3.** A  $\tau$ -HN automaton consists of the following data.

- (1) A directed graph  $\Theta$  together with a  $G$ -labelling  $\ell: \Theta_1 \rightarrow G$ .
- (2) A  $\Theta$ -subset  $S \subset \text{ob } \mathcal{C}$ .
- (3) A  $\Theta$ -representation  $M$  of  $\mathbf{Z}$ -modules that assigns the module  $\mathbf{Z}^\Sigma$  to every vertex such that the HN-multiplicity vector

$$S \xrightarrow{\text{HN}_\tau} M$$

defines a map of  $\Theta$ -sets.

Let us make condition (3) explicit. Let  $\alpha: v \rightarrow w$  be an edge of  $\Theta$ . Then condition (3) means that the following diagram commutes:

$$\begin{array}{ccc} S_v & \xrightarrow{\text{HN}_\tau} & \mathbf{Z}^\Sigma \\ \downarrow \ell(\alpha) & & \downarrow M(\alpha) \\ S_w & \xrightarrow{\text{HN}_\tau} & \mathbf{Z}^\Sigma. \end{array}$$

Thus, for the objects in  $S_v$ , the matrix  $M(\alpha)$  captures the evolution of HN multiplicities under the application of  $\ell(\alpha)$ .

We obtain the mass of an object by applying a linear function to the HN-multiplicity vector. Hence, an HN automaton also allows us to understand the evolution of the mass.

*Remark 5.4.* We establish some terminology for HN automata. Consider a  $\tau$ -HN automaton as explained in Definition 5.3. Following standard conventions in the theory of finite automata, we call the vertices of  $\Theta$  the *states* of the automaton. Given a state  $v$ , we say that  $S_v$  is the set of objects *supported at the state*  $v$ . Note that an object may be supported at multiple states. We say that an object is *recognised by*  $\Theta$  if it is supported at some state.

We say that an expression  $g = g_n \cdots g_1$  in  $G$  is *recognised by*  $\Theta$  if there is a path  $(\alpha_1, \dots, \alpha_n)$  in  $\Theta$  such that  $g_i = \ell(\alpha_i)$ , and hence

$$g = \ell((\alpha_1, \dots, \alpha_n)).$$

Finally, let  $x$  be an object of  $\mathcal{C}$  supported at a state  $v$ . Consider an expression  $g_n \cdots g_1 \cdot x$  for some group elements  $g_n, \dots, g_1$ . Suppose that  $g_n \cdots g_1$  is a recognised expression via a path starting at the state  $v$ . Then we say that the expression  $g_n \cdots g_1 \cdot x$  is *recognised by*  $\Theta$ .

## 6. THE CATEGORY $\mathcal{C}_\Gamma$

In this section we recall the definition and properties of 2-Calabi–Yau categories associated to graphs. Fix  $\Gamma$ , a connected undirected graph without self-loops or multiple edges.

**Definition 6.1.** The *Artin–Tits braid group associated to*  $\Gamma$ , denoted by  $B_\Gamma$  is the group generated by  $\sigma_i$ , where  $i$  ranges over the set of vertices of  $\Gamma$ , modulo the following relations:

$$\begin{aligned} \sigma_i \sigma_j \sigma_i &= \sigma_j \sigma_i \sigma_j & \text{if there is an edge between } i \text{ and } j, \\ \sigma_i \sigma_j &= \sigma_j \sigma_i & \text{otherwise.} \end{aligned}$$

**Definition 6.2.** The *Coxeter group associated to*  $\Gamma$ , denoted by  $W_\Gamma$ , is the quotient of  $B_\Gamma$  by the following additional relations. For each vertex  $i$  of  $\Gamma$ , set

$$\sigma_i^2 = 1.$$

Denote the image of  $\sigma_i$  in  $W_\Gamma$  by  $s_i$ .

Let  $V_\Gamma$  be the  $\mathbf{Z}$ -module with basis vectors  $v_i$  indexed by the vertices of  $\Gamma$ . Define a bilinear form on  $V_\Gamma$  by the following formula:

$$\langle v_i, v_j \rangle = \begin{cases} 2 & \text{if } i = j, \\ -1 & \text{if } i \text{ and } j \text{ are connected by an edge,} \\ 0 & \text{otherwise.} \end{cases}$$

**Definition 6.3.** The *standard representation* of  $W_\Gamma$  is the action on  $V_\Gamma$  defined by the formula

$$s_i(v_j) = v_j - \langle v_i, v_j \rangle v_i.$$

We describe a category  $\mathcal{C}_\Gamma$  with a (weak) action of  $B_\Gamma$  that categorifies the standard representation of  $W_\Gamma$ . The construction is via the zig-zag algebra, which was introduced in [19]. A construction of  $\mathcal{C}_\Gamma$  for type  $A_n$  graphs is in [36, 39].

We first recall the zig-zag algebra. A version of this construction is also in [2, §2.3], but in that version, we have not been careful with signs; see Remarks 6.6, 6.10 and 6.11.

Let  $\Gamma^{\text{dbl}}$  denote the doubled quiver of  $\Gamma$ . This is the directed graph obtained by replacing each edge of  $\Gamma$  by a pair of oppositely oriented arrows between its endpoints. For each original edge  $e$  in  $\Gamma$  between  $i$  and  $j$ , we let  $e_{ij}$  and  $e_{ji}$  be the corresponding arrows in  $\Gamma^{\text{dbl}}$  from  $i$  to  $j$  and  $j$  to  $i$  respectively. Let  $\text{Path}(\Gamma^{\text{dbl}})$  denote the path algebra of  $\Gamma^{\text{dbl}}$ . As a vector space over  $\mathbf{k}$ , the algebra  $\text{Path}(\Gamma^{\text{dbl}})$  is spanned by all oriented paths in  $\Gamma^{\text{dbl}}$ , and the multiplication is given by concatenation. The length function on paths that declares arrows to have length one induces a non-negative grading on  $\text{Path}(\Gamma^{\text{dbl}})$ .

We define the zigzag algebra of  $\Gamma$ , as follows.

**Definition 6.4.** Suppose  $\Gamma$  has at least two vertices. Fix a sign  $s_{ij} \in \{+1, -1\}$  for each arrow  $e_{ij}$  in  $\Gamma^{\text{dbl}}$  such that  $s_{ij} = -s_{ji}$ . Let  $A(\Gamma) = A_s(\Gamma)$  be the quotient of the path algebra  $\text{Path}(\Gamma^{\text{dbl}})$  by the two-sided ideal generated by the following elements:

- (1) all length three paths,
- (2) length two paths whose source and target differ,
- (3) for each triple of vertices  $i, j, k$  of  $\Gamma$  such that there are arrows  $e_{ij}$  and  $e_{ik}$ , the element

$$s_{ij}e_{ij}e_{ji} - s_{ik}e_{ik}e_{ki}.$$

If  $\Gamma$  has one vertex, (type  $A_1$ ), set  $A(\Gamma) = \mathbf{C}[x]/x^2$ . We call  $A(\Gamma)$  the *zigzag algebra* of  $\Gamma$ .

Note that the third relations in the definition above imply that all degree-two paths starting and ending at the same vertex are equal up to sign. We call one such nonzero path a “loop” at a vertex. When  $\Gamma$  has at least two vertices, the relations of  $A(\Gamma)$  are all homogeneous, and so the natural grading on  $\text{Path}(\Gamma^{\text{dbl}})$  descends to a grading on  $A(\Gamma)$ . When  $\Gamma$  has a single vertex, grade  $A(\Gamma) = \mathbf{C}[x]/x^2$  by setting  $\deg(x) = 2$ .

The choice of signs does not matter.

**Proposition 6.5.** Consider another choice of signs  $s'_{ij} \in \{+1, -1\}$  such that  $s'_{ij} = -s'_{ji}$ . Then the map

$$e_{ij} \mapsto s_{ij}s'_{ij}e_{ij}$$

induces an isomorphism of graded algebras  $A_s(\Gamma) \cong A_{s'}(\Gamma)$ .



*Proof.* We leave the proof to the reader.  $\square$

*Remark 6.6.* In [2], we gave a definition of the zig-zag algebra that ignored the choice of signs in Definition 6.4, in effect taking  $s_{ij} = s_{ji} = 1$ . This unsigned version also appears in [19] and [39] for type  $A$ . Let  $B(\Gamma)$  denote this unsigned version of the zig-zag algebra, namely the algebra with the same generators and relations as  $A(\Gamma)$  from Definition 6.4, but in which we set  $s_{ij} = 1$ .

We prefer to work with the algebra  $A(\Gamma)$  in this paper, as it gives rise to a strongly 2-Calabi–Yau category  $\mathcal{C}_\Gamma$ ; see Remarks 6.10 and 6.11. However, if the undirected graph  $\Gamma$  is bipartite, the two definitions are equivalent, as stated in Proposition 6.7. In particular, the definitions are equivalent for finite type  $ADE$ .

The following is easy to check; we omit the proof.

**Proposition 6.7.** *Let  $\Gamma = \{\Gamma_1, \Gamma_2\}$  be an undirected bipartite graph without self-loops or multiple edges, in which each edge connects a vertex of  $\Gamma_1$  with a vertex of  $\Gamma_2$ . Let  $A(\Gamma)$  be the zig-zag algebra as in Definition 6.4 and  $B(\Gamma)$  the unsigned zig-zag algebra. Consider the map from  $A(\Gamma)$  to  $B(\Gamma)$  in which*

$$e_{ij} \mapsto \begin{cases} s_{ij}e_{ij} & \text{if } i \in \Gamma_1 \text{ and } j \in \Gamma_2, \\ e_{ij} & \text{if } i \in \Gamma_2 \text{ and } j \in \Gamma_1. \end{cases}$$

*This map is an algebra isomorphism.*

The minimal idempotents of  $A(\Gamma)$  are the length zero paths  $(i)$  for  $i \in \Gamma$ . The modules  $P_i = A(\Gamma)(i)$  are indecomposable projective left  $A(\Gamma)$  modules. Any finitely-generated graded projective left  $A(\Gamma)$  module is isomorphic to a direct sum of grading shifts of the modules  $P_i$ .

The category of complexes of graded projective left  $A(\Gamma)$  modules admits two “shift” functors, namely the homological shift and the grading shift. We work in a simpler setting where the two shifts are identified. We construct the desired category as follows.

Regard  $A(\Gamma)$  as a differential graded algebra (dga) where all differentials are zero. Let  $K(\text{dgmod-}A)$  be the category of finite-dimensional differential graded modules (dgms) over the dga  $A(\Gamma)$ . Morphisms in  $K(\text{dgmod-}A)$  are homotopy classes of chain maps that are compatible with the action of  $A(\Gamma)$ . Then  $K(\text{dgmod-}A)$  is a triangulated category (see, e.g. [38, §22.8]).

**Definition 6.8.** Set  $\mathcal{C}_\Gamma$  to be the smallest full and strict triangulated subcategory generated by the objects  $P_i$  as  $i$  ranges over the vertices of  $\Gamma$ .

*Remark 6.9.* The categories considered in [36] and [39] are described in a slightly different way. However, [2, Proposition 2.1] shows that they coincide with our category  $\mathcal{C}_\Gamma$ . The category considered in [19] is the category of graded projective  $A(\Gamma)$ -modules; see [2, § 2.3.3] for a discussion on how it is related to  $\mathcal{C}_\Gamma$ .

We state the salient properties of  $\mathcal{C}_\Gamma$  from [2, § 2.3].

- (1)  $\mathcal{C}_\Gamma$  is a  $\mathbf{k}$ -linear triangulated category.
- (2)  $\mathcal{C}_\Gamma$  is (strongly) 2-Calabi–Yau. That is, for every pair of objects  $x, y \in \mathcal{C}$  and  $i \in \mathbf{Z}$ , we have a natural non-degenerate graded symmetric pairing

$$\text{Hom}^i(x, y) \otimes \text{Hom}^{2-i}(y, x) \rightarrow \mathbf{k}.$$

That is, the pairing is symmetric for even  $i$  and skew-symmetric for odd  $i$ .

(3)  $\mathcal{C}_\Gamma$  is classically generated by the objects  $P_i$ . These objects satisfy

$$\begin{aligned} \mathrm{Hom}(P_i, P_i[n]) &= \begin{cases} \mathbf{k} & \text{if } n = 0 \text{ or } n = 2, \\ 0 & \text{otherwise;} \end{cases} \\ \mathrm{Hom}(P_i, P_j[n]) &= \begin{cases} \mathbf{k} & \text{if } n = 1 \text{ and } i \text{ and } j \text{ are neighbours,} \\ 0 & \text{otherwise, for } i \neq j. \end{cases} \end{aligned}$$

(4) The extension closure of the objects  $P_i$  is an abelian category, which is the heart of a bounded t-structure on  $\mathcal{C}_\Gamma$ . We call this t-structure the *standard t-structure* on  $\mathcal{C}_\Gamma$ , and refer to its heart as the *standard heart*  $\heartsuit_{\mathrm{std}}$ . The simple objects of  $\heartsuit_{\mathrm{std}}$  are  $P_1, \dots, P_n$ . Every object of  $\heartsuit_{\mathrm{std}}$  has a Jordan–Hölder filtration of finite length.

The Grothendieck group  $K_\Gamma$  of  $\mathcal{C}_\Gamma$  is a  $\mathbf{Z}$ -module with a pairing given by

$$\langle [x], [y] \rangle = \bigoplus_i (-1)^i \dim \mathrm{Hom}^i(x, y).$$

It is easy to check that we have an isomorphism  $V_\Gamma \rightarrow K_\Gamma$  compatible with the pairing, defined by  $v_i \mapsto [P_i]$ .

*Remark 6.10.* The pairing

$$\mathrm{Hom}^i(x, y) \otimes \mathrm{Hom}^{2-i}(y, x) \rightarrow \mathbf{k}$$

is defined exactly as in [2, § 2.3]. The choice of signs in Definition 6.4 gives the graded symmetry, and hence a strongly 2-Calabi–Yau category (see [28]).

*Remark 6.11.* The zig-zag algebra  $A(\Gamma)$  arises naturally as follows. Let  $\{P_i \mid i \in \Gamma\}$  be a collection of spherical objects in a strongly 2-CY triangulated category. Assume that for  $i \neq j$ , we have

$$\mathrm{Hom}(P_i, P_j[1]) \cong \begin{cases} \mathbf{k} & \text{if } i-j \text{ is an edge of } \Gamma, \\ 0 & \text{otherwise.} \end{cases}$$

Set  $P = \bigoplus_{i \in \Gamma} P_i$ . Then the endomorphism algebra of  $P$  is isomorphic to the zig-zag algebra  $A(\Gamma)$  as defined in Definition 6.4. We describe an explicit isomorphism

$$\psi: A(\Gamma) \rightarrow \mathrm{End}(P).$$

Choose a sign  $s_{ij} \in \{\pm 1\}$  for each edge  $e_{ij}$  of  $\Gamma^{\mathrm{dbl}}$  such that  $s_{ij} = -s_{ji}$ . For edge  $e_{ij}$  with  $s_{ij} = 1$ , let  $\psi(e_{ij}) \in \mathrm{Hom}^1(P_i, P_j)$  be any nonzero element and  $\psi(e_{ji})$  be the unique element such that

$$\langle \psi(e_{ij}), \psi(e_{ji}) \rangle = 1.$$

Then it is easy to check that  $\psi: A(\Gamma) \rightarrow \mathrm{End}(P)$  is an isomorphism of graded algebras.

Recall that the indecomposable projective modules  $P_i$  in  $\mathcal{C}_\Gamma$  are spherical. Furthermore, the spherical twists in the  $P_i$  satisfy the defining relations of the Artin–Tits braid group  $B_\Gamma$  associated to  $\Gamma$  (see, e.g., [19]). That is,

$$\begin{aligned} \sigma_{P_i} \sigma_{P_j} \sigma_{P_i} &\cong \sigma_{P_j} \sigma_{P_i} \sigma_{P_j} \text{ when } i \text{ and } j \text{ are connected by an edge of } \Gamma, \\ \sigma_{P_i} \sigma_{P_j} &\cong \sigma_{P_j} \sigma_{P_i} \text{ when } i \text{ and } j \text{ are not connected by an edge of } \Gamma. \end{aligned}$$

Via the homomorphism  $B_\Gamma \rightarrow \mathrm{Aut}(\mathcal{C}_\Gamma)$  defined by  $\sigma_i \mapsto \sigma_{P_i}$ , we have a (weak) action of  $B_\Gamma$  on  $\mathcal{C}_\Gamma$ . This action is known to be faithful in some cases—for example when  $\Gamma$  is an ADE Dynkin diagram [6, 29]—and is conjecturally faithful for all  $\Gamma$ . The induced action on the Grothendieck group  $K_\Gamma$  is the standard (geometric) representation of the Coxeter group associated to  $\Gamma$ .

Let us introduce some notation for later use. Given an edge  $e_{ij}$  in  $\Gamma^{\text{dbl}}$ , let  $P_{ij}$  denote the cone of a nonzero morphism  $P_i[-1] \rightarrow P_j$ . Since  $\Gamma$  has no multiple edges, we can check that  $P_{ij} \cong \sigma_i(P_j)$ . Thus  $P_{ij}$  is spherical, and fits in an exact sequence

$$0 \rightarrow P_j \rightarrow P_{ij} \rightarrow P_i \rightarrow 0$$

in the standard heart.

**6.1. Injectivity of the mass map.** We now fix the undirected graph  $\Gamma$  without self-loops or multiple edges. Let  $\mathcal{C} = \mathcal{C}_\Gamma$  be the associated strongly 2-CY category. We say that a stability condition  $\tau$  is *standard* if its  $[0, 1)$  heart is the standard heart  $\heartsuit_{\text{std}}$ . It is easy to see that the set of standard stability conditions is connected. Let  $\text{Stab}^\circ \mathcal{C} \subset \text{Stab} \mathcal{C}$  be the connected component containing the standard stability conditions.

Let  $\mathbf{S}$  be the set of spherical objects  $x$  of  $\mathcal{C}$  such that  $x$  is stable with respect to some stability condition in  $\text{Stab}^\circ \mathcal{C}$ . Note that  $\mathbf{S}$  satisfies Conditions 3.11 for  $U = \text{Stab}^\circ(\mathcal{C}_\Gamma)$ .

Fix some  $q > 0$ . The goal of Section 6.1 is to show that the map  $m_q: \text{Stab}^\circ(\mathcal{C})/\mathbf{C} \rightarrow \mathbf{P}^{\mathbf{S}}$  is injective. We need two lemmas.

**Lemma 6.12.** *Let  $\tau$  be a standard stability condition on  $\mathcal{C} = \mathcal{C}_\Gamma$ . Let  $\tau'$  be any stability condition in  $\text{Stab}^\circ(\mathcal{C})$  such that for all  $x \in \mathbf{S}$ , we have*

$$m_{q,\tau}(x) = m_{q,\tau'}(x).$$

*Label the vertices of the graph  $\Gamma$  so that the vertices labelled 1 and 2 form a subgraph of type  $A_2$ . Suppose that  $\phi_\tau(P_1) \neq \phi_\tau(P_2)$  and  $\phi_{\tau'}(P_1) = \phi_\tau(P_1)$ . Then  $\phi_{\tau'}(P_2) = \phi_\tau(P_2)$ .*

*Proof.* Swapping the labels if required, suppose that  $\phi_\tau(P_1) < \phi_\tau(P_2)$ . Thus the argument of the central charge of  $P_{21}$  lies strictly between these two; that is

$$\phi_\tau(P_1) < \arg(Z_\tau(P_{21})) < \phi_\tau(P_2).$$

Since the only proper sub-object of  $P_{21}$  in the standard heart is  $P_1$ , we observe that  $P_{21}$  is  $\tau$ -stable. The previous inequality gives us

$$(29) \quad \phi_\tau(P_1) < \phi_\tau(P_{21}) < \phi_\tau(P_1) < \phi_\tau(P_1) + 1.$$

Recall from Corollary 3.13 that  $\tau$  and  $\tau'$  have the same stable spherical objects. In particular,  $P_{21}$  is also  $\tau'$ -stable. Since we know that

$$\text{Hom}(P_1, P_{21}) \neq 0, \quad \text{Hom}(P_{21}, P_2) \neq 0, \quad \text{and} \quad \text{Hom}(P_2, P_1[1]) \neq 0,$$

the analogous inequalities hold for  $\phi_{\tau'}$ :

$$(30) \quad \phi'_\tau(P_i) < \phi'_\tau(P_{ji}) < \phi'_\tau(P_j) < \phi'_\tau(P_i) + 1.$$

We also know that

$$m_{q,\tau'}(P_1) = m_{q,\tau}(P_1), \quad m_{q,\tau'}(P_{21}) = m_{q,\tau}(P_{21}), \quad m_{q,\tau'}(P_2) = m_{q,\tau}(P_2).$$

By the  $q$ -SSS lemma [1, Lemma 5.2], it follows that the pair of complex numbers  $(Z_\tau(P_1), Z_\tau(P_2))$  is equal to the pair  $(Z_{\tau'}(P_1), Z_{\tau'}(P_2))$  possibly after a rotation and a reflection. The inequalities (29) and (30) imply that no reflection is necessary, and the equality  $\phi_\tau(P_1) = \phi_{\tau'}(P_1)$  implies that no rotation is necessary. We conclude that  $Z_\tau(P_2) = Z_{\tau'}(P_2)$ ; call this number  $z$ . Since both  $\phi = \phi_\tau(P_2)$  and  $\phi = \phi_{\tau'}(P_2)$  have the property that  $e^{i\pi\phi}$  lies on the same real ray as  $z$ , and both lie in the interval  $[\phi_\tau(P_1), \phi_\tau(P_1) + 1)$ , we conclude that they are equal.  $\square$

**Lemma 6.13.** *Let  $\tau$  be a standard stability condition on  $\mathcal{C}_\Gamma$ . Let  $\tau'$  be any stability condition such that for all  $x \in \mathbf{S}$ , we have*

$$m_{q,\tau}(x) = m_{q,\tau'}(x).$$

*Label the vertices of the graph  $\Gamma$  so that vertices 1 and 2 form a subgraph of type  $A_2$ . Moreover, suppose  $\phi_\tau(P_1) = \phi_\tau(P_2)$ . Then  $\phi_{\tau'}(P_1) = \phi_{\tau'}(P_2)$ . In particular, if  $\phi_{\tau'}(P_1) = \phi_\tau(P_1)$ , then we also have  $\phi_{\tau'}(P_2) = \phi_\tau(P_2)$ .*

*Proof.* Recall from Corollary 3.13 that  $\tau$  and  $\tau'$  have the same stable spherical objects. In particular  $P_1$  and  $P_2$  are also  $\tau'$ -stable.

Let  $x = P_{12}$ . We claim that the  $\tau'$ -stable factors in the  $\tau'$ -HN filtration of  $x$  are precisely  $P_1$  and  $P_2$ . To see this, consider a filtration

$$(31) \quad 0 = x_0 \rightarrow x_1 \rightarrow \cdots \rightarrow x_n = x$$

whose factors  $a_i$  are  $\tau'$ -stable and appear in the order of non-increasing phase. By the Mukai lemma [8, Lemma 12.2] and [21, Corollary 2.3], the  $a_i$  are spherical. By Corollary 3.13, they are also  $\tau$ -stable. We have

$$m_{q,\tau'}(x) = \sum_i m_{q,\tau'}(a_i),$$

and hence

$$m_{q,\tau}(x) = \sum_i m_{q,\tau}(a_i).$$

But  $x$  is  $\tau$ -semistable. Therefore by Proposition 3.3, we conclude that the  $\tau$ -stable objects  $a_i$  all have  $\tau$ -phase  $\phi = \phi_\tau(x)$ . Since the  $a_i$  are  $\tau$ -stable, they are simple objects of the category  $\mathcal{C}_{\tau,\phi}$ . Thus, (31) is a Jordan–Hölder filtration of  $x$  in  $\mathcal{C}_{\tau,\phi}$ . But we know that  $x$  has a unique Jordan–Hölder filtration in  $\mathcal{C}_{\tau,\phi}$ , namely

$$(32) \quad \begin{array}{ccccc} 0 & \xrightarrow{\quad} & P_2 & \xrightarrow{\quad} & x. \\ & \swarrow \text{dashed} & \searrow & \swarrow \text{dashed} & \searrow \\ & & P_2 & & P_1 \end{array}$$

Therefore, the filtration (31) must coincide with (32). That is,  $P_1$  and  $P_2$  are the  $\tau'$ -stable HN factors of  $x$ , with

$$\phi_{\tau'}(P_2) \geq \phi_{\tau'}(P_1).$$

By the same argument, the  $\tau'$ -stable factors of  $y = P_{21}$  are also  $P_1$  and  $P_2$ , with

$$\phi_{\tau'}(P_1) \geq \phi_{\tau'}(P_2).$$

It follows that  $\phi_{\tau'}(P_1) = \phi_{\tau'}(P_2)$ . □

**Proposition 6.14** (Injectivity). *Let  $\Gamma$  be a finite connected graph without self-loops or multiple edges. Let  $q > 0$  be a real number. Then the mass map  $m_q: \text{Stab}^\circ(\mathcal{C}_\Gamma)/\mathbf{C} \rightarrow \mathbf{P}^{\mathbf{S}}$  is injective.*

*Proof.* Let  $\tau$  be a stability condition in  $\text{Stab}^\circ(\mathcal{C})$ . By [23, Proposition 4.13],  $\tau$  is in the braid group orbit of a standard stability condition up to the action of  $\mathbf{C}$ . (Although [23] treats the case of preprojective algebras, the same proof works in our setting.) By applying a braid and an element of  $\mathbf{C}$ , suppose that  $\tau$  is standard. Since the objects  $P_i$  are simple objects of the standard heart, they are  $\tau$ -stable.

Let  $\tau'$  be another stability condition in  $\text{Stab}^\circ(\mathcal{C})$  such that  $m_q(\tau) = m_q(\tau') \in \mathbf{P}^{\mathbf{S}}$ . By scaling the central charge of  $\tau'$ , we may assume that  $m_{q,\tau} = m_{q,\tau'}$  on  $\mathbf{S}$ . By Corollary 3.13,  $\tau$  and  $\tau'$  have the same spherical stable objects.

Label the vertices of  $\Gamma$  by  $\{1, \dots, n\}$  so that for each  $i > 1$  there is some  $j < i$  such that the subgraph induced by  $\{i, j\}$  is of type  $A_2$ . By translating the slicing of  $\tau'$  if necessary, assume that  $\phi_{\tau'}(P_1) = \phi_{\tau}(P_1)$ . Rescale the central charge again to ensure that  $m_{q,\tau} = m_{q,\tau'}$  continues to hold on  $\mathbf{S}$ . Then  $Z_{\tau}(P_1) = Z_{\tau'}(P_1)$ .

We now prove by induction on  $i$  that  $\phi_{\tau}(P_i) = \phi_{\tau'}(P_i)$  and  $Z_{\tau}(P_i) = Z_{\tau'}(P_i)$ . The base case  $i = 1$  holds by construction. For the induction step, let  $i > 1$ . Consider  $j < i$  such that the subgraph induced by  $\{i, j\}$  is of type  $A_2$ . By the induction hypothesis, we have  $\phi_{\tau'}(P_j) = \phi_{\tau}(P_j)$  and  $Z_{\tau'}(P_j) = Z_{\tau}(P_j)$ . Depending on whether or not the quantities  $\phi_{\tau}(P_i)$  and  $\phi_{\tau}(P_j)$  are equal, we are now in the setting of one of Lemmas 6.12 and 6.13. In each case, we conclude that  $\phi_{\tau'}(P_i) = \phi_{\tau}(P_i)$ . Since we already have  $m_{q,\tau}(P_i) = m_{q,\tau'}(P_i)$ , we conclude that  $Z_{\tau}(P_i) = Z_{\tau'}(P_i)$ .

Since  $\phi_{\tau}(P_i) = \phi_{\tau'}(P_i)$  and  $0 \leq \phi_{\tau}(P_i) < 1$ , the same holds for  $\phi_{\tau'}(P_i)$ . In particular,  $P_i$  lies in the  $[0, 1)$  heart of  $\tau'$ . As a result,  $\heartsuit_{\text{std}}$  is contained in the  $[0, 1)$ -heart of  $\tau'$ . Since both are hearts of bounded  $t$ -structures, they must be equal. Furthermore, since the  $P_i$  span the Grothendieck group and  $Z_{\tau}(P_i) = Z_{\tau'}(P_i)$ , we have  $Z_{\tau} = Z_{\tau'}$ . So  $\tau'$  is a stability condition with the same heart and central charge as  $\tau$ . We conclude that  $\tau' = \tau$ .  $\square$

**6.2. Homeomorphism onto the image.** Having proved injectivity, we take up the question of  $m_q$  being a homeomorphism onto its image.

As before, let  $\mathcal{C}$  be the 2-CY category associated to a finite connected graph without self-loops or multiple edges, and fix a real number  $q > 0$ . We begin by analysing  $m_q$  on standard stability conditions. Let  $U \subset \text{Stab}(\mathcal{C})/\mathbf{C}$  be the image of the set of standard stability conditions.

**Lemma 6.15.** *We have the following.*

- (1) *Let  $\tau \in \text{Stab}(\mathcal{C})$ . The image of  $\tau$  in  $\text{Stab}(\mathcal{C})/\mathbf{C}$  lies in  $U$  if and only if all  $P_i$  are  $\tau$ -semistable and their  $\tau$ -phases lie in an open interval of length 1. In this case, the  $P_i$  are in fact  $\tau$ -stable.*
- (2)  *$U \subset \text{Stab}(\mathcal{C})/\mathbf{C}$  is open.*

*Proof.* We prove statement (1) of the lemma. If the image of  $\tau$  lies in  $U$ , then by changing  $\tau$  by the  $\mathbf{C}$  action, we may assume that  $\tau$  is standard. Then the  $P_i$  are stable and their phases lie in  $[0, 1)$ . Since we have finitely many  $P_i$ , we may in fact choose the phases to lie in  $(0, 1)$ , as required. Conversely, suppose all  $P_i$  are  $\tau$ -semistable and their phases lie in an open interval of length 1. Using the  $\mathbf{C}$  action, we may assume that the interval is  $(0, 1)$ . Then the  $[0, 1)$ -heart of  $\tau$  contains the standard heart and hence must be the standard heart. As a result,  $\tau$  is standard. Since the  $P_i$  are simple objects of the standard heart, they are  $\tau$ -stable.

We now prove statement (2) of the lemma. Given a stability condition  $\tau$  and an object  $x$ , denote by  $\phi^+(x, \tau)$  (resp.  $\phi^-(x, \tau)$ ) the highest (resp. lowest) phase among the  $\tau$ -HN factors of  $x$ . Then  $\phi^{\pm}(x, -)$  are continuous functions on  $\text{Stab}(\mathcal{C})$  [7, Proposition 8.1]. The difference of two such functions is a continuous function on  $\text{Stab}(\mathcal{C})/\mathbf{C}$ .

We claim that the subset  $U \subset \text{Stab}(\mathcal{C})/\mathbf{C}$  is defined by

$$(33) \quad \max_i(\phi^+(P_i, -)) - \min_i(\phi^-(P_i, -)) < 1.$$

It follows that  $U$  is open. To see the claim, note that statement (1) implies that all points of  $U$  satisfy (33). Conversely, let  $\tau \in \text{Stab}(\mathcal{C})$  be such that its image  $[\tau] \in \text{Stab}(\mathcal{C})/\mathbf{C}$  satisfies (33). By changing  $\tau$  by the  $\mathbf{C}$  action if necessary, we can arrange that  $\phi^{\pm}(P_i, \tau) \in (0, 1)$ . But then the  $P_i$  lie in the  $[0, 1)$ -heart of  $\tau$ . Therefore, the  $[0, 1)$ -heart of  $\tau$  contains

the standard heart, and hence is the standard heart. As a result  $\tau$  is standard, and hence  $[\tau]$  lies in  $U$ .  $\square$

Let  $W \subset \mathbf{C}^n$  be the set of complex numbers  $(z_1, \dots, z_n)$  satisfying:

- (1)  $z_i \neq 0$  for all  $i$ ,
- (2)  $z_1, \dots, z_n$  lie in an open half plane in  $\mathbf{C}$ .

We have an action of  $\mathbf{C}^\times$  on  $W$  by scaling. We have a map  $h: W/\mathbf{C}^\times \rightarrow U$  defined as follows. Given  $w \in W/\mathbf{C}^\times$ , choose a representative  $(z_1, \dots, z_n) \in W$  so that all  $z_i$  lie in the upper half plane. Let  $h(w)$  be the standard stability condition such that  $Z(P_i) = z_i$ . It is easy to check that  $h: W/\mathbf{C}^\times \rightarrow U$  is a homeomorphism.

Recall that for edge  $e_{ij}$  in  $\Gamma^{\text{dbl}}$ , we denote by  $P_{ij}$  the spherical object in the standard heart defined by a (non-split) exact sequence  $0 \rightarrow P_i \rightarrow P_{ij} \rightarrow P_j \rightarrow 0$ . Let  $F \subset \mathbf{S}$  be the finite set

$$F = \{P_i \mid i \text{ is a vertex of } \Gamma\} \cup \{P_{ij} \mid ij \text{ is an edge of } \Gamma\}.$$

**Proposition 6.16.** *The map  $m_q \circ h: W/\mathbf{C}^\times \rightarrow \mathbf{P}^F$  is a homeomorphism onto its image.*

*Proof.* The basic idea is the same as in Proposition 6.14: for a connected graph without self-loops or multiple edges, the masses of  $P_i$ 's and  $P_{ij}$ 's suffice to determine the central charge. The only additional ingredient is to keep track of the topology, which we do using compactness. In this proof, we denote the complex number  $\sqrt{-1}$  by  $\mathbf{i}$  to distinguish it from the index  $i$ .

Let  $W_1 \subset W$  be the set of  $(z_1, \dots, z_n)$  where  $z_1 = 1$ . Then  $W_1 \rightarrow W/\mathbf{C}^\times$  is a homeomorphism. Likewise, let  $M \subset \mathbf{R}_{\geq 0}^F$  be the set of  $m: F \rightarrow \mathbf{R}$  where  $m(P_1) = 1$ . Then  $M \rightarrow \mathbf{P}^F$  is a homeomorphism onto an open subset. The map  $W/\mathbf{C}^\times \rightarrow \mathbf{P}^F$  naturally lifts to a map  $W_1 \rightarrow M$ . It suffices to prove that  $W_1 \rightarrow M$  is a homeomorphism.

For  $(z_1 = 1, z_2, \dots, z_n) \in W_1$ , all  $z_i$  must be away from the negative real axis. We have the homeomorphism

$$(34) \quad \mathbf{R}_{>0} \times (-1, 1) \rightarrow \mathbf{C} - \mathbf{R}_{\leq 0} \text{ given by } (r, \theta) \mapsto re^{i\pi\theta}.$$

Consider  $V \subset (\mathbf{R}_{>0} \times (-1, 1))^n$  consisting of  $((r_1, \theta_1), \dots, (r_n, \theta_n))$  where  $(r_1, \theta_1) = (1, 0)$  and  $\max(\theta_i) - \min(\theta_i) < 1$ . Then (34) gives a homeomorphism  $V \rightarrow W_1$ . We prove that  $f: V \rightarrow M$ , induced by  $m_q \circ h$ , is a homeomorphism onto its image.

Let us write  $f$  explicitly. Given  $((r_1, \theta_1), \dots, (r_n, \theta_n)) \in V$ , let  $m \in M$  be its image under  $f$ . Then  $m$  is the mass function of the stability condition associated to  $((r_i, \theta_i))$ . So

$$(35) \quad \begin{aligned} m(P_i) &= r_i q^{\theta_i}, \text{ and} \\ m(P_{ij}) &= \begin{cases} r_i q^{\theta_i} + r_j q^{\theta_j} & \text{if } \theta_i \geq \theta_j, \\ r q^\theta, \text{ where } re^{i\pi\theta} = r_i e^{i\pi\theta_i} + r_j e^{i\pi\theta_j} \text{ and } \theta \in (-1, 1) & \text{if } \theta_i < \theta_j. \end{cases} \end{aligned}$$

For an  $\epsilon > 0$ , let  $M_\epsilon \subset M$  be the open subset containing  $m$  such that for  $i = 2, \dots, n$ , we have  $\epsilon < m(P_i) < \epsilon^{-1}$ , and let  $V_\epsilon \subset V$  be the pre-image of  $M_\epsilon$ . Since  $M$  is the union of all  $M_\epsilon$ , as  $\epsilon$  varies, it suffices to prove that for every  $\epsilon$ , the map  $V_\epsilon \rightarrow M_\epsilon \subset M$  is a homeomorphism onto its image.

Consider the closure  $\overline{V}_\epsilon$  of  $V_\epsilon$  in  $(\mathbf{R}_{>0} \times [-1, 1])^n$ . This is the set of  $((r_1, \theta_1), \dots, (r_n, \theta_n))$  satisfying  $(r_1, \theta_1) = (1, 0)$  together with  $\epsilon \leq r_i q^{\theta_i} \leq \epsilon^{-1}$  and  $\max \theta_i - \min \theta_i \leq 1$ . Note that  $\overline{V}_\epsilon$  is compact. The formulas in (35) allow us to extend the map  $f: V_\epsilon \rightarrow M$  to  $\overline{f}: \overline{V}_\epsilon \rightarrow M$ .

We claim that  $\overline{f}$  is injective. Indeed, suppose we have two points  $((r_i, \theta_i))$  and  $((r'_i, \theta'_i))$  in  $\overline{V}_\epsilon$  with images  $m$  and  $m'$  under  $\overline{f}$  with  $m = m'$ . By definition, we have  $(r_1, \theta_1) = (r'_1, \theta'_1) =$

$(1, 0)$ . Furthermore, if we have  $(r_i, \theta_i) = (r'_i, \theta'_i)$  for some  $i$ , and  $e_{ij}$  is an edge of  $\Gamma^{\text{dbl}}$ , then we show that  $(r_j, \theta_j) = (r'_j, \theta'_j)$ . To do so, suppose  $\theta_i < \theta_j$ . By the  $q$ -triangle inequality (Lemma 3.4), we have  $m(P_{ij}) < m(P_{ji})$  and hence  $m'(P_{ij}) < m'(P_{ji})$ . The latter implies  $\theta'_i < \theta'_j$ . So both  $m(P_{ij})$  and  $m'(P_{ij})$  are defined by the second case in (35).

Define the complex numbers  $z_1 = r_i e^{i\pi\theta_i}$  and  $z_2 = r_j e^{i\pi\theta_j}$ , and likewise  $z'_1$  and  $z'_2$ . We know that  $z_1 = z'_1$ . Applying the  $q$ -SSS lemma [1, Lemma 5.2] gives  $z'_2 = z_2$ , and hence  $(r_j, \theta_j) = (r'_j, \theta'_j)$ . (In [1, Lemma 5.2] the  $z_i$  are assumed to lie in a semi-closed upper half plane, but the statement holds with the same proof for  $z_i$  lying in the closed upper half plane.) The case  $\theta_i > \theta_j$  is analogous, using  $P_{ji}$ . The case  $\theta_i = \theta_j$  is easier, and uses only the  $q$ -triangle inequality. Since  $\Gamma$  is connected, we conclude by induction that for all  $i$ , we have  $(r_i, \theta_i) = (r'_i, \theta'_i)$ .

Since  $\overline{V}_\epsilon$  is compact and  $\overline{f}: \overline{V}_\epsilon \rightarrow M$  is injective, it must be a homeomorphism onto its image. Then the same is true for its restriction  $f: V_\epsilon \rightarrow M$ .  $\square$

Recall that  $U \subset \text{Stab}(\mathcal{C})/\mathbf{C}$  is the set of standard stability conditions.

**Proposition 6.17.** *The map  $m_q: U \rightarrow \mathbf{P}^{\mathbf{S}}$  is a homeomorphism onto its image.*

*Proof.* Set  $I = m_q(U)$ . Since  $h: W/\mathbf{C}^\times \rightarrow U$  is a homeomorphism, it suffices to prove that  $m_q \circ h: W/\mathbf{C}^\times \rightarrow I$  is a homeomorphism. Let  $F$  be as in Proposition 6.16. Note that the projection  $\mathbf{P}^{\mathbf{S}} \dashrightarrow \mathbf{P}^F$  is well-defined on  $I$ . Let  $J$  be the image of  $I$  under this projection. The inverse of  $W/\mathbf{C}^\times \rightarrow I$  is given by the projection  $I \rightarrow J$  followed by the inverse of the homeomorphism  $W/\mathbf{C}^\times \rightarrow J$  from Proposition 6.16.  $\square$

To prove an analogue of Proposition 6.17 for all stability conditions, we need to prove that  $m_q(U)$  is open in  $m_q(\text{Stab}(\mathcal{C})/\mathbf{C})$ . We introduce hypotheses that imply this. They are used when we show that  $m_q$  is a homeomorphism onto its image in type  $A_2$  and  $\widehat{A}_1$ .

**Definition 6.18.** Fix some  $q > 0$ . Let  $\tau$  be a stability condition. We say that  $\tau$  *collapses* a distinguished triangle  $x \rightarrow y \rightarrow z \xrightarrow{+1}$  if

$$m_{q,\tau}(y) = m_{q,\tau}(x) + m_{q,\tau}(z).$$

If  $\tau$  does not collapse  $x \rightarrow y \rightarrow z \xrightarrow{+1}$ , then we have a strict triangle inequality

$$m_{q,\tau}(y) < m_{q,\tau}(x) + m_{q,\tau}(z).$$

**Proposition 6.19.** *Let  $\tau$  be a standard stability condition. Suppose there exists a finite set  $T$  of distinguished triangles of objects in  $\mathbf{S}$  with the following properties:*

- (1) *no triangle in  $T$  is collapsed by  $\tau$ ;*
- (2) *if  $\tau'$  is any stability condition such that no triangle in  $T$  is collapsed by  $\tau'$ , then  $\tau'$  is standard up to the action of  $\mathbf{C}$ .*

*Then  $m_q: \text{Stab}(\mathcal{C})/\mathbf{C} \rightarrow \mathbf{P}^{\mathbf{S}}$  is a local homeomorphism onto its image at  $\tau$ .*

*Proof.* We abbreviate  $m_q$  by  $m$ . Let  $V \subset m(\text{Stab}(\mathcal{C})/\mathbf{C})$  be the set of points  $[f] \in \mathbf{P}^{\mathbf{S}}$  that satisfy the inequalities

$$f(y) < f(x) + f(z)$$

for each triangle  $x \rightarrow y \rightarrow z \xrightarrow{+1}$  in  $T$ . Then  $V$  is an open subset of  $m(\text{Stab}(\mathcal{C})/\mathbf{C})$ . By the first hypothesis, it contains  $m(\tau)$ . By the second hypothesis, we have  $m^{-1}(V) \subset U$ , the set of standard stability conditions. By Proposition 6.17,  $m: U \rightarrow m(U)$  is a homeomorphism. So its restriction  $m^{-1}(V) \rightarrow V$  is also a homeomorphism. Thus  $m$  gives a homeomorphism

of an open neighbourhood of  $\tau$  in  $\text{Stab}(\mathcal{C})/\mathbf{C}$  with an open neighbourhood of  $m(\tau)$  in  $m(\text{Stab}(\mathcal{C})/\mathbf{C})$ .  $\square$

*Remark 6.20* (Existence of triangles). We prove the existence of the triangles  $T$  required in Proposition 6.19 in the  $A_2$  and  $\widehat{A}_1$  cases. We expect that such sets of triangles exist more generally.

*Remark 6.21* (Functionals in the closure). It follows from Corollary 4.14 that the  $\overline{\text{hom}}$  functionals are in the closure of the image of the mass map. Whether these functionals are dense in the boundary depends on  $q$ . In the next two sections we answer this question for  $q = 1$  in the  $A_2$  and  $\widehat{A}_1$  cases. [1] answers this question for other values of  $q$  in the  $A_2$  case.

## 7. THE $A_2$ CASE

The aim of this section is to understand the Thurston compactification at  $q = 1$  of the stability space for the 2-Calabi–Yau category  $\mathcal{C} = \mathcal{C}_\Gamma$  where  $\Gamma$  is the  $A_2$  graph

$$\begin{array}{c} \bullet \text{ --- } \bullet \\ 1 \quad \quad 2 \end{array}.$$

Before we dive into the details, we explain the results. Recall from Section 6 that the category  $\mathcal{C}$  admits an action of the three strand braid group  $B_3$ . The Artin generators  $\sigma_1$  and  $\sigma_2$  of  $B_3$  act by spherical twists in the objects  $P_1$  and  $P_2$ . In particular, we have  $\sigma_1(P_1) = P_1[-1]$  and  $\sigma_1(P_2) = P_{12}$ . Similarly,  $\sigma_2(P_1) = P_{21}$  and  $\sigma_2(P_2) = P_2[-1]$ . Set  $\sigma_X = \sigma_2\sigma_1\sigma_2^{-1}$ ; it acts by the spherical twist in  $X = P_{21}$ . Let  $\mathbf{S}$  be the set of spherical objects of  $\mathcal{C}$ , up to isomorphism and shift.

The picture of  $\text{Stab}(\mathcal{C})/\mathbf{C}$  and its compactification is as follows (see Fig. 2). The space  $\text{Stab}(\mathcal{C})/\mathbf{C}$  is homeomorphic to a 2-dimensional disk, tiled by  $B_3$ -translates of a triangle  $\Lambda$  minus its three vertices. The mass map embeds  $\text{Stab}(\mathcal{C})/\mathbf{C}$  homeomorphically onto a subspace  $M \subset \mathbf{P}^{\mathbf{S}}$ . The closure of  $M$  is thus a compactification of  $\text{Stab}(\mathcal{C})/\mathbf{C}$ . In the compactification, the missing vertices in the tiling appear on the boundary as the  $\overline{\text{hom}}$  functionals of spherical objects. The closure  $P$  of the set of  $\overline{\text{hom}}$  functionals is homeomorphic to a circle, disjoint from  $M$ . The closure of  $M$  is  $M \cup P$  and is homeomorphic to a closed disk.

**7.1. The automaton.** Recall that a stability condition  $\tau \in \text{Stab}(\mathcal{C})$  is standard if its  $[0, 1]$ -heart is the standard heart  $\heartsuit_{\text{std}}$ , or equivalently, if  $P_1$  and  $P_2$  are  $\tau$ -stable of phase in  $[0, 1]$ . Let  $\tau$  be a standard stability condition. We say that  $\tau$  is of *type I* if  $\phi_\tau(P_1) \leq \phi_\tau(P_2)$  and of *type II* if  $\phi_\tau(P_2) \leq \phi_\tau(P_1)$ . We say that  $\tau$  is *on-the-wall* if  $\phi_\tau(P_1) = \phi_\tau(P_2)$ ; otherwise, it is *off-the-wall* (see Fig. 3).

Fix an off-the-wall standard stability condition  $\tau$  of type I. Let  $X = P_{21}$  and  $\Sigma = \{P_1, P_2, X\}$ . Note that the objects of  $\Sigma$  are exactly the  $\tau$ -stable spherical objects in  $\heartsuit_{\text{std}}$  (see, for example, [39, Theorem 5.5]). Define the braid  $\gamma \in B_3$  by

$$\gamma = \sigma_2\sigma_1 = \sigma_X\sigma_2 = \sigma_1\sigma_X.$$

Note that  $\gamma(X) = P_2[-1]$ ,  $\gamma(P_1) = X[-1]$ , and  $\gamma(P_2) = P_1$ . Fig. 4 describes a  $\tau$ -HN automaton  $\Theta$ .

Formally (following Definition 5.3), the automaton  $\Theta$  is defined by the  $B_3$ -labeled graph shown in Fig. 4. The states are called  $[X, P_1]$ ,  $[P_1, P_2]$ , and  $[P_2, X]$ . The  $\Theta$ -set  $S$  is defined by setting  $S_{[a,b]}$  to be the set of spherical objects whose  $\tau$ -stable HN factors are shifts of  $a$  and  $b$ . For example, the object  $\sigma_1\sigma_1(P_2)$  has stable HN factors  $P_1[-1]$ ,  $P_1$ , and  $P_2$ , and hence is supported at the state  $[P_1, P_2]$ . The objects  $P_1$ ,  $P_2$ , and  $X$  are each supported at



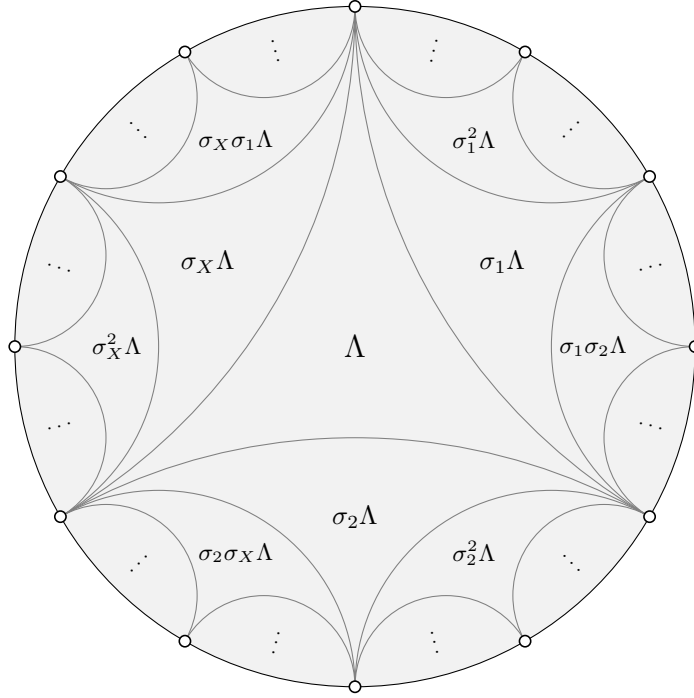


FIGURE 2.  $\text{Stab}(\mathcal{C})/\mathbf{C}$  (the shaded open disk) is tiled by the images of  $\Lambda$  under the braid group. The boundary circle is  $P$ , the closure of the set of  $\overline{\text{hom}}$  functionals, some of which are shown as white circles.

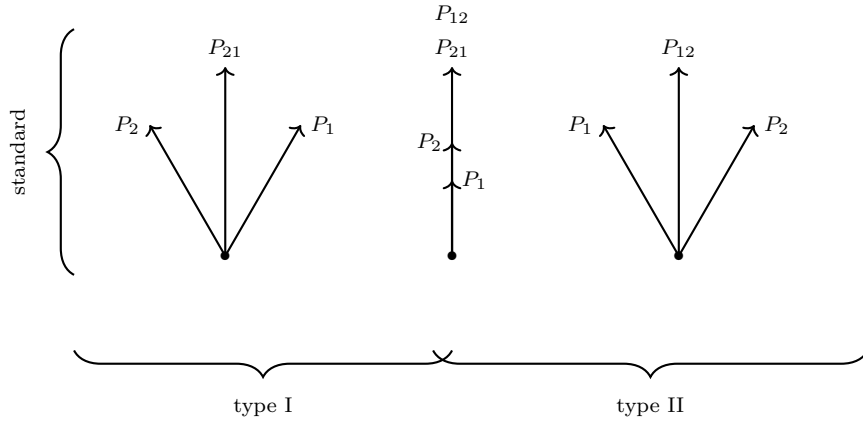


FIGURE 3. Central charges in a standard stability condition of types I and II. The intersection of the two types are on-the-wall stability conditions.

two states. Recall that the representation  $M$  associates the rank 3 module  $\mathbf{Z}^\Sigma$  to each state. However, by construction, at the state  $[a, b]$ , the image of  $\text{HN}_\tau: S \rightarrow \mathbf{Z}^\Sigma$  lands in the rank 2 sub-module  $\mathbf{Z}^{\{a, b\}}$ . We replace  $\mathbf{Z}^\Sigma$  by this sub-module. Then the maps  $M(e)$  are given by

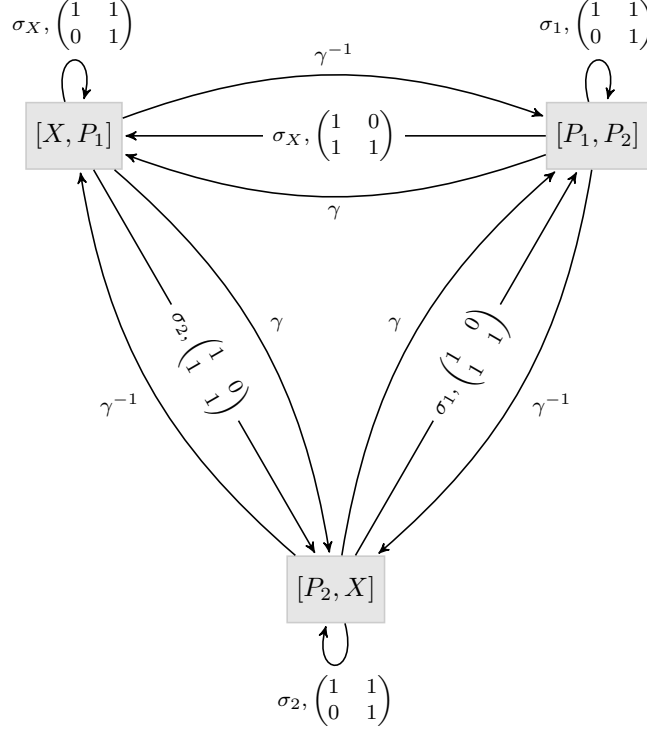


FIGURE 4. An automaton describing the dynamics of HN filtrations in a stability condition with stable objects  $P_1$ ,  $P_2$ , and  $X = P_{21}$ .

$2 \times 2$  matrices in the standard basis, as shown in Fig. 4. The arrows labelled  $\gamma^{\pm 1}$  correspond to the identity matrix, which we omit in the figure.

To check that this setup defines an HN automaton, we need to check that an arrow takes objects supported at its source to objects supported at its target. We also need to check that the HN multiplicities transform according to the matrix associated to the arrow, as in Definition 5.3. We carry out both these checks in the next proposition.

**Proposition 7.1.** *Let  $v = [a, b]$  and  $w = [c, d]$  be two states of  $\Theta$ . Let  $e: v \rightarrow w$  be an arrow labelled by the element  $g \in B_3$ . Let  $x \in \mathcal{C}$  be any object whose stable  $\tau$ -HN factors are shifts of  $a$  and  $b$ . The following hold.*

- (1) *The stable  $\tau$ -HN factors of  $g \cdot x$  are shifts of  $c$  and  $d$ .*
- (2) *The following diagram commutes.*

$$\begin{array}{ccc}
 x & \xrightarrow{\text{HN}_\tau} & \text{HN}_\tau(x) \in \mathbf{Z}^{\{a,b\}} \\
 \downarrow g & & \downarrow M(e) \\
 gx & \xrightarrow{\text{HN}_\tau} & \text{HN}_\tau(gx) \in \mathbf{Z}^{\{c,d\}}
 \end{array}$$

*In other words,*

$$\text{HN}_\tau(gx) = M(e) \cdot \text{HN}_\tau(x).$$

*Proof.* Let

$$0 \rightarrow x_0 \rightarrow x_1 \rightarrow \cdots \rightarrow x_n = x$$

be a filtration of  $x$  whose factors  $z_i$  are stable and appear in non-increasing order of phase. Then, by hypothesis each  $z_i$  is a shift of  $a$  or  $b$ . By applying  $g$ , we get a filtration

$$(36) \quad 0 \rightarrow gx_0 \rightarrow gx_1 \rightarrow \cdots \rightarrow gx_n = gx$$

with factors  $gz_i$ .

We check that the filtration (36) is rectifiable in the sense of Definition A.10. We use Theorem A.22.

We outline the case  $v = [P_1, P_2]$ , leaving the others to the reader. Denote by  $\lceil x \rceil$  (resp.  $\lfloor x \rfloor$ ) the HN factor of  $x$  of highest (resp. lowest) phase. To apply Theorem A.22 to the filtration (36), we must check the following: for  $i < j$ , either

- (1)  $\text{Hom}(gz_j, gz_i[1]) = \text{Hom}(gz_i, gz_j[1]) = 0$ , or
- (2)  $\lfloor gz_i \rfloor \geq \lceil gz_j \rceil$ .

If  $z_i \cong z_j$ , then (1) holds. Let us assume otherwise. Since our stability condition is off-the-wall, we have  $\phi(z_i) > \phi(z_j)$ . Since we are in a 2-CY category,  $\text{Hom}(gz_j, gz_i[1])$  vanishes if and only if  $\text{Hom}(gz_i, gz_j[1])$  does. We enumerate the pairs  $(z_i, z_j)$  such that

- $z_i$  and  $z_j$  are shifts of  $P_1$  or  $P_2$ ,
- $\phi(z_i) > \phi(z_j)$ , and
- $\text{Hom}(z_j, z_i[1]) \neq 0$ .

Up to a simultaneous shift, the only such pairs are  $(P_1[1], P_1)$ ,  $(P_2[1], P_2)$ , and  $(P_2, P_1)$ . The outgoing edges  $e$  from  $v$  are labeled  $\sigma_1$ ,  $\sigma_X$ ,  $\gamma$ , or  $\gamma^{-1}$ . For each of these elements  $g$  and for each object  $x \in \{P_1, P_2\}$ , we tabulate  $\lfloor gx \rfloor$  and  $\lceil gx \rceil$  as follows.

|               | $\lfloor gP_1 \rfloor$ | $\lceil gP_1 \rceil$ | $\lfloor gP_2 \rfloor$ | $\lceil gP_2 \rceil$ |
|---------------|------------------------|----------------------|------------------------|----------------------|
| $\sigma_1$    | $P_1[-1]$              | $P_1[-1]$            | $P_2$                  | $P_1$                |
| $\sigma_X$    | $X[-1]$                | $P_1$                | $P_1[1]$               | $P_1[1]$             |
| $\gamma$      | $X[-1]$                | $X[-1]$              | $P_1$                  | $P_1$                |
| $\gamma^{-1}$ | $P_2$                  | $P_2$                | $X[1]$                 | $X[1]$               |

It is now a simple check that for each of these labels and each of the enumerated pairs, condition (2) above holds.

Since the filtration (36) is rectifiable, it follows from Theorem A.19 that

$$\text{HN}_\tau(gx) = \sum \text{HN}_\tau(gz_i).$$

Therefore, it suffices to prove the proposition for  $x = z_i$ , and hence, for  $x = a$  and  $x = b$ . This is another straightforward calculation.  $\square$

The automaton recognises every spherical object in the sense of Remark 5.4.

**Proposition 7.2.** *The following hold.*

- (1) Every  $\beta \in B_3$  has an expression of the form

$$\beta = \gamma^n \sigma_{a_1}^{m_1} \sigma_{a_2}^{m_2} \cdots \sigma_{a_k}^{m_k},$$

where  $n$  is an integer,  $k$  is a non-negative integer, the  $m_i$  are positive integers, and the sequence  $(a_1, a_2, \dots, a_k)$  is a contiguous subsequence of the sequence

$$(\dots, X, 1, 2, X, 1, 2, \dots).$$

(2) Every spherical object  $s \in \mathcal{C}$  has an expression

$$s = \beta_1 \cdots \beta_n P_1$$

that is recognised by  $\Theta$ .

*Proof.* For (1), we repeatedly use the commutation relations

$$\gamma \sigma_2 \gamma^{-1} = \sigma_1, \quad \gamma \sigma_X \gamma^{-1} = \sigma_2, \quad \gamma \sigma_1 \gamma^{-1} = \sigma_X.$$

Begin by writing  $\beta$  as any product of the generators  $\sigma_1$  and  $\sigma_2$ , along with their inverses. Eliminate the inverses of the generators by rewriting as follows:

$$\sigma_1^{-1} = \sigma_X \gamma^{-1}, \quad \sigma_2^{-1} = \sigma_1 \gamma^{-1}.$$

Next, use the commutation relations to rewrite

$$(37) \quad \begin{aligned} \gamma^i \sigma_X &= \sigma_2 \gamma^i, & \gamma^i \sigma_1 &= \sigma_X \gamma^i, & \gamma^i \sigma_2 &= \sigma_1 \gamma^i \text{ for } i > 0, \text{ and} \\ \gamma^i \sigma_X &= \sigma_1 \gamma^i, & \gamma^i \sigma_1 &= \sigma_2 \gamma^i, & \gamma^i \sigma_2 &= \sigma_X \gamma^i \text{ for } i < 0, \end{aligned}$$

and thus move all powers of  $\gamma$  to the left. The rest of  $\beta$  is now a product of elements from  $\{\sigma_1, \sigma_2, \sigma_X\}$ . Replace any occurrences of  $\sigma_2 \sigma_1$ ,  $\sigma_1 \sigma_X$ , or  $\sigma_X \sigma_2$  by  $\gamma$  and again move  $\gamma$  to the left as before. It is easy to see that eventually,  $\beta$  reaches the desired form.

For (2), we recall that every spherical object is in the  $B_3$ -orbit of  $P_1$  (see, for example, [2, Theorem 1.1] and the references therein). begin by writing  $s = \beta P_1$  for some  $\beta \in B_3$ . Write

$$\beta = \gamma^n \sigma_{a_1}^{m_1} \sigma_{a_2}^{m_2} \cdots \sigma_{a_k}^{m_k}$$

as in (1). Note that  $P_1$  is supported at two states, namely  $[P_1, P_2]$  and  $[X, P_1]$ . Each of the letters  $\{\sigma_1, \sigma_2, \sigma_X\}$  is applicable at least at one of the two states above, so we can apply  $\sigma_{a_k}$ . The condition on our cyclic writing guarantees that we can apply each subsequent letter. Therefore,  $\Theta$  recognises the expression

$$s = \gamma^n \sigma_{a_1}^{m_1} \sigma_{a_2}^{m_2} \cdots \sigma_{a_k}^{m_k} P_1.$$

□

As a consequence of Proposition 7.2, every spherical object is supported on at least one of the three states. In particular, its HN factors involve (up to shift) at most 2 out of the 3 objects  $P_1$ ,  $P_2$ , and  $X$ .

**Proposition 7.3.** *Let  $s$  be a spherical object supported at the state  $[P_1, P_2]$ . Assume that  $s$  is not a shift of  $P_1$  or  $P_2$ . Then*

- (1) *we can write  $s = \sigma_1^r t$  for some  $r > 0$  and an object  $t$  supported at  $[P_2, X]$ ;*
- (2) *we can write  $s = \sigma_2^{-r} t$  for some  $r > 0$  and an object  $t$  supported at  $[X, P_1]$ .*

*Analogous statements hold for the other two states.*

*Proof.* Write a recognised expression  $s = \gamma^n \sigma_{a_1}^{m_1} \sigma_{a_2}^{m_2} \cdots \sigma_{a_k}^{m_k} P_1$  as in the proof of Proposition 7.2. Since  $s$  is not a shift of  $P_1, P_2$ , or  $X$ , we have  $k \geq 1$ . Using the commutation relations (37), move  $\gamma$  to the right to obtain another expression  $s = \sigma_{b_1}^{m_1} \sigma_{b_2}^{m_2} \cdots \sigma_{b_k}^{m_k} \gamma^n P_1$ , which is also recognised by  $\Theta$ . Observe that the state at which the recognising path ends is dictated by  $b_1$ . For  $b_1 = 1$ , the end state is  $[P_1, P_2]$ ; for  $b_1 = 2$ , it is  $[P_2, X]$ ; and for  $b_1 = X$ , it is  $[X, P_1]$ . By assumption, we must have  $b_1 = 1$ .

Taking  $r = m_1$  and  $t = \sigma_{b_2}^{m_2} \cdots \sigma_{b_k}^{m_k} \gamma^n P_1$  yields the first assertion of the proposition. For the second assertion, consider

$$\sigma_2 s = \sigma_2 \sigma_{b_1}^{m_1} \cdots \sigma_{b_k}^{m_k} \gamma^n P_1.$$

Write  $\sigma_2\sigma_{b_1} = \sigma_2\sigma_1$  as  $\gamma$  and propagate  $\gamma$  to the right using the commutation relations (37), to obtain a recognised expression for  $\sigma_2s$ , as follows:

$$\sigma_2s = \sigma_{b'_1}^{m_1-1}\sigma_{b'_2}^{m_2}\dots\sigma_{b'_k}^{m_k}\gamma^{n+1}P_1,$$

where  $\sigma_{b'_i} = \gamma\sigma_{b_i}\gamma^{-1}$ . In particular,  $\sigma_{b'_1} = \sigma_X$ . Depending on the values of  $m_1$  and  $k$ , the leftmost letter of this expression is one of  $\gamma$ ,  $\sigma_X$ , or  $\sigma_1$ . In the first two cases, we stop; in the last case we consider

$$\sigma_2^2s = \sigma_2^2\sigma_{b_1}^{m_1}\dots\sigma_{b_k}^{m_k}\gamma^n P_1$$

and continue as above. After finitely many steps say  $r$ , we obtain  $t = \sigma_2^r s$ , which has a recognised expression whose leftmost letter is either  $\gamma$  or  $\sigma_X$ .

If the leftmost letter is  $\gamma$ , then  $t = \gamma^{n+r}P_1$ , and so  $t$  is one of  $P_1$ ,  $P_2$ , or  $X$ , up to shift. By assumption,  $s$  is not a shift of  $P_2$ , so neither is  $t$ . In particular,  $t$  is supported at  $[X, P_1]$ . If the leftmost letter is  $\sigma_X$ , then  $t$  is supported at  $[X, P_1]$  as observed earlier in the proof.  $\square$

As an application of the automaton, we record when a braid takes a stable object to a stable object. Fix a standard off-the-wall stability condition  $\tau$  of type I. Let  $\beta = \gamma^n\sigma_{a_1}^{m_1}\dots\sigma_{a_k}^{m_k}$  be a cyclic writing for  $\beta \in B_3$  as in Proposition 7.2. Then the expression  $\gamma^n\sigma_{a_1}^{m_1}\dots\sigma_{a_k}^{m_k}(P_1)$  is recognised by  $\Theta$  starting at the state  $[P_1, P_2]$  if  $a_k \in \{1, X\}$  or at  $[X, P_1]$  if  $a_k \in \{X, 2\}$ .

**Proposition 7.4.** *In the setup above,  $\beta P_1$  is  $\tau$ -stable if and only if the cyclic writing for  $\beta$  is one of:*

- (1)  $\beta = \gamma^n$ , or
- (2)  $\beta = \gamma^n\sigma_1^{m_1}$ , or
- (3)  $\beta = \gamma^n\sigma_{a_1}\dots\sigma_{a_k}$  where  $a_k = 2$ .

*Proof.* The non-negativity of the matrices in  $\Theta$  implies that with each transition, the number of HN factors does not decrease. If  $\beta(P_1)$  is stable, then at each step in the evaluation of  $\gamma^n\sigma_{a_1}^{m_1}\dots\sigma_{a_k}^{m_k}(P_1)$  by  $\Theta$ , the total number of HN factors must remain 1. It is easy to check that the enumerated writings are the only ones with this property.  $\square$

Modulo signs, we can identify the  $\Theta$ -set  $M$  with a  $\Theta$ -set arising from a linear representation. To describe this identification, recall that we have a surjective homomorphism  $B_3 \rightarrow \mathrm{PSL}_2(\mathbf{Z})$  given by

$$(38) \quad \sigma_1 \mapsto \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \sigma_2 \mapsto \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}.$$

Its kernel is the center  $Z(B_3)$ , generated by  $\gamma^3$ . Via  $B_3 \rightarrow \mathrm{PSL}_2(\mathbf{Z})$ , the group  $B_3$  acts on  $\mathbf{Z}^2/\pm$ . As a result,  $\mathbf{Z}^2/\pm$  becomes a  $\Theta$ -set. The  $\Theta$ -representation  $M$  from Fig. 4 gives rise to the  $\Theta$ -set  $M/\pm$ . For each of the three states  $v$  of  $\Theta$ , define a linear map  $\phi_v: M_v \rightarrow \mathbf{Z}^2$  as follows:

$$\begin{aligned} \phi_{[P_1, P_2]} &: (1, 0) \mapsto (1, 0) \text{ and } (0, 1) \mapsto (0, 1), \\ \phi_{[P_2, X]} &: (1, 0) \mapsto (0, 1) \text{ and } (0, 1) \mapsto (1, -1), \\ \phi_{[X, P_1]} &: (1, 0) \mapsto (1, -1) \text{ and } (0, 1) \mapsto (1, 0). \end{aligned}$$

**Proposition 7.5.** *The maps  $\phi_v$  give an isomorphism of  $\Theta$ -sets  $M/\pm \rightarrow \mathbf{Z}^2/\pm$ .*

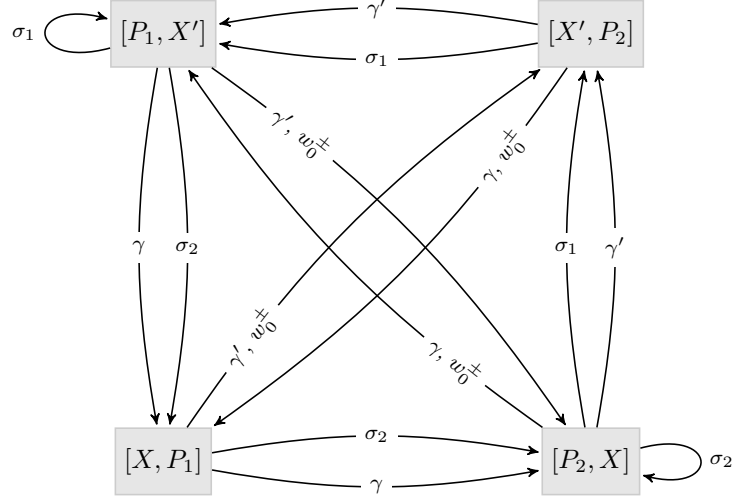


FIGURE 5. An automaton describing the dynamics of HN filtrations in an on-the-wall stability condition with semistable objects  $P_1$ ,  $P_2$ ,  $X = P_{21}$ , and  $X' = P_{12}$ . Here  $\gamma = \sigma_2\sigma_1$ ;  $\gamma' = \sigma_1\sigma_2$ ; and  $w_0 = \sigma_1\sigma_2\sigma_1 = \sigma_2\sigma_1\sigma_2$ .

*Proof.* All three maps  $\phi_v$  are clearly bijective. It remains to check that they are  $\Theta$ -equivariant. That is, up to sign, for every edge  $e: v \rightarrow w$ , we have

$$e \circ \phi_v = \phi_w \circ M(e).$$

We omit this straightforward verification.  $\square$

*Remark 7.6.* Let  $\tau$  be an on-the-wall standard stability condition. Fig. 5 describes a  $\tau$ -HN automaton that recognises all spherical objects. Since the automaton in Fig. 4 suffices for mass computations, we omit the details.

*Remark 7.7.* For every off-the-wall stability condition, we have an HN automaton obtained by applying an appropriate braid to the automaton in (4). Likewise, for an on-the-wall stability condition, we have an HN automaton obtained by applying a braid to the automaton in (5). Thus, we have an automaton at every point of the stability space. Observe that these automata exhibit a wall-and-chamber behaviour: they remain constant in a chamber, get more complicated along a wall, and flip to a different automaton as we cross the wall. It would be interesting to understand the wall-crossing formulas for recognized expressions of a fixed braid in different chambers.

**7.2. Spherical objects, their HN filtrations, and hom functionals.** Using the automaton, we analyse HN filtrations of spherical objects in  $\mathcal{C}$ . We prove that the closure  $P$  of the image of the map  $h: \mathbf{S} \rightarrow \mathbf{P}^{\mathbf{S}}$  is a circle. Along the way, we also give a simpler proof of a theorem of Rouquier and Zimmermann [35, Proposition 4.8].

Recall from (38) that we have a surjective homomorphism  $B_3 \rightarrow \mathrm{PSL}_2(\mathbf{Z})$  whose kernel is generated by  $\gamma^3$ . The braid  $\gamma^3$  acts on the category as the shift  $x \rightarrow x[2]$ . As a result, the action of  $B_3$  on  $\mathbf{S}$  induces an action of  $\mathrm{PSL}_2(\mathbf{Z})$  on  $\mathbf{S}$ . For  $\beta \in B_3$ , let  $\bar{\beta}$  be the image in  $\mathrm{PSL}_2(\mathbf{Z})$ .

**Proposition 7.8.** *The stabiliser in  $\mathrm{PSL}_2(\mathbf{Z})$  of  $P_1 \in \mathbf{S}$  is generated by  $\bar{\sigma}_1$ .*

*Proof.* Let  $\beta \in B_3$  be a braid such that  $\beta(P_1)$  is a shift of  $P_1$ . Since  $\beta(P_1)$  is stable, Proposition 7.4 implies that  $\beta = \gamma^n$  or  $\beta = \gamma^n \sigma_1^m$  or  $\beta$  has the cyclic writing  $\gamma^n \sigma_{a_1} \cdots \sigma_{a_k}$  where  $a_k = 2$ . Using further that  $\beta(P_1)$  is a shift of  $P_1$ , in the first two cases,  $n$  must be divisible by 3, and in the third case  $\beta = \gamma^{3m}(\sigma_X \sigma_1 \sigma_2)^k$  or  $\gamma^{3m-1} \sigma_2(\sigma_X \sigma_1 \sigma_2)^k$  or  $\gamma^{3m-2} \sigma_1 \sigma_2(\sigma_X \sigma_1 \sigma_2)^k$ . In each case, it is easy to check that  $\beta$  maps to a power of  $\bar{\sigma}_1$  in  $\mathrm{PSL}_2(\mathbf{Z})$ .  $\square$

Consider  $\mathbf{P}^1(\mathbf{R})$  with its standard  $\mathrm{PSL}_2(\mathbf{Z})$  action. The stabiliser of  $[1 : 0] \in \mathbf{P}^1(\mathbf{R})$  is generated by  $\bar{\sigma}_1 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ . Therefore, we have a  $\mathrm{PSL}_2(\mathbf{Z})$ -equivariant injection

$$(39) \quad i: \mathbf{S} \rightarrow \mathbf{P}^1(\mathbf{R})$$

that sends  $P_1$  to  $[1 : 0]$ . Since  $\mathbf{S}$  is a single  $\mathrm{PSL}_2(\mathbf{Z})$ -orbit, the image of  $i$  is the  $\mathrm{PSL}_2(\mathbf{Z})$  orbit of  $[1 : 0]$ , which is  $\mathbf{P}^1(\mathbf{Z}) = \{[a : c] \mid a, c \in \mathbf{Z} \text{ with } \gcd(a, c) = 1\}/\pm$ .

Given  $[a : c] \in \mathbf{P}^1(\mathbf{Z})$ , we can find its pre-image in  $\mathbf{S}$  under  $i$  as follows. Assuming  $c \neq 0$ , write  $a/c$  as a continued fraction with an odd number of terms:

$$\frac{a}{c} = n_0 + \frac{1}{n_2 + \frac{1}{\ddots + \frac{1}{n_{2k}}}}.$$

In the standard affine coordinate on  $\mathbf{P}^1(\mathbf{R})$ , the element  $\sigma_1$  acts by  $z \mapsto z + 1$  and  $\sigma_2^{-1}$  by  $z \mapsto z/(z + 1)$ . Therefore, the continued fraction expansion gives

$$\frac{a}{c} = \sigma_1^{n_0} \sigma_2^{-n_1} \cdots \sigma_2^{-n_{2k-1}} \sigma_1^{n_{2k}}(0).$$

Using  $[0 : 1] = i(P_2)$ , we get

$$(40) \quad [a : c] = i(\sigma_1^{n_0} \sigma_2^{-n_1} \cdots \sigma_1^{n_{2k}}(P_2)).$$

For example, using (40) we get that under  $i$ ,

$$P_1 \mapsto [1 : 0], \quad P_2 \mapsto [0 : 1], \quad P_{21} \mapsto [1 : -1], \quad \text{and} \quad P_{12} \mapsto [1 : 1].$$

We now interpret  $i(s) \in \mathbf{P}^1(\mathbf{Z})$  in terms of HN filtrations. Fix a standard off-the-wall type I stability condition  $\tau$ . For a state  $v$  in the automaton, recall the map  $\phi_v$  from Proposition 7.5.

**Proposition 7.9.** *For a spherical object  $s$  supported at the state  $v$ , we have the equality*

$$i(s) = \phi_v(\mathrm{HN}_\tau(s)) \in \mathbf{P}^1(\mathbf{Z}).$$

*Moreover  $\phi_v(\mathrm{HN}_\tau(s)) \in \mathbf{Z}^2/\pm$  is the unique representative of  $i(s)$  whose coordinates are relatively prime.*

*Proof.* The equation evidently holds for  $s = P_1$ . If it holds for  $s$  supported at  $v$ , and  $e: v \rightarrow w$  is an edge of  $\Theta$  with label  $\beta \in B_3$ , then using the  $\Theta$ -equivariance of  $i$ ,  $\phi_v$ , and  $\mathrm{HN}_\tau$ , we see that it also holds for  $\beta s$ . By Proposition 7.2, it holds for all spherical objects.

By Proposition 7.2, the vector  $\mathrm{HN}_\tau(s)$  is obtained by applying a sequence of matrices in Fig. 4 to the vector  $(1, 0)$  or  $(0, 1)$ . Since all these matrices have determinant 1, we see that the coordinates of  $\mathrm{HN}_\tau(s)$  are relatively prime. Since the linear maps defining  $\phi$  are also of determinant  $\pm 1$ , the same is true for  $\phi_v(\mathrm{HN}_\tau(s))$ .  $\square$

Proposition 7.9 leads to a simpler proof of a theorem of Rouquier and Zimmermann [35, Proposition 4.8]. To explain it, we first introduce some basic notation.

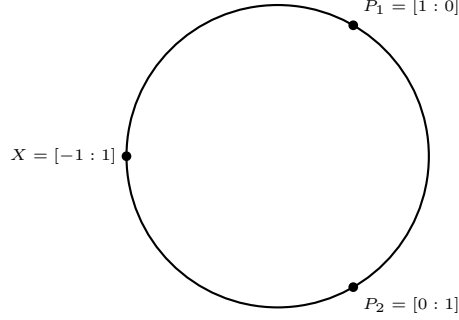


FIGURE 6. The points  $P_1$ ,  $P_2$ , and  $X = P_{21}$  divide  $\mathbf{P}^1(\mathbf{R})$  into three arcs. The HN factors of an object (up to shift) only include the two endpoints of the arc on which the object lies.

**Definition 7.10.** Let  $x$  be an object of  $\mathcal{C}$  belonging to the standard heart. The *Jordan–Hölder (JH) multiplicity* of  $P_1$  (resp.  $P_2$ ) in  $x$  is the multiplicity of  $P_1$  (resp.  $P_2$ ) in any Jordan–Hölder filtration of  $x$ . More generally, the JH multiplicity of  $P_i$  in an arbitrary  $x$  is the sum of the JH multiplicities of  $P_i$  in each cohomology of  $x$ , where cohomology is taken with respect to the  $t$ -structure whose heart is the standard heart.

**Proposition 7.11.** Let  $s$  be a spherical object with  $i(s) = [a : c] \in \mathbf{P}^1(\mathbf{Z})$ , where  $a, c$  are relatively prime integers. Then the JH multiplicities of  $P_1$  and  $P_2$  in  $s$  are  $|a|$  and  $|c|$  respectively.

*Proof.* Since the  $\tau$ -HN filtration refines the cohomology filtration with respect to the standard heart, the JH multiplicities of  $P_1$  and  $P_2$  are linear functions of the  $\tau$ -HN multiplicities. Up to sign, these linear functions are precisely the maps  $\phi$  in Proposition 7.5. The result now follows by Proposition 7.9.  $\square$

*Remark 7.12.* Rouquier and Zimmermann work in the homotopy category of complexes of projective modules over the zig-zag algebra, which is formally a different category from our 2-CY category  $\mathcal{C}$ . As explained, e.g., in [2, §2.3.3], the effect of passing from their category to ours is to collapse one grading. Since [35, Proposition 4.8] involves forgetting the grading, Proposition 7.11 implies [35, Proposition 4.8].

The set  $\mathbf{S}$  is the union of the three sets supported at the three vertices of  $\Theta$ . Under the map  $i: \mathbf{S} \rightarrow \mathbf{P}^1(\mathbf{R})$ , this division corresponds to a geometric division of the circle  $\mathbf{P}^1(\mathbf{R})$ , which we now describe. The three points  $i(P_1) = [1 : 0]$  and  $i(P_2) = [0 : 1]$ , and  $i(X) = [-1 : 1]$  divide  $\mathbf{P}^1(\mathbf{R})$  into three closed arcs (see Fig. 6). We denote these arcs by  $[P_1, P_2]$ ,  $[P_2, X]$ , and  $[X, P_1]$ . While this notation is identical to the state labels of our HN automaton, the following proposition justifies this choice.

**Proposition 7.13.** The map  $i$  sends the objects of  $\mathbf{S}$  supported at the state  $[P_1, P_2]$  to the arc  $[P_1, P_2] \subset \mathbf{P}^1(\mathbf{R})$ , and likewise for the other two states.

*Proof.* By Proposition 7.9, the map  $i: \mathbf{S} \rightarrow \mathbf{P}^1(\mathbf{R})$  is equal to  $s \mapsto \phi(\text{HN}_\tau(s))$ . For  $s$  supported at  $[P_1, P_2]$ , the vector  $\text{HN}_\tau(s)$  lies in the non-negative cone in  $\mathbf{Z}^{\{P_1, P_2\}}$ , which is mapped by  $\phi_{[P_1, P_2]}$  to the arc  $[P_1, P_2]$ . The argument for the other two states is similar.  $\square$



Consider the diagram (without the dashed arrow)

$$(41) \quad \begin{array}{ccc} \mathbf{S} & \xrightarrow{h} & \mathbf{P}^{\mathbf{S}} \\ \downarrow i & \nearrow & \\ \mathbf{P}^1(\mathbf{R}) & & \end{array}$$

Our next goal is to prove that we have an embedding  $\mathbf{P}^1(\mathbf{R}) \hookrightarrow \mathbf{P}^{\mathbf{S}}$  (the dashed arrow) that extends  $h: \mathbf{S} \rightarrow \mathbf{P}^{\mathbf{S}}$ . To do so, we relate  $i(x)$  and the  $\overline{\text{hom}}$  functionals on  $x$  in the next two propositions.

**Proposition 7.14.** *Let  $x$  be a spherical object with  $i(x) = [a : c] \in \mathbf{P}^1(\mathbf{Z})$ , where  $a, c$  are relatively prime integers. Then  $\overline{\text{hom}}(P_2, x) = |a|$  and  $\overline{\text{hom}}(P_1, x) = |c|$ .*

*Proof.* We prove the assertion for  $\overline{\text{hom}}(P_1, x)$ . The other case follows by applying  $\gamma$ . It is easy to check the result for  $x = P_1, P_2$ , and  $X$ , or their shifts, by hand. So we assume that  $x$  is not a shift of  $P_1, P_2$ , or  $X$ .

Fix a standard off-the-wall stability condition. By Proposition 7.11,  $|c|$  gives the JH-multiplicity of  $P_2$ , which is a linear function of the HN-multiplicities. The key idea is to prove that  $\overline{\text{hom}}(P_1, -)$  is also a linear function of the HN-multiplicities. Then the cases  $x = P_1, P_2$ , and  $X$  imply the result for all  $x$ .

We now prove the linearity. Begin with an object  $x$  supported at  $[P_2, X]$ . Let

$$0 = x_0 \rightarrow \cdots \rightarrow x_n = x.$$

be the HN filtration with factors  $z_i$ . We must prove that

$$\overline{\text{hom}}(P_1, x) = \sum \overline{\text{hom}}(P_1, z_i).$$

Note that since neither  $x$  nor  $z_i$  is a shift of  $P_1$ , we have  $\overline{\text{hom}} = \dim \text{Hom}^\bullet$ .

Apply  $\text{Hom}^\bullet(P_1, -)$  to the HN filtration of  $x$  to get the following filtration in the bounded derived category of graded vector spaces:

$$(42) \quad 0 = \text{Hom}^\bullet(P_1, x_0) \rightarrow \cdots \rightarrow \text{Hom}^\bullet(P_1, x_n) = \text{Hom}^\bullet(P_1, x).$$

The factors in (42) are  $\text{Hom}^\bullet(P_1, z_i)$ . We argue that the connecting map

$$\text{Hom}^\bullet(P_1, z_i) \rightarrow \text{Hom}^\bullet(P_1, x_{i-1}[1])$$

vanishes. To see this, consider the map  $z_i \rightarrow x_{i-1}[1]$ . It suffices to prove that for every HN factor  $y$  of  $x_{i-1}[1]$  and every map  $z_i \rightarrow y$ , the induced map

$$\text{Hom}^\bullet(P_1, z_i) \rightarrow \text{Hom}^\bullet(P_1, y)$$

vanishes. But the HN factors of  $x_{i-1}[1]$  are simply  $z_j[1]$  for  $j < i$ . Using that both  $z_i$  and  $z_j$  are shifts of  $P_2$  or  $X$  and that  $\phi(z_j) \geq \phi(z_i)$ , we see that (up to isomorphisms and shifts) the only nonzero possibilities for  $z_i \rightarrow z_j[1]$  are

$$P_2 \rightarrow P_2[2], \quad X \rightarrow X[2], \quad \text{and} \quad P_2 \rightarrow X[2].$$

All three are killed by the functor  $\text{Hom}^\bullet(P_1, -)$ .

Since the connecting maps in (42) vanish, we get

$$\text{Hom}^\bullet(P_1, x) \cong \bigoplus \text{Hom}^\bullet(P_1, z_i),$$

and by taking dimensions

$$\dim \text{Hom}^\bullet(P_1, x) = \sum \dim \text{Hom}^\bullet(P_1, z_i).$$

We now treat the case of  $x$  supported at the other two states. Instead of directly proving linearity, we argue using the group action. By Proposition 7.11, the JH multiplicities of  $P_2$  in  $x$  and  $\sigma_1^r x$  are equal. We also have

$$\overline{\text{hom}}(P_1, x) = \overline{\text{hom}}(P_1, \sigma_1^r x).$$

Hence, the proposition for  $x$  implies the proposition for  $\sigma_1^r x$ . By Proposition 7.3, any  $x$  supported at  $[P_1, P_2]$  that is not a shift of  $P_1$  or  $P_2$  can be written as  $x = \sigma_1^r y$  for some  $r \geq 0$  and  $y$  supported at  $[P_2, X]$ . Using the analogous statements for the other states, any  $x$  supported at  $[P_2, X]$  that is not a shift of  $P_2$  or  $X$  can be written as  $x = \sigma_1^r y$  for some  $r \leq 0$  and  $y$  supported at  $[P_2, X]$ . Thus we reduce to the previous case.  $\square$

Consider the action of  $B_3$  on  $\mathbf{R}^2/\pm$  via the homomorphism  $B_3 \rightarrow \text{PSL}_2 \mathbf{Z}$ . Let  $\pi: \mathbf{R}^2 \rightarrow \mathbf{R}$  be the second projection.

**Proposition 7.15.** *Let  $\beta \in B_3$ . Set  $s = \beta P_1$  and let  $x$  be any spherical object with  $i(x) = [a : c]$ , where  $a, c \in \mathbf{Z}$  are relatively prime. Then*

$$\overline{\text{hom}}(s, x) = |\pi(\beta^{-1}(a, c))|.$$

*In particular, we have*

$$\overline{\text{hom}}(P_1, x) = |c|, \quad \overline{\text{hom}}(P_2, x) = |a|, \quad \overline{\text{hom}}(P_{21}, x) = |a + c|, \quad \text{and} \quad \overline{\text{hom}}(P_{12}, x) = |a - c|.$$

*Proof.* We have

$$\overline{\text{hom}}(s, x) = \overline{\text{hom}}(\beta P_1, x) = \overline{\text{hom}}(P_1, \beta^{-1} x).$$

By Proposition 7.14, the last quantity is  $|\pi(\beta^{-1}(a, c))|$ , as required. Taking  $\beta = 1, \gamma^{-1}, \sigma_2$ , and  $\sigma_2^{-1}$  yield the four equations.  $\square$

We now have the tools to construct the extension asserted in (41). Recall from (28) that  $h: \mathbf{S} \rightarrow \mathbf{P}^{\mathbf{S}}$  is defined by  $h: x \mapsto [\overline{\text{hom}}(x, -)]$ . Consider  $\mathbf{S}$  as a subset of  $\mathbf{P}^1(\mathbf{R})$  via the bijection  $i: \mathbf{S} \rightarrow \mathbf{P}^1(\mathbf{Z})$  from (39).

**Proposition 7.16.** *The map  $h: \mathbf{S} \rightarrow \mathbf{P}^{\mathbf{S}}$  extends to a continuous map  $\mathbf{P}^1(\mathbf{R}) \rightarrow \mathbf{P}^{\mathbf{S}}$ , which is a homeomorphism onto its image.*

*Proof.* For an  $s \in \mathbf{S}$ , chose a  $\beta_s \in B_3$  such that  $s = \beta_s P_1$ . Define  $\tilde{h}: \mathbf{P}^1(\mathbf{R}) \rightarrow \mathbf{P}^{\mathbf{S}}$  by

$$\tilde{h}: [a : c] \mapsto [|\pi(\beta_s^{-1}(a, c))|].$$

Then  $\tilde{h}$  is evidently continuous and by Proposition 7.15, it extends  $h$ .

We check that  $\tilde{h}$  is a homeomorphism onto its image. Since the domain is compact and the target is Hausdorff, it suffices to check that  $\tilde{h}$  is injective. Consider the composition of  $\tilde{h}$  with the projection onto the homogenous coordinates corresponding to the objects  $P_1, P_2$ , and  $P_{21}$ . By Proposition 7.15, the composition is given by

$$[a : c] \mapsto [|a| : |c| : |a + c|].$$

This map is injective, and hence so is  $\tilde{h}$ .  $\square$

**7.3. Stability conditions and the mass map.** In this section we study the action of the braid group  $B_3$  on the space of stability conditions  $\text{Stab}(\mathcal{C})/\mathbf{C}$ . We show that the mass map  $m: \text{Stab}(\mathcal{C})/\mathbf{C} \rightarrow \mathbf{P}^{\mathbf{S}}$  is a homeomorphism onto its image  $M$ .

Recall that  $X = P_{21}$  and  $\gamma = \sigma_2\sigma_1$ . Let  $\Lambda^\circ \subset \text{Stab}(\mathcal{C})/\mathbf{C}$  be the image in of the set of off-the-wall standard stability conditions of type I.

**Proposition 7.17.** *The subset  $\Lambda^\circ \subset \text{Stab}(\mathcal{C})/\mathbf{C}$  is open. It consists of points  $[\tau] \in \text{Stab}(\mathcal{C})/\mathbf{C}$  such that  $P_1, P_2, X$  are  $\tau$ -stable and*

$$\phi_\tau(P_1) < \phi_\tau(X) < \phi_\tau(P_2) < \phi_\tau(P_1) + 1.$$

Furthermore,  $\gamma\Lambda^\circ = \Lambda^\circ$ .

*Proof.* By Lemma 6.15, the image  $U$  of the set of standard stability conditions is open. The set  $\Lambda^\circ \subset U$  consists of  $[\tau] \in U$  for which  $\phi_\tau(P_1) < \phi_\tau(P_2)$ . It follows that it is open in  $\text{Stab}(\mathcal{C})/\mathbf{C}$ . The second assertion also follows from Lemma 6.15. For the last assertion, observe that  $\gamma(P_1) = X[-1]$  and  $\gamma(X) = P_2[-1]$  and  $\gamma(P_2) = P_1$ . It follows that the condition that  $P_1, P_2, X$  are semi-stable with

$$\phi(P_1) < \phi(X) < \phi(P_2) < \phi(P_1) + 1$$

is  $\gamma$ -invariant. □

Let  $\Lambda \subset \text{Stab}(\mathcal{C})/\mathbf{C}$  be the closure of  $\Lambda^\circ$ . Since  $\Lambda^\circ$  is  $\gamma$ -invariant, so is  $\Lambda$ .

**Proposition 7.18.** *The set  $\Lambda$  consists of  $[\tau] \in \text{Stab}(\mathcal{C})/\mathbf{C}$  such that  $P_1, P_2, X$  are  $\tau$ -semistable and*

$$(43) \quad \phi_\tau(P_1) \leq \phi_\tau(X) \leq \phi_\tau(P_2) \leq \phi_\tau(P_1) + 1.$$

*The subset  $\Lambda - \Lambda^\circ$  consists of the following three disjoint connected components:*

- (1) *the set  $\partial$  consisting of the images of standard on-the-wall stability conditions; these satisfy  $\phi(P_1) = \phi(X) = \phi(P_2)$ ;*
- (2) *the set  $\gamma(\partial)$ ; these satisfy  $\phi(P_2) = \phi(X) = \phi(P_1) + 1$ ;*
- (3) *the set  $\gamma^{-1}(\partial)$ ; these satisfy  $\phi(P_1) = \phi(X) = \phi(P_2) - 1$ .*

*Proof.* If  $[\tau] \in \Lambda$ , it is clear by continuity that  $P_1, P_2, X$  are  $\tau$ -semistable and their phases satisfy (43). Conversely, suppose  $[\tau] \in \text{Stab}(\mathcal{C})/\mathbf{C}$  is such that  $P_1, P_2, X$  are  $\tau$ -semistable and their phases satisfy (43). If the inequalities are strict, then  $[\tau] \in \Lambda^\circ$ .

If the inequalities are not strict, then  $Z_\tau(P_1), Z_\tau(P_2)$ , and  $Z_\tau(X) = Z_\tau(P_1) + Z_\tau(P_2)$  lie on a (real) line in  $\mathbf{C}$ . If all three lie on the same ray, then  $\phi_\tau(P_1) = \phi_\tau(X) = \phi_\tau(P_2)$ . In this case, then up to the  $\mathbf{C}$  action,  $\tau$  is a standard on-the-wall stability condition. It is a limit of a standard type I off-the-wall stability condition, so  $[\tau] \in \Lambda$ . The set of such  $[\tau]$  is  $\partial$ . If  $Z_\tau(P_2)$  and  $Z_\tau(X)$  lie on the same ray, but  $Z_\tau(P_1)$  lies on the opposite ray, then  $\phi_\tau(X) = \phi_\tau(P_2) = \phi_\tau(P_1) + 1$ . In this case,  $\gamma^{-1}(\tau)$  lies in  $\partial$ . If  $Z_\tau(P_1)$  and  $Z_\tau(X)$  lie on the same ray, but  $Z_\tau(P_2)$  lies on the opposite ray, then  $\phi_\tau(P_1) = \phi_\tau(X) = \phi_\tau(P_2) - 1$ . In this case,  $\gamma(\tau)$  lies in  $\partial$ . □

The manifold  $\text{Stab}(\mathcal{C})/\mathbf{C}$  is tiled by the translates of  $\Lambda$  in the following sense.

**Proposition 7.19.**  $\text{Stab}(\mathcal{C})/\mathbf{C}$  is the union of the translates  $g(\Lambda)$  for  $g \in B_3$ . Furthermore,

$$\Lambda \cap g(\Lambda) = \begin{cases} \Lambda & \text{if } g \in \langle \gamma \rangle \\ \partial & \text{if } g \in \sigma_1 \langle \gamma \rangle \\ \gamma \partial & \text{if } g \in \sigma_X \langle \gamma \rangle \\ \gamma^{-1} \partial & \text{if } g \in \sigma_2 \langle \gamma \rangle \\ \emptyset & \text{otherwise.} \end{cases}$$

See Fig. 2 for an illustration of the tiling.

*Proof.* It follows from e.g. [25, 26] that  $\text{Stab}(\mathcal{C})/\mathbf{C}$  is connected, and is the union of the  $B_3$ -translates of the set  $U$  of standard stability conditions. But it is easy to check that  $U = \Lambda \cup \sigma_1 \Lambda$ . So  $\text{Stab}(\mathcal{C})/\mathbf{C}$  is the union of  $B_3$ -translates of  $\Lambda$ .

To understand the intersections of the translates, we recall that the only semi-stable spherical objects (up to shift) of a standard stability condition of type I are  $\{P_1, P_2, X\}$  if it is off-the-wall and  $\{P_1, P_2, X, P_{12}\}$  if it is on-the-wall.

Suppose  $g \in B_3$  is such that  $\Lambda \cap g(\Lambda)$  is non-empty. If the intersection contains a point of  $\Lambda^\circ$ , then we conclude that  $g$  must permute  $\{P_1, P_2, X\}$  (up to shift). Using Proposition 7.4 (and its analogue for  $P_2$  and  $X$ ), we get  $g = \gamma^n$ . In this case,  $g(\Lambda) = \Lambda$ .

If  $\Lambda \cap g(\Lambda)$  is non-empty but does not contain a point of  $\Lambda^\circ$ , then we conclude that  $g$  must permute  $\{P_1, P_2, X, P_{12}\}$ . In this case,  $g$  must map at least two of  $\{P_1, P_2, X\}$  to  $\{P_1, P_2, X\}$  (up to shift). Again, Proposition 7.4 implies that  $g = \gamma^n \sigma_1$  or  $\gamma^n \sigma_2$  or  $\gamma^n \sigma_X$ . The three cases give the next three possibilities for  $\Lambda \cap g(\Lambda)$ .  $\square$

Having established basic results about  $\text{Stab}(\mathcal{C})/\mathbf{C}$ , we turn to the mass map

$$m: \text{Stab}(\mathcal{C})/\mathbf{C} \rightarrow \mathbf{P}^{\mathbf{S}}.$$

Our goal is to prove that it is an embedding. Recall from Definition 6.18 that  $x \rightarrow y \rightarrow z \xrightarrow{+1}$  is a collapsed triangle if  $m(y) = m(x) + m(z)$ .

**Proposition 7.20.** *Consider the following collection of distinguished triangles:*

$$T = \{P_1 \rightarrow X \rightarrow P_2 \xrightarrow{+1}, \quad X \rightarrow P_2 \rightarrow P_1[1] \xrightarrow{+1}, \quad P_2[-1] \rightarrow P_1 \rightarrow X \xrightarrow{+1}\}.$$

*If  $\tau$  is an off-the-wall standard stability condition of type I, then no triangle in  $T$  is collapsed by  $\tau$ . Furthermore, if  $\tau'$  is any stability condition such that no triangle in  $T$  is collapsed by  $\tau'$ , then  $\tau'$  is standard of type I, up to the action of  $\mathbf{C}$ .*

*Proof.* It is immediate from Proposition 3.3 (or a direct check) that no triangle in  $T$  is collapsed by any off-the-wall standard stability condition of type I.

Let  $\tau'$  be any stability condition. Recall that  $\tau'$  is in the braid group orbit of a standard stability condition  $\tau$  of type I. Write  $\tau' = \beta \tau''$  for some braid  $\beta$  and some standard stability condition  $\tau''$  of type I.

If  $\beta$  is a power of  $\gamma$ , then  $\tau'$  is already standard of type I, up to the action of  $\mathbf{C}$ . Otherwise, we exhibit a triangle of  $T$  that is collapsed by  $\tau'$ . Equivalently, we exhibit a triangle of  $\beta^{-1}T$  that is collapsed by  $\tau''$ .

Suppose that  $\tau''$  is off-the-wall. Consider a cyclic writing of  $\beta^{-1}$  as described in Proposition 7.2. Since  $\beta$  is not a power of  $\gamma$ , this writing has rightmost letter either  $\sigma_1$ ,  $\sigma_2$ , or  $\sigma_X$ . Suppose the rightmost letter of  $\beta^{-1}$  is  $\sigma_1$ . Consider the second triangle in  $T$ , namely

$X \rightarrow P_2 \rightarrow P_1[1] \xrightarrow{+1}$ . Observe from the automaton in Fig. 4 that the objects  $P_1$ ,  $P_2$ , and  $X$  are all sent to the state  $[P_1, P_2]$  after applying  $\sigma_1$ . Furthermore, we have the equation

$$\mathrm{HN}_{\tau''}(\sigma_1(P_2)) = \mathrm{HN}_{\tau''}(\sigma_1(X)) + \mathrm{HN}_{\tau''}(\sigma_1(P_1)).$$

The equation above persists by linearity as we apply each successive letter of the cyclic writing of  $\beta^{-1}$ , resulting in the equation

$$\mathrm{HN}_{\tau''}(\beta^{-1}(P_2)) = \mathrm{HN}_{\tau''}(\beta^{-1}(X)) + \mathrm{HN}_{\tau''}(\beta^{-1}(P_1)),$$

and hence

$$m_{\tau''}(\beta^{-1}(P_2)) = m_{\tau''}(\beta^{-1}(X)) + m_{\tau''}(\beta^{-1}(P_1)).$$

Hence the triangle  $\beta^{-1}X \rightarrow \beta^{-1}P_2 \rightarrow \beta^{-1}P_1[1] \xrightarrow{+1}$  is collapsed by  $\tau''$ . Similarly, we can check that if the cyclic writing of  $\beta^{-1}$  ends in  $\sigma_2$  (respectively  $\sigma_X$ ), then the first (respectively third) triangle of  $T$  is collapsed by  $\tau'$ .

If  $\tau''$  is on-the-wall, then it is a limit of on-the-wall stability conditions. By continuity of the masses, we still conclude that  $\tau' = \beta\tau''$  collapses one of the triangles in  $T$ .  $\square$

**Proposition 7.21.** *Consider the following collection of distinguished triangles:*

$$T = \{X \rightarrow P_2 \rightarrow P_1[1] \xrightarrow{+1}, \quad P_2[-1] \rightarrow P_1 \rightarrow X \xrightarrow{+1}, \quad P_{12} \rightarrow P_1 \rightarrow P_2[1] \xrightarrow{+1}\}.$$

*If  $\tau$  is an on-the-wall standard stability condition, then no triangle in  $T$  is collapsed by  $\tau$ . Furthermore, if  $\tau'$  is any stability condition such that no triangle in  $T$  is collapsed by  $\tau'$ , then  $\tau'$  is standard of type I, up to the action of  $\mathbf{C}$ .*

*Proof.* Analogous to the proof of Proposition 7.20.  $\square$

**Proposition 7.22.** *The map  $m: \mathrm{Stab}(\mathcal{C})/\mathbf{C} \rightarrow \mathbf{P}^{\mathbf{S}}$  is a homeomorphism onto its image.*

*Proof.* Proposition 6.14 shows that  $m$  is injective. Proposition 6.19 together with Proposition 7.20 and Proposition 7.21 shows that  $m$  is a local homeomorphism at a standard stability condition of type I. Since every  $\tau \in \mathrm{Stab}(\mathcal{C})/\mathbf{C}$  is in the braid group orbit of a standard stability condition of type I, it follows that  $m$  is a local homeomorphism at every  $\tau$ . As a result,  $m$  is a homeomorphism onto its image.  $\square$

**Remark 7.23.** The complement of  $\Lambda$  in  $\mathrm{Stab}(\mathcal{C})/\mathbf{C}$  has three connected components. We can describe these three components in two ways, as follows.

- (1) Write  $\tau'$  in the complement of  $\Lambda$  as  $\tau' = \beta\tau$ , where  $\beta \in B_3$  and  $\tau \in \Lambda$ . Write  $\beta$  in a form recognised by the automaton in Fig. 4. Then the three components correspond to the three possible end states of the automaton when it reads the writing of  $\beta$ .
- (2) Recall from the proof of Proposition 7.20 that the three numbers

$$m_{\tau'}(P_1), m_{\tau'}(P_2), \text{ and } m_{\tau'}(X)$$

satisfy a collapsed triangle inequality—one is the sum of the other two. The three connected components correspond to the ways in which the inequality collapses.

**7.4. Closure of the stability manifold.** Let  $M \subset \mathbf{P}^{\mathbf{S}}$  be the homeomorphic image of  $\mathrm{Stab}(\mathcal{C})/\mathbf{C}$ . In this section, we study the closure  $\overline{M}$  and prove that it is the disjoint union of  $M$  and  $P$ .

**Definition 7.24** (Gromov coordinates). Let  $\tau$  be a standard stability condition of type I. Since the three positive real numbers  $m_{\tau}(P_1)$ ,  $m_{\tau}(P_2)$ , and  $m_{\tau}(X)$  satisfy the triangle inequalities, there exist non-negative real numbers  $x, y, z$  such that

$$m_{\tau}(P_1) = y + z, \quad m_{\tau}(P_2) = z + x, \quad m_{\tau}(X) = x + y.$$

We call the  $x, y, z$ , the *Gromov coordinates* of  $\tau$ .

More generally, let  $\tau$  be an arbitrary stability condition. Then there are three  $\tau$  semistable objects, say  $A, B, C$ , whose classes in the Grothendieck group are (up to sign) the classes of  $P_1, P_2$ , and  $X$ . (These objects are obtained simply by applying an appropriate braid to  $P_1, P_2$ , and  $X$ .) Define the Gromov coordinates  $x, y, z$  for  $\tau$  by the condition

$$m_\tau(A) = y + z, \quad m_\tau(B) = x + z, \quad m_\tau(C) = x + y.$$

Note that if  $\tau$  is off-the-wall, then the Gromov coordinates are all positive. If  $\tau$  is on-the-wall, then one Gromov coordinate is zero.

Recall that  $m_\tau: \mathbf{S} \rightarrow \mathbf{R}$  is the mass function associated to  $\tau$ . Let  $\overline{\text{hom}}(s): \mathbf{S} \rightarrow \mathbf{R}$  be the function  $\overline{\text{hom}}(s, -)$  as defined in Definition 4.9.

**Proposition 7.25** (Linearity). *Let  $\tau$  be a stability condition. Let  $A, B, C$  be  $\tau$  semi-stable objects of class  $[P_1], [P_2], [X]$  and let  $x, y, z$  be the Gromov coordinates of  $\tau$ . Then we have*

$$m_\tau = x \overline{\text{hom}}(A) + y \overline{\text{hom}}(B) + z \overline{\text{hom}}(C).$$

*Proof.* It suffices to treat the case of standard stability conditions of type I. The general statement follows by applying the  $B_3$ -action.

Let  $s \in \mathbf{S}$  be a spherical object. Suppose  $\tau$  is off-the-wall. Observe that the HN filtration of  $s$  is the same for all off-the-wall type I stability conditions, and its stable factors are  $P_1, P_2$ , and  $X$ , up to shift. Denoting by  $a(s), b(s)$ , and  $c(s)$  the multiplicities of these three, we have

$$m_\tau(s) = (y + z)a(s) + (x + z)b(s) + (x + y)c(s).$$

In particular,  $m_\tau(s)$  is linear in the Gromov coordinates. Using Proposition 7.11 and Proposition 7.14, we have

$$\begin{aligned} b(s) + c(s) &= \text{JH multiplicity of } P_2 \text{ in } s \\ &= \overline{\text{hom}}(P_1, s); \\ a(s) + c(s) &= \text{JH multiplicity of } P_1 \text{ in } s \\ &= \overline{\text{hom}}(P_2, s). \end{aligned}$$

By applying  $\gamma$  to the first equality, we get  $a(s) + b(s) = \overline{\text{hom}}(X, s)$ . It follows that

$$m_\tau(s) = x \overline{\text{hom}}(P_1, s) + y \overline{\text{hom}}(P_2, s) + z \overline{\text{hom}}(X, s).$$

The case of an on-the-wall  $\tau$  follows by continuity.  $\square$

Let  $\overline{\Delta} = (\mathbf{R}_{\geq 0}^3 \setminus 0) / \mathbf{R}_{>0}$ . Then  $\overline{\Delta}$  is a closed triangle. Let  $\Delta \subset \overline{\Delta}$  be the complement of the three vertices. Note that  $\Delta$  consists of points  $[x : y : z]$  where at least two of the three coordinates are non-zero.

Recall from Section 7.3 that  $\Lambda \subset \text{Stab}(\mathcal{C})/\mathbf{C}$  is the closure of the set of standard stability conditions of type I. We have a homeomorphism  $\Delta \rightarrow m(\Lambda) \subset \mathbf{P}^{\mathbf{S}}$  induced by

$$(44) \quad (x, y, z) \mapsto x \cdot \overline{\text{hom}}(P_1) + y \cdot \overline{\text{hom}}(P_2) + z \cdot \overline{\text{hom}}(X).$$

To ease notation, we identify  $\Lambda$  with  $m(\Lambda) \subset \mathbf{P}^{\mathbf{S}}$ .

**Proposition 7.26** (Closure of  $\Lambda$ ). *The closure of  $\Lambda$  in  $\mathbf{P}^{\mathbf{S}}$  is*

$$\overline{\Lambda} = \Lambda \cup \{h(P_1), h(P_2), h(X)\}.$$

*The homeomorphism  $\Delta \rightarrow \Lambda$  in (44) extends to a homeomorphism  $\overline{\Delta} \rightarrow \overline{\Lambda}$ .*

*Proof.* Define  $\overline{\Delta} \rightarrow \mathbf{P}^{\mathbf{S}}$  using the same formula (44). Since  $\overline{\Delta}$  is compact, the image is closed. Evidently, the image is the union of  $\Lambda$  and the three additional points  $h(P_1), h(P_2)$ , and  $h(X)$ . The projection  $\overline{\Delta} \rightarrow \mathbf{P}^{\{P_1, P_2, X\}}$  is

$$[x : y : z] \mapsto [y + z : x + z : x + y],$$

which is injective. So  $\overline{\Delta} \rightarrow \mathbf{P}^{\mathbf{S}}$  is also injective and hence a homeomorphism onto its image. Since  $\Delta \subset \overline{\Delta}$  is dense,  $\Lambda$  is dense in the image of  $\overline{\Delta}$ . So the image of  $\overline{\Delta}$  is the closure  $\overline{\Lambda}$ .  $\square$

*Remark 7.27.* Observe that the points  $h(P_1)$  and  $h(P_2)$  in  $\overline{M}$  are the two end-points of the closure of the edge  $\partial \subset \Lambda$ . Consider the graph whose vertices are  $\mathbf{S}$  and whose edges are the  $\mathrm{PSL}_2(\mathbf{Z})$  translates of the edge  $P_1 P_2$ . Identify  $\mathbf{S}$  with  $\mathbf{P}^1(\mathbf{Z})$  by the bijection  $i$  in (39). Then the graph is the Farey graph on  $\mathbf{P}^1(\mathbf{Z})$ : two points  $[a : c]$  and  $[b : d]$  (expressed in reduced form) are adjacent if and only if  $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \pm 1$ .

Having understood the closure of  $\Lambda$ , we now turn to the closure of  $M$ . It is convenient to have a measure of closeness for two elements of a projective space. Let  $n$  be a positive integer. Given two nonzero vectors in  $\mathbf{R}^n$ , denote by  $\angle(v, w)$  the (acute) angle between the lines they span. By a slight abuse of notation, we use  $\angle(v, w)$  also for two points  $v, w \in \mathbf{P}^{n-1}$ ; this is just the angle between any two representatives in  $\mathbf{R}^n$ .

Given a finite subset  $T \subset \mathbf{S}$ , let  $h_T : \mathbf{S} \rightarrow \mathbf{P}^T$  be the composition of  $h : \mathbf{S} \rightarrow \mathbf{P}^{\mathbf{S}}$  and the projection onto the  $T$ -coordinates.

**Proposition 7.28** (Group action contracts). *Fix two elements  $x, y \in \mathbf{S}$  and a finite subset  $T \subset \mathbf{S}$ . Given an  $\epsilon > 0$ , for all but finitely many elements  $g$  of  $\mathrm{PSL}_2(\mathbf{Z})$ , we have*

$$\angle(h_T(gx), h_T(gy)) < \epsilon.$$

*Proof.* Recall the bijection  $i : \mathbf{S} \rightarrow \mathbf{P}^1(\mathbf{Z})$ . Set  $a = i(x)$  and  $b = i(y)$ . The map  $h_T : \mathbf{P}^1(\mathbf{R}) \rightarrow \mathbf{P}^T$  is continuous, and hence uniformly continuous. Therefore, it suffices to check that for all but finitely many  $g$ , the angle between  $ga$  and  $gb$  in  $\mathbf{R}^2$  is small. Let  $X$  be the matrix with columns  $a$  and  $b$ . Then,

$$|\sin(\angle(ga, gb))| = \frac{|\det g| \cdot |\det(X)|}{|ga| \cdot |gb|} = \frac{|\det(X)|}{|ga| \cdot |gb|}.$$

Since  $g \in \mathrm{PSL}_2(\mathbf{Z})$  has integer entries, the quantity  $|ga| \cdot |gb|$  is greater than  $\epsilon^{-1} |\det(X)|$  for all but finitely many  $g$ . The claim follows.  $\square$

Recall that  $M = m(\mathrm{Stab}(\mathcal{C})/\mathbf{C}) \subset \mathbf{P}^{\mathbf{S}}$  and  $P \subset \mathbf{P}^{\mathbf{S}}$  is the closure of  $h(\mathbf{S})$ .

**Proposition 7.29** (Closure of  $M$ ). *The sets  $M$  and  $P$  are disjoint and their union is the closure of  $M$ .*

*Proof.* Recall that we have an embedding  $i : \mathbf{S} \rightarrow \mathbf{P}^1(\mathbf{R})$  and  $P = h(\mathbf{P}^1(\mathbf{R}))$ . Let  $[a : c] \in \mathbf{P}^1(\mathbf{R})$  be arbitrary. From Proposition 7.15, the  $\{P_1, P_2, P_{21}, P_{12}\}$  coordinates of  $h[a : c]$  are  $(|c|, |a|, |a + c|, |a - c|)$ , respectively. Note that the first three coordinates always form a degenerate triangle. On the other hand, for  $[\tau] \in \Lambda^\circ$ , the  $P_1, P_2, P_{21}$  coordinates of  $m_\tau$  form a non-degenerate triangle. So  $h[a : c] \notin \Lambda^\circ$ . For  $[\tau]$  on the side  $\partial \subset \Lambda$ , we have

$$m_\tau(P_{12}) = m_\tau(P_{21}) \text{ and } m_\tau(P_1), m_\tau(P_2) > 0.$$

But  $|a + c| = |a - c|$  only holds if  $a = 0$  or  $c = 0$ . It follows that  $h[a : c] \notin \partial$ . So  $P$  is disjoint from  $\Lambda^\circ \cup \partial$ . Since  $P$  is  $B_3$ -invariant and  $M$  is the union of  $B_3$ -translates of  $\Lambda^\circ \cup \partial$  (Proposition 7.19), we conclude that  $M$  and  $P$  are disjoint.

By Proposition 7.26, we know that  $P$  is contained in the closure of  $M$ . We now prove that  $M \cup P$  is closed. Let  $\tau_n$  be a sequence in  $\text{Stab}(\mathcal{C})/\mathbf{C}$  whose images  $m(\tau_n)$  in  $M$  approach a limit  $t \in \mathbf{P}^{\mathbf{S}}$ . We show that  $t$  lies in  $M \cup P$ .

Write  $\tau_n = \beta_n \tau'_n$ , where  $\beta_n$  is a braid and  $\tau'_n \in \Lambda$  is a standard stability condition of type I. Let  $\overline{\beta_n}$  be the image of  $\beta_n$  in  $\text{PSL}_2(\mathbf{Z})$ . We have two possibilities: the set  $\{\overline{\beta_n}\}$  is finite or infinite.

If  $\{\overline{\beta_n}\}$  is finite, then the sequence  $\{\tau_n\}$  lies in the union of finitely many translates of  $\Lambda$ . By Proposition 7.26, the closure of  $\Lambda$  is contained in  $M \cup P$ . Hence, the closure of the union of finitely many translates of  $\Lambda$  is also contained in  $M \cup P$ . So the limit  $t$  lies in  $M \cup P$ .

Suppose  $\{\overline{\beta_n}\}$  is infinite. Set  $a_n = \beta_n(P_1)$ ,  $b_n = \beta_n(P_2)$ , and  $c_n = \beta_n(X)$ . Since  $P$  is compact, we may assume, after passing to a subsequence if necessary, that  $h(a_n)$ ,  $h(b_n)$ , and  $h(c_n)$  have limits in  $P$ , say  $a$ ,  $b$ , and  $c$ .

We first prove that  $a = b = c$ . To see this, let  $T \subset \mathbf{S}$  be an arbitrary finite set, and use the subscript  $T$  to denote projections to  $\mathbf{P}^T$ . It suffices to show that  $a_T = b_T = c_T$ . Since  $\{\overline{\beta_n}\}$  is infinite, by Proposition 7.28, we note that the angles between  $h_T(a_n)$ ,  $h_T(b_n)$ , and  $h_T(c_n)$  approach 0 as  $n$  approaches  $\infty$ . Therefore, the three sequences have the same limits. But these limits are  $a_T$ ,  $b_T$ , and  $c_T$ .

Next, we claim that  $t = h(a) = h(b) = h(c)$ . Indeed, by linearity (Proposition 7.25), we know that there exist non-negative real numbers  $x_n, y_n, z_n$  such that

$$m_T(\tau_n) = x_n h_T(A_n) + y_n h_T(B_n) + z_n h_T(C_n).$$

Taking the limit as  $n \rightarrow \infty$  in projective space yields

$$t = h(a) = h(b) = h(c).$$

In particular,  $t$  lies in  $M \cup P$ . The proof is thus complete.  $\square$

**7.5. Homeomorphism to the closed disk.** In this section, we give a homeomorphism from  $\overline{M}$  to the closed disk, that sends  $M$  to the open disk, and  $P$  to the boundary circle. The homeomorphism uses unprojectivised, but suitably normalised, mass and  $\overline{\text{hom}}$  functions.

Letting  $X = P_{21}$ , set

$$T = \{P_1, P_2, X\}.$$

Define a map

$$\mu: \text{Stab}(\mathcal{C})/i\mathbf{R} \rightarrow \mathbf{R}^T = \mathbf{R}^3$$

by

$$\mu: \tau \mapsto (m_\tau(P_1), m_\tau(P_2), m_\tau(X)).$$

Similarly, define

$$\eta: \mathbf{S} \rightarrow \mathbf{R}^T = \mathbf{R}^3$$

by

$$\eta(s) = (\overline{\text{hom}}(s, P_1), \overline{\text{hom}}(s, P_2), \overline{\text{hom}}(s, X)).$$

Thinking of  $\mathbf{S}$  as  $\mathbf{P}^1(\mathbf{Z}) \subset \mathbf{P}^1(\mathbf{R})$ , it is easy to see that  $\eta$  extends to a continuous map

$$\eta: \mathbf{P}^1(\mathbf{R}) \rightarrow \mathbf{R}^3.$$

Indeed, using Proposition 7.15, we get

$$\eta: [a : c] \mapsto (|c|, |a|, |a + c|).$$



Let  $\tau$  be a stability condition with (semi)-stable objects  $A$ ,  $B$ , and  $C$  of class  $[P_1]$ ,  $[P_2]$ , and  $[X]$  in the Grothendieck group. Let  $x, y, z$  be the Gromov coordinates of  $\tau$ , namely the non-negative real numbers such that

$$m_\tau(A) = y + z, \quad m_\tau(B) = z + x, \quad m_\tau(C) = x + y.$$

Denote by  $|\cdot|$  the standard Euclidean norm in  $\mathbf{R}^3$ . We say that  $\tau$  is *normalised* if

$$(45) \quad x|\eta(A)| + y|\eta(B)| + z|\eta(C)| = 1.$$

*Remark 7.30.* We point out one subtle aspect of the definition. If  $\tau$  is on-the-wall, then one of the  $A$ ,  $B$ , or  $C$  is not uniquely determined (even up to shift), since on-the-wall stability conditions have non-isomorphic semi-stable objects with the same class in the Grothendieck group (for example,  $P_{21}$  and  $P_{12}$ ). In this case, however, the Gromov coordinate corresponding to the ambiguous object is 0. So (45) is unambiguous.

The normalisation yields a continuous section of

$$\text{Stab}(\mathcal{C})/i\mathbf{R} \rightarrow \text{Stab}(\mathcal{C})/\mathbf{C}.$$

That is, for every stability condition  $\tau$  up to rotation and scaling, there is a unique normalised stability condition  $\tau^\nu$  up to rotation, and furthermore the map  $\tau \rightarrow \tau^\nu$  is continuous.

We now identify  $\text{Stab}(\mathcal{C})/\mathbf{C}$  with its image  $M$  in  $\mathbf{P}^{\mathbf{S}}$  and  $\mathbf{P}^1(\mathbf{R})$  with its image  $P$  in  $\mathbf{P}^{\mathbf{S}}$ . Define

$$(46) \quad \pi: \overline{M} = M \cup P \rightarrow \mathbf{R}^3$$

as follows. For  $\tau \in M$ , set

$$(47) \quad \pi(\tau) = \mu(\tau^\nu),$$

and for  $s \in P$ , set

$$\pi(s) = \eta(s)/|\eta(s)|.$$

By construction,  $\pi([a : c])$  is the unit vector in the direction of

$$\eta([a : c]) = (|a|, |c|, |a + c|).$$

That is,  $\pi$  maps  $P$  to the unit sphere in  $\mathbf{R}^3$ . It is easy to see that  $\pi|_P$  is injective, and hence a homeomorphism onto its image. The image consists of three circular arcs, one for each pair of end-points from the three points

$$\frac{1}{\sqrt{2}}(0, 1, 1), \quad \frac{1}{\sqrt{2}}(1, 0, 1), \quad \text{and} \quad \frac{1}{\sqrt{2}}(1, 1, 0).$$

The arc joining each pair is a geodesic arc on the unit sphere; that is, the plane it spans passes through the origin.

Eq. (45) and the triangle inequality imply that for every stability condition  $\tau$ , we have  $|\pi(\tau)| \leq 1$ . In fact, for every stability condition, at least two of the Gromov coordinates are nonzero. Therefore, we actually have a strict inequality  $|\pi(\tau)| < 1$ .

Recall that  $\Lambda$  is the closure of the type I stability conditions in  $\text{Stab}(\mathcal{C})/\mathbf{C}$ . Recall also that the  $B_3$ -translates of  $\Lambda$  tile  $\text{Stab}(\mathcal{C})/\mathbf{C}$ . To get a better understanding of  $\pi$ , let us study it on the translates of  $\Lambda$ . Let  $\beta$  be a braid. Consider the translate  $\beta\bar{\Lambda}$ . Set

$$A = \beta(P_1), \quad B = \beta(P_2), \quad C = \beta(X).$$

These are the (semi)-stable objects for the stability conditions in  $\beta\bar{\Lambda}$ . Recall that  $\beta\bar{\Lambda}$  is homeomorphic to the projectivised octant  $\bar{\Delta} = (\mathbf{R}_{\geq 0}^3 \setminus 0) / \mathbf{R}_{>0}$ ; the homeomorphism is

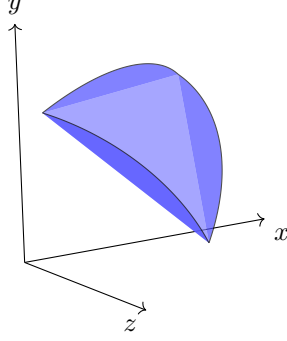


FIGURE 7. The homeomorphic image  $\Phi \subset \mathbf{R}^3$  of the compactified stability manifold.

given by the Gromov coordinates. The normalisation condition (45) is a linear condition on the Gromov coordinates. It cuts out an affine hyperplane slice of the octant, and gives a section of the projectivisation map. Since  $\pi$  is linear in the Gromov coordinates for normalised stability conditions, it maps  $\beta\bar{\Lambda}$  linearly, and homeomorphically, onto the triangle in  $\mathbf{R}^3$  with vertices  $\pi(A)$ ,  $\pi(B)$ , and  $\pi(C)$ .

Let  $\Phi \subset \mathbf{R}^3$  be the union of the triangle

$$\{(x, y, z) \mid x + y + z = \sqrt{2}, x + y - z \geq 0, y + z - x \geq 0, z + x - y \geq 0\},$$

and the three circular segments, each bounded by an edge of the triangle above and the arc in the image of  $\pi(P)$  with the same end-points as the edge (see Fig. 7).

Observe that  $\pi$  maps  $\bar{\Lambda}$  to the central triangle of  $\Phi$ . On the other hand, the translates of  $\bar{\Lambda}$  on the three sides of the identity in Fig. 2 are mapped to the three circular segments. Indeed, the segment to which  $\pi$  sends a non-standard stability condition  $\tau$  depends on how the triangle inequality collapses among the  $\tau$ -masses of  $P_1$ ,  $P_2$ , and  $X$ .

**Proposition 7.31.** *The map  $\pi: \bar{M} \rightarrow \Phi$  is a homeomorphism, where  $\bar{M}$  is given the subspace topology from  $\mathbf{P}^S$ .*

*Proof.* Let us first show that  $\pi$  is continuous. Since  $\pi$  is linear on the translates of  $\Lambda$ , and these translates cover  $M$ , it follows that  $\pi$  is continuous on  $M$ . The restriction of  $\pi$  to  $P$  is given by

$$\pi: [a : c] \mapsto \frac{1}{\sqrt{a^2 + c^2 + (a + c)^2}} (|a|, |c|, |a + c|),$$

which is continuous. We must show that  $\pi$  is continuous on  $\bar{M}$  at a point  $p \in P$ .

Let  $\tau_n$  be a sequence of normalised stability conditions converging to  $p \in P$ . We already know that  $m_T(\tau_n)$  converges to  $h_T(p)$  in the projective space  $\mathbf{P}^T$ . Therefore, it suffices to show that  $\pi(\tau_n)$  approaches a vector of norm 1 in  $\mathbf{R}^3$ .

Let  $\epsilon > 0$  be given. Write  $\tau_n = \beta_n \tau'_n$  for some braid  $\beta_n$  and standard stability condition  $\tau'_n$ . Denote by  $\bar{\beta}_n$  the image of  $\beta_n$  in  $\mathrm{PSL}_2(\mathbf{Z})$ . If the set  $\{\bar{\beta}_n\}$  is finite, then the sequence  $\tau_n$  lies in finitely many translates of  $\Lambda$ . By passing to a subsequence, we may assume that it lies in one translate, say  $\beta\Lambda$ . Recall that  $\pi: \bar{\beta}\Lambda \rightarrow \mathbf{R}^3$  is linear in the Gromov coordinates and hence continuous. Therefore,  $\pi(\tau_n) \rightarrow \pi(p)$  as  $n \rightarrow \infty$ .

The harder case is when the set  $\{\bar{\beta}_n\}$  is infinite. Set  $A_n = \beta_n(P_1)$ ,  $B_n = \beta_n(P_2)$ ,  $C_n = \beta_n(X)$ , and let  $x_n, y_n, z_n$  be the Gromov coordinates of  $\tau_n$ . In this case, we know

by Proposition 7.28 that the angle between  $\eta(A_n)$ ,  $\eta(B_n)$ , and  $\eta(C_n)$  approaches 0 as  $n$  approaches  $\infty$ . Therefore, the difference between

$$|x_n\eta(A_n) + y_n\eta(B_n) + z_n\eta(C_n)| \text{ and } x_n|\eta(A_n)| + y_n|\eta(B_n)| + z_n|\eta(C_n)|$$

approaches 0 as  $n$  approaches  $\infty$ . Since  $\tau_n$  is normalised, the right-hand quantity is 1, and the left-hand quantity is  $|\pi(\tau_n)|$ .

We have now proved that  $\pi: \overline{M} \rightarrow \Phi$  is continuous. Since  $\overline{M}$  is compact,  $\pi$  is a homeomorphism once we know that it is a bijection. Let  $\partial\Phi = \Phi \cap \{v \mid |v| = 1\}$  be the boundary and  $\Phi^\circ = \Phi \cap \{v \mid |v| < 1\}$  the interior. We also know that  $M$  is the union of the translates of  $\Lambda$ , glued as in the Farey triangulation (see Remark 7.27). It is easy to see that  $\pi$  maps a Farey triangle homeomorphically to the Euclidean triangle spanned by the images of the three vertices. We also know that  $\pi: P \rightarrow \partial\Phi$  is a homeomorphism. Then it follows that  $\pi: M \rightarrow \Phi^\circ$ , and hence  $\overline{M} \rightarrow \Phi$ , is a bijection.  $\square$

## 8. THE $\hat{A}_1$ CASE

The aim of this section is to construct the Thurston compactification at  $q = 1$  of stability space for the 2-Calabi–Yau category  $\mathcal{C} := \mathcal{C}_\Gamma$ , where  $\Gamma$  is the  $\hat{A}_1$  graph. Since the  $\hat{A}_1$  graph has a double edge, we give need to slightly modify the constructions from Section 6 to define the 2-Calabi–Yau category and to construct the Artin–Tits braid group action on it. Before giving the details of this modification, we briefly describe the final picture of  $\text{Stab}(\mathcal{C})/\mathbf{C}$  and its compactification, illustrated in Fig. 8. The space  $\text{Stab}(\mathcal{C})/\mathbf{C}$  is homeomorphic to

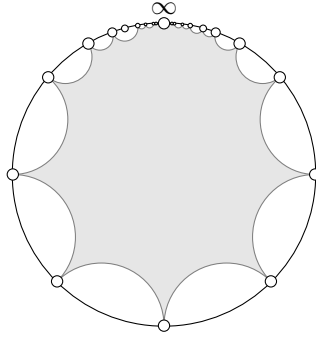


FIGURE 8. The space  $\text{Stab}(\mathcal{C})/\mathbf{C}$  is tiled by translates of an infinite-sided polygon (shaded in grey) lying in an open disk, with endpoints on the boundary circle. The vertices of this polygon have a single accumulation point, thought of as the point at infinity. The boundary of  $\text{Stab}(\mathcal{C})/\mathbf{C}$  is the boundary circle.

an open 2-dimensional disk, tiled by  $B_\Gamma$ -translates of a polygon  $\Lambda$  without its vertices. Whereas in the  $A_2$  case the relevant polygon was a triangle, here  $\Lambda$  has countably many vertices, one of which is distinguished. Here  $B_\Gamma$  is the Artin–Tits braid group of type  $\hat{A}_1$ , which is isomorphic to the free group on two generators.

The mass map embeds  $\text{Stab}(\mathcal{C})/\mathbf{C}$  homeomorphically onto a subspace  $M \subset \mathbf{P}^S$ . The closure of  $M$  is thus a compactification of  $\text{Stab}(\mathcal{C})/\mathbf{C}$ . In the compactification, the missing vertices from the tiles all appear on the boundary. Except for the distinguished ones, the boundary functionals corresponding to vertices are the hom functionals of spherical objects.

Though they also appear in the boundary, the distinguished vertex of each tile is not the  $\overline{\text{hom}}$  functional of a spherical object. In fact, the distinguished vertices correspond to a semi-rigid object of the category  $\mathcal{C}$ , although we do not include the precise statement here.

As was the case in  $A_2$ , the closure  $P$  of the set of  $\overline{\text{hom}}$  functionals is homeomorphic to a circle, disjoint from  $M$ . The closure of  $M$  is  $M \cup P$ , and we conjecture that this union is homeomorphic to a closed disk.

As was the case in type  $A_2$ , to prove the above claims we use an HN automaton. Verifying the basic properties of this automaton requires work which is independent from that done in the previous section. However, once the basic properties of this automaton are established, the rest of the arguments proceed largely as in the  $A_2$  case.

**8.1. The 2-Calabi–Yau category of type  $\hat{A}_1$ .** Since the  $\hat{A}_1$  graph has a double edge, we modify the constructions from Section 6 to handle this case.

**8.1.1. The Coxeter system of type  $\hat{A}_1$ .** Let  $\Gamma$  be the Dynkin diagram of type  $\hat{A}_1$ , which has two vertices, called 0 and 1, and two (unoriented) edges connecting them.



**Definition 8.1.** The *Artin–Tits braid group associated to  $\Gamma$*  is the free group on two letters  $F_2$ :

$$B_\Gamma \cong F_2 = \langle \sigma_0, \sigma_1 \rangle.$$

**Definition 8.2.** The *Coxeter group associated to  $\Gamma$*  is isomorphic to the quotient of  $B_\Gamma$  by the relations

$$\sigma_0^2 = \sigma_1^2 = 1.$$

We denote this group by  $W_\Gamma$ , and denote the images of  $\sigma_i$  in  $W_\Gamma$  by  $s_i$ .

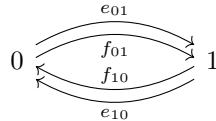
Let  $V_\Gamma$  be the rank two  $\mathbf{Z}$ -module with basis vectors  $v_i$  indexed by the vertices of  $\Gamma$ . Define a bilinear form on  $V_\Gamma$  by the following formula:

$$\langle v_i, v_j \rangle = \begin{cases} 2 & i = j, \\ -2 & i \neq j. \end{cases}$$

**Definition 8.3.** The *standard representation* of  $W_\Gamma$  is the action on  $V_\Gamma$  defined by the formula

$$s_i(v_j) = v_j - \langle v_i, v_j \rangle v_i.$$

**8.1.2. The zigzag algebra of type  $\hat{A}_1$ .** Let  $\Gamma^{dbl}$  denote the doubled quiver of  $\Gamma$ . If the two edges of  $\Gamma$  are denoted  $e$  and  $f$ , then there are four oriented edges denoted  $e_{01}, e_{10}, f_{01}, f_{10}$  in  $\Gamma^{dbl}$ , where the subscript  $ij$  means that the arrow points from  $i$  to  $j$ . Fix a sign  $s_{ij} \in \{\pm 1\}$  for each arrow  $e_{ij}$ , and a sign  $t_{ij} \in \{\pm 1\}$  for each arrow  $f_{ij}$ , such that  $s_{ij} = -s_{ji}$  and  $t_{ij} = -t_{ji}$ .



The zigzag algebra is the quotient of the path algebra  $\text{Path}(\Gamma^{dbl})$  by the two-sided ideal generated by

- (1) the elements  $e_{ij}f_{ji}$  and  $f_{ij}e_{ji}$ , for  $i \neq j$ , and

(2) the elements

$$s_{ij}e_{ij}e_{ji} - t_{ij}f_{ij}f_{ji},$$

for  $i \neq j$ .

The algebras corresponding to the two kinds of sign choices  $s_{ij} = t_{ij}$  and  $s_{ij} = -t_{ij}$  are isomorphic, as was the case in simply laced type.

8.1.3. *The 2-Calabi–Yau category of type  $\widehat{A}_1$ .* The strongly 2-Calabi–Yau category associated to the  $\widehat{A}_1$  graph, which we denote as  $\mathcal{C}$  in what follows, is now defined exactly as in simply-laced type in Section 6, using the above  $\widehat{A}_1$  zigzag algebra. The category  $\mathcal{C}$  is a graded,  $\mathbf{k}$ -linear triangulated category classically generated by two spherical objects, which we denote  $P_0$  and  $P_1$ . The hom spaces between these generators are given as follows:

$$\begin{aligned} \mathrm{Hom}^\bullet(P_i, P_i[n]) &= \begin{cases} \mathbf{k} & \text{if } n = 0 \text{ or } n = 2, \\ 0 & \text{otherwise;} \end{cases} \\ \mathrm{Hom}^\bullet(P_i, P_j[n]) &= \begin{cases} \mathbf{k}^2 & \text{if } n = 1 \text{ and } i \neq j, \\ 0 & \text{otherwise, for } i \neq j. \end{cases} \end{aligned}$$

The extension closure of  $P_0$  and  $P_1$  in  $\mathcal{C}$  is an abelian category; it is the heart of a (bounded)  $t$ -structure. We refer to it as the *standard heart*  $\heartsuit_{\mathrm{std}}$ .

8.2. **The automaton.** Denoting the spherical twist in  $P_i$  as  $\sigma_{P_i}$ , we have a weak action of  $B_\Gamma \cong F_2$  on  $\mathcal{C}$  given by

$$\sigma_i \rightarrow \sigma_{P_i}.$$

The image of  $B_\Gamma$  in  $\mathrm{Aut}(\mathcal{C})$  is precisely the subgroup generated by the spherical twists  $\sigma_{P_0}$  and  $\sigma_{P_1}$ . It is not difficult to check directly, via e.g. a ping-pong argument, that the action of  $B_\Gamma$  on  $\mathcal{C}$  is faithful, so we can regard  $B_\Gamma$  as a subgroup of  $\mathrm{Aut}(\mathcal{C})$  [33].

Set  $\gamma = \sigma_1 \sigma_0$ . One can check that for any integer  $n$ , the object  $\sigma_1 \gamma^n(P_0)$  is the cone of a morphism from a direct sum of  $|2n+2|$  copies of  $P_1[-1]$  to a direct sum of  $|2n+1|$  copies of  $P_0$ . Similarly, the object  $\gamma^n P_1$  is the cone of a morphism from a direct sum of  $|2n+1|$  copies of  $P_1[-1]$  to a direct sum of  $|2n|$  copies of  $P_0$ . In other words, we may write

$$\begin{aligned} \sigma_1 \gamma^n(P_0) &= \mathrm{Cone}(P_1^{\oplus |2n+2|}[-1] \rightarrow P_0^{\oplus |2n+1|}), \text{ and} \\ \gamma^n P_1 &= \mathrm{Cone}(P_1^{\oplus |2n+1|}[-1] \rightarrow P_0^{\oplus |2n|}). \end{aligned}$$

The objects obtained as above are all spherical, and their union consists of each of the objects  $\mathrm{Cone}(P_1^{\oplus n} \rightarrow P_0^{\oplus (n \pm 1)})$  exactly once, as  $n$  ranges over the integers. For convenience, we set

$$P_n = \mathrm{Cone}(P_1^{\oplus |n|} \rightarrow P_0^{\oplus |n-1|}) \text{ for } n \in \mathbf{Z}.$$

The objects  $P_n$  fit into distinguished triangles as follows:

$$\begin{aligned} (48) \quad P_{2i+1}[-1] &\rightarrow P_{2i} \oplus P_{2i} \rightarrow P_{2i-1} \xrightarrow{+1}, \\ P_{2i+2} &\rightarrow P_{2i+1} \oplus P_{2i+1} \rightarrow P_{2i}[1] \xrightarrow{\pm}. \end{aligned}$$

Regard  $\mathbf{Z}$  as the subset of  $\mathbf{P}^1(\mathbf{Z})$  consisting of points of the form  $[k : 1]$ . We can write a homomorphism  $B_\Gamma \rightarrow \mathrm{PSL}_2(\mathbf{Z})$ , as follows:

$$(49) \quad \sigma_0 \mapsto \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}, \quad \sigma_1 \mapsto \begin{pmatrix} -3 & 2 \\ -2 & 1 \end{pmatrix}.$$

Let  $\mathbf{S}$  be the set of all spherical objects, up to shift, that are stable with respect to some stability condition in  $\text{Stab}^\circ(\mathcal{C})$ . It can be checked, for example using the explicit construction of spherical stable objects in [2, § 4], that  $\mathbf{S}$  is the  $B_\Gamma$  orbit of  $\{P_0, P_1\}$ . We have a  $B_\Gamma$ -equivariant map  $i: \mathbf{S} \rightarrow \mathbf{P}^1(\mathbf{Z})$  characterised by  $P_n \mapsto [n : 1]$ . It is easy to check that  $i$  is injective. For convenience, let  $P_\infty$  be any one of the indecomposable extensions of  $P_0$  by  $P_1$ . We extend the embedding above to  $\mathbf{S} \cup \{P_\infty\}$  by sending  $P_\infty$  to  $[1 : 0]$ , which is the point at infinity.

The element  $\gamma = \sigma_1 \sigma_0$  acts on the objects  $P_n$  by an infinite-order cyclic rotation by two steps, lowering the phases. More precisely,

$$\begin{aligned} \gamma: P_n &\mapsto P_{n+2} \text{ for } n \neq 0, -1, \infty, \\ \gamma: P_n &\mapsto P_{n+2}[-1] \text{ for } n = 0, -1, \\ \gamma: P_\infty &\mapsto P_\infty. \end{aligned}$$

For each  $k \in \mathbf{Z}$ , set

$$\sigma_{2k} = \gamma^k \sigma_0 \gamma^{-k} \text{ and } \sigma_{2k+1} = \gamma^k \sigma_1 \gamma^{-k}.$$

It is easy to see that the image of  $\sigma_k$  in  $\text{Aut}(\mathcal{C})$  is the spherical twist in the object  $P_k$ . The image of  $\sigma_k$  in  $\text{PSL}_2(\mathbf{Z})$  is given by

$$(50) \quad \sigma_k \mapsto \begin{pmatrix} 2k+1 & -2k^2 \\ 2 & -2k+1 \end{pmatrix}.$$

**8.2.1. Standard stability conditions.** We say that a stability condition  $\tau$  is *standard* if its  $[0, 1)$  heart is the standard heart  $\heartsuit_{\text{std}}$ . Let  $\phi$  denote the phase function for  $\tau$ . Then we either have  $\phi(P_0) \leq \phi(P_1)$  or  $\phi(P_1) \leq \phi(P_0)$ . In the first case, we say  $\tau$  is standard of *type I*, and in the latter case, of *type II*. Note that, in type I, all objects  $P_n$  are  $\tau$ -semistable.

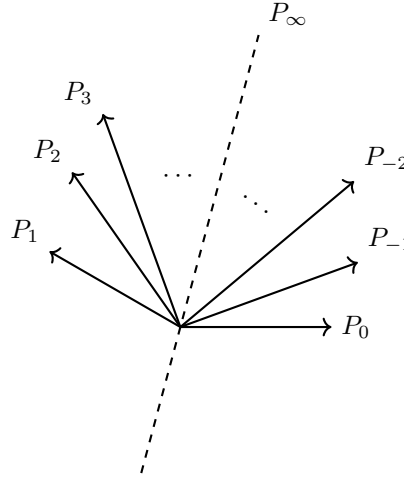


FIGURE 9. The central charges of the objects  $P_i$  in a standard off-the-wall stability condition of type I.

Let  $\Lambda \subset \text{Stab}(\mathcal{C})/\mathbf{C}$  be the closure of the set of standard stability conditions of type I. Then  $\Lambda$  includes non-standard stability conditions where  $\phi(P_1) = \phi(P_0) + 1$ . We continue to call these stability conditions as of *type I*, reserving the adjective standard for those where

$\phi(P_1) < \phi(P_0) + 1$ . Fig. 9 shows the central charges of the objects  $P_n$  in a standard off-the-wall stability condition of type I. The dotted line is the direction of the sum of the central charges of  $P_0$  and  $P_1$ , which is also the central charge of  $P_\infty$ .

8.2.2. *The automaton of an off-the-wall stability condition.* Fix an off-the-wall standard stability condition  $\tau$  of type I. Recall that  $\Sigma$  is the set of indecomposable  $\tau$ -semistable objects in the standard heart, which contains  $P_n$  for  $n \in \mathbf{Z}$  as well as the objects  $P_\infty$ . Fig. 10 describes a  $\tau$ -HN-automaton  $\Theta$  by illustrating the incoming and outgoing edges at one of the vertices.

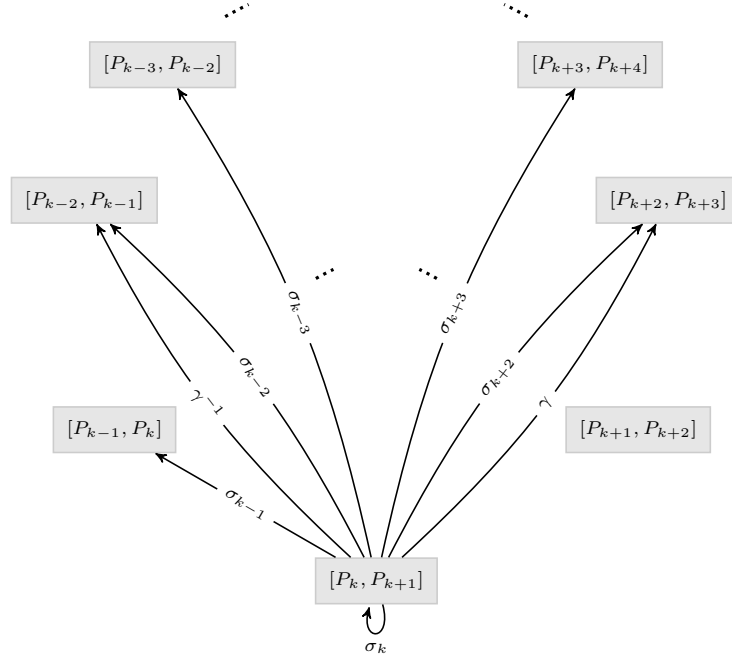


FIGURE 10. An HN automaton for an off-the-wall type I stability condition for  $\hat{A}_1$ , showing the outgoing edges from  $[P_k, P_{k+1}]$ .

Here is a more formal description of the automaton, following Definition 5.3.

- (1) We have a  $B_\Gamma$ -labelled graph with vertices indexed by  $\mathbf{Z}$  and edges as indicated in Fig. 10: an edge from state  $k$  to state  $j$  labelled  $\sigma_j = \sigma_{P_j}$  (unless  $j = k + 1$ ); an edge from state  $k$  to state  $(k + 2)$  labelled  $\gamma$ ; and an edge from state  $k$  to state  $(k - 2)$  labelled  $\gamma^{-1}$ .
- (2) The  $k$ th state is labelled  $[P_k, P_{k+1}]$ . The set  $S_{[P_k, P_{k+1}]}$  consists of the spherical objects of  $\mathcal{C}$  whose HN factors are shifts of  $P_k$  and  $P_{k+1}$ .
- (3) The representation  $M$  associates the free abelian group  $\mathbf{Z}^\Sigma$  to each state. However, as we show in Proposition 8.4, the image of the HN multiplicity map from  $S$  to  $M$  will only be supported on the sub-representation that assigns  $\mathbf{Z}^{\{P_k, P_{k+1}\}}$  to the state labelled as  $[P_k, P_{k+1}]$ . So we simplify notation by setting  $M$  to be this sub-representation. The map associated to the edge  $\sigma_j: k \rightarrow j$  is represented by the

$2 \times 2$  matrix

$$(51) \quad \begin{pmatrix} |j-k-1| & |j-k-2| \\ |j-k| & |j-k-1| \end{pmatrix}$$

in the standard bases. The maps associated to the edges labelled  $\gamma^\pm$  are represented by the identity matrix.

Analogously to Proposition 7.1, the next proposition shows that the description above fits into the framework of Definition 5.3.

**Proposition 8.4.** *Let  $e: [P_k, P_{k+1}] \rightarrow [P_j, P_{j+1}]$  be an edge of the automaton labelled by the group element  $g \in B_\Gamma$ . Let  $x \in \mathcal{C}$  be an object whose HN filtration consists of shifts of  $P_k$  and  $P_{k+1}$ .*

- (1) *The HN filtration of  $gx$  consists of shifts of  $P_j$  and  $P_{j+1}$ .*
- (2) *The following diagram commutes.*

$$\begin{array}{ccc} x & \xrightarrow{\text{HN}_\tau} & \text{HN}_\tau(x) \in \mathbf{Z}\{P_k, P_{k+1}\} \\ \downarrow e & & \downarrow M(e) \\ gx & \xrightarrow{\text{HN}_\tau} & \text{HN}_\tau(gx) \in \mathbf{Z}\{P_j, P_{j+1}\} \end{array}$$

In other words,

$$\text{HN}_\tau(gx) = M(e) \cdot \text{HN}_\tau(x).$$

*Proof.* By an argument similar to Proposition 7.1, the image under  $e$  of the HN filtration of  $x$  is rectifiable. So it suffices to prove the proposition for  $x = P_k$  and  $x = P_{k+1}$ .

Suppose  $g = \gamma^\pm$  so that  $j = k \pm 2$ . Since  $\gamma^\pm$  sends  $P_i$  to  $P_{i \pm 2}$  (up to shift), preserving the ordering of the phases, it is clear that  $e \cdot x$  is supported at  $[P_j, P_{j+1}]$ , and that the HN multiplicities transform by the identity matrix.

Otherwise,  $g = \sigma_j$ . Suppose that  $k = 0$ . We can check explicitly that if  $j \neq 1$ , we have

$$(52) \quad \begin{aligned} \sigma_j P_0 &= \text{Cone}(P_j^{\oplus |j-1|}[-1] \rightarrow P_{j+1}^{\oplus |j|}), \text{ and} \\ \sigma_j P_1 &= \text{Cone}(P_j^{\oplus |j-2|}[-1] \rightarrow P_{j+1}^{\oplus |j-1|}). \end{aligned}$$

As a result, both  $\sigma_j(P_0)$  and  $\sigma_j(P_1)$  are both supported at state  $j$  and their HN multiplicities transform according to the matrix  $M(e)$ .

Let  $k$  be arbitrary. Observe that we have

$$\sigma_j(P_k) = \begin{cases} \gamma^m \sigma_{j-2m} P_0, & \text{if } k = 2m \text{ is even,} \\ \gamma^m \sigma_{j-2m} P_1, & \text{if } k = 2m + 1 \text{ is odd.} \end{cases}$$

Using the above, we see that for any  $k$  except  $k = j - 1$ , we have

$$(53) \quad \sigma_j P_k = \text{Cone}(P_j^{\oplus |j-k-1|}[-1] \rightarrow P_{j+1}^{\oplus |j-k|}).$$

As a result,  $\sigma P_k$  and  $\sigma P_{k+1}$  are both supported at state  $j$  and their HN multiplicities transform according to the matrix  $M(e)$ .  $\square$

The next proposition shows that the automaton recognises every braid and every object of  $\mathbf{S}$ .

**Proposition 8.5.** (1) *Every  $\beta \in B_\Gamma$  has an expression of the form*

$$\beta = \gamma^n \sigma_{a_1} \sigma_{a_2} \cdots \sigma_{a_k},$$

*such that  $a_i - a_{i+1} \neq 1$  for each  $i$ .*



(2) Every object  $s \in \mathbf{S}$  has an expression

$$s = \beta_1 \cdots \beta_n P_i$$

where  $i \in \{0, 1\}$ , which is recognised by the automaton in Fig. 10.

*Proof.* For (1), first write  $\beta$  as any expression in the generators  $\{\sigma_0^{\pm 1}, \sigma_1^{\pm 1}\}$ . For any  $i \in \mathbf{Z}$ , we have

$$\sigma_i^{-1} = \sigma_{i-1} \gamma^{-1}, \quad \gamma^{-1} \sigma_i = \sigma_{i-2} \gamma^{-1}, \quad \gamma = \sigma_i \sigma_{i-1}.$$

Using these relations, we first rewrite  $\beta$  solely in terms of the elements  $\sigma_i$  for  $i \in \mathbf{Z}$  and  $\gamma^{\pm}$ . Then we successively replace any instances of  $\sigma_i \sigma_{i-1}$  by  $\gamma$ , and commute all powers of  $\gamma$  to the left. This process terminates, resulting in an expression of the desired form.

For (2), write  $s = \beta P_i$  where  $i$  is either 0 or 1. Assume that  $i = 0$ ; the other case is similar. Rewrite

$$\beta = \gamma^n \sigma_{a_1} \sigma_{a_2} \cdots \sigma_{a_k}$$

as in (1). Note that the object  $P_0$  is supported at two states, namely  $[P_0, P_1]$  and  $[P_{-1}, P_0]$ . We can apply  $\sigma_{a_k}$  at least at one of these two states. From the condition on our expression for  $\beta$ , it is clear that all subsequent letters are applicable. To finish, any power of  $\gamma$  is applicable at any vertex. Therefore  $s$  is recognised by the automaton.  $\square$

As a consequence of Proposition 8.5, every spherical object is supported on at least one of the states of the automaton in Fig. 10. In particular, its HN factors involve (up to shift) at most the objects  $P_i$  and  $P_{i+1}$  for some integer  $i$ .

**Proposition 8.6.** *Let  $x$  be a spherical object of  $\mathcal{C}$  supported at the state  $[P_k, P_{k+1}]$ . Assume that  $x$  is not a shift of  $P_k$  or  $P_{k+1}$ . Then*

- (1) *we can write  $s = \sigma_k^r t$  for some  $r > 0$  and an object  $t$  supported at  $[P_{k+1}, P_{k+2}]$ ;*
- (2) *we can write  $s = \sigma_{k+1}^{-r} t$  for some  $r > 0$  and an object  $t$  supported at  $[P_{k-1}, P_k]$ .*

*Proof.* Analogous to the proof of Proposition 7.3.  $\square$

Recall that we set  $M$  to be the  $\Theta$ -representation that associates the free abelian group  $\mathbf{Z}^{\{P_k, P_{k+1}\}}$  to the state  $[P_k, P_{k+1}]$ . As in the  $A_2$ -case, the  $\Theta$ -set  $M/\pm$  is isomorphic to  $\mathbf{Z}^2/\pm$  with the standard action of  $\mathrm{PSL}_2(\mathbf{Z})$ , as follows. Corresponding to each state  $[k, k+1]$  of  $\Theta$ , we define a linear map  $\phi_k: M_{[k, k+1]} \rightarrow \mathbf{Z}^2$  by

$$\phi_k: (1, 0) \mapsto (k, 1), \quad (0, 1) \mapsto (k+1, 1).$$

**Proposition 8.7.** *The maps  $\phi_k$  give an isomorphism of  $\Theta$ -sets  $M/\pm \rightarrow \mathbf{Z}^2/\pm$ .*

*Proof.* Let us check that  $\phi$  is  $\Theta$ -equivariant. It is evident that  $\phi_k$  intertwines the edges labelled  $\gamma^{\pm}$ . To check that it also intertwines  $\sigma_j: k \rightarrow j$ , we must check that

$$\sigma_j \circ \phi_k = \phi_j \circ M(e).$$

Recalling the matrix of  $\sigma_{P_j}$  from (50) and  $M(e)$  from (51), we must check

$$\begin{pmatrix} 2j+1 & -2j^2 \\ 2 & -2j+1 \end{pmatrix} \begin{pmatrix} k & k+1 \\ 1 & 1 \end{pmatrix} = \pm \begin{pmatrix} j-k-1 & j-k-2 \\ j-k & j-k-1 \end{pmatrix} \begin{pmatrix} j & j+1 \\ 1 & 1 \end{pmatrix}.$$

This is straightforward.  $\square$

*Remark 8.8.* Let  $\tau$  be an on-the-wall standard stability condition. Set  $P'_i$  to be the object  $\sigma_0 P_{i-1}$ . The objects  $P'_i$  are also semistable in  $\tau$ , in addition to the objects  $P_i$ . Note that up to shift, we have  $P'_1 = P_0$  and  $P'_0 = P_1$ . Recall that  $\gamma = \sigma_1 \sigma_0$ . Let  $\gamma' = \sigma_0 \sigma_1$ .

Fig. 11 shows an automaton that computes  $\tau$ -HN multiplicities for an on-the-wall stability condition  $\tau$ . Only the outgoing arrows labelled  $\sigma_0$  and  $\sigma_1$  are depicted. The remaining possible labels for outgoing arrows are  $\gamma^\pm$  and  $\gamma'^\pm$ , which act as follows:

$$\begin{aligned} \gamma: [P_i, P_{i+1}] &\rightarrow [P_{i+2}, P_{i+3}] \text{ unless } i = -2, & \gamma: [P'_i, P'_{i+1}] &\rightarrow [P_1, P_2]; \\ \gamma^{-1}: [P_i, P_{i+1}] &\rightarrow [P_{i-2}, P_{i-3}], \text{ unless } i = 2, & \gamma^{-1}: [P'_i, P'_{i+1}] &\rightarrow [P_{-2}, P_{-1}]; \\ \gamma': [P'_i, P'_{i+1}] &\rightarrow [P'_{i+2}, P'_{i+3}] \text{ unless } i = -2, & \gamma': [P_i, P_{i+1}] &\rightarrow [P'_1, P'_2]; \\ \gamma'^{-1}: [P'_i, P'_{i+1}] &\rightarrow [P'_{i-2}, P'_{i-3}] \text{ unless } i = 2, & \gamma'^{-1}: [P_i, P_{i+1}] &\rightarrow [P'_{-2}, P'_{-1}]. \end{aligned}$$

Since  $\Theta$  suffices for mass computations, we omit further details.

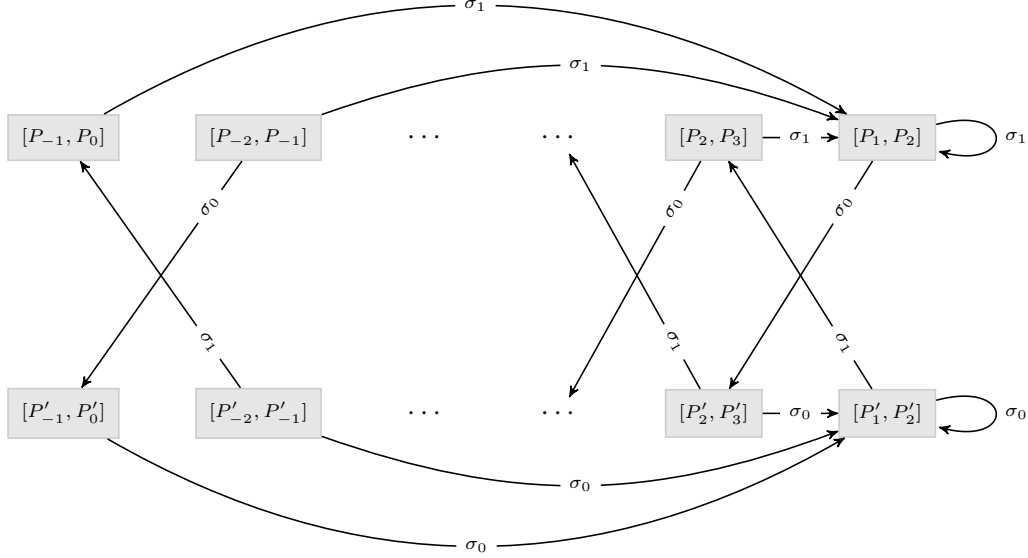


FIGURE 11. An HN automaton for an on-the-wall standard stability condition, depicting outgoing arrows labelled  $\sigma_0$  and  $\sigma_1$ .

**8.3. Spherical objects, their HN filtrations, and hom functionals.** Using the automaton constructed above, the analysis of HN filtrations of spherical objects in  $\mathcal{C}$  proceeds largely the same as in the  $A_2$  case. For example, we prove that the closure  $P$  of the image of the map  $h: \mathbf{S} \rightarrow \mathbf{P}^{\mathbf{S}}$  is a circle, and we establish an  $\hat{A}_1$  analog of Rouquier and Zimmermann [35, Proposition 4.8].

Recall that we have a  $B_{\Gamma}$ -equivariant map  $i: \mathbf{S} \rightarrow \mathbf{P}^1(\mathbf{Z})$ . The following proposition identifies  $i(s)$  explicitly in terms of the HN multiplicities.

**Proposition 8.9.** *For a spherical object  $s$  supported at the  $k$ th state, we have the equality*

$$i(s) = \phi_k(\text{HN}_{\tau}(s)) \in \mathbf{P}^1(\mathbf{Z}).$$

Moreover,  $\phi_k(\text{HN}_\tau(s)) \in \mathbf{Z}^2/\pm$  is the unique representative of  $i(s)$  whose coordinates are relatively prime.

*Proof.* Analogous to the proof of Proposition 7.9  $\square$

As a consequence of the automaton, we also obtain a Rouquier–Zimmermann–type theorem for  $\widehat{A}_1$ . Just as in type  $A_2$ , this theorem relates the parametrization of rational points  $[a : c] \in \mathbf{P}^1(\mathbf{Z})$  with Jordan–Hölder multiplicities of the corresponding spherical object.

Recall the notion of Jordan–Hölder (or JH) multiplicity from Definition 7.10. We use the same definition for the  $\widehat{A}_1$  category, with respect to the standard heart.

**Proposition 8.10.** *Let  $x \in \mathbf{S}$  such that  $i(x) = [a : c] \in \mathbf{P}^1(\mathbf{Z})$ , with  $a$  and  $c$  relatively prime integers. Then the JH multiplicities of  $P_0$  and  $P_1$  in  $x$  are exactly  $|a - c|$  and  $|a|$  respectively.*

*Proof.* Analogous to the proof of Proposition 7.11, using the result of Proposition 8.7 in lieu of Proposition 7.5.  $\square$

The following statement computes values of the  $\overline{\text{hom}}$  function of a spherical object from the JH multiplicities of  $P_0$  and  $P_1$  in that object.

**Proposition 8.11.** *Let  $x \in \mathbf{S}$  such that the JH multiplicities of  $P_0$  and  $P_1$  in  $x$  are  $n_0$  and  $n_1$  respectively. Then  $\overline{\text{hom}}(x, P_1) = 2n_0$  and  $\overline{\text{hom}}(x, P_0) = 2n_1$ .*

*Proof.* We prove the assertion for  $\overline{\text{hom}}(x, P_1)$ . The proof for  $\overline{\text{hom}}(x, P_0)$  is similar. We can compute directly from the definitions that  $\overline{\text{hom}}(P_1, P_k) = 2|k - 1|$  for each  $k \in \mathbf{Z}$ .

Fix a standard off-the-wall stability condition of type I. The key idea, as in the proof of Proposition 7.14, is to prove that  $\overline{\text{hom}}(-, P_1)$  is a linear function of the HN multiplicities. Then the cases  $x = P_i$  imply the result for all  $x$ .

Suppose that  $x$  is supported at  $[P_k, P_{k+1}]$  for  $k \neq 0, 1$ . Consider the Harder–Narasimhan filtration of  $x$ :

$$0 = x_0 \rightarrow x_1 \rightarrow \cdots \rightarrow x_n = x,$$

with factors  $z_i = \text{Cone}(x_i \rightarrow x_{i+1})$ . Each  $z_i$  is either  $P_k$  or  $P_{k+1}$ , up to shift.

By applying  $\text{Hom}^\bullet(P_1, -)$ , we obtain the following filtration in the bounded derived category of graded vector spaces:

$$(54) \quad 0 = \text{Hom}^\bullet(P_1, x_0) \rightarrow \cdots \rightarrow \text{Hom}^\bullet(P_1, x_n) = \text{Hom}^\bullet(P_1, x).$$

As in the first case in the proof of Proposition 7.14, we obtain

$$\text{Hom}^\bullet(P_1, x) = \bigoplus \text{Hom}^\bullet(P_1, z_i).$$

We now treat the cases where  $x$  is supported either at  $[P_0, P_1]$  or at  $[P_1, P_2]$ . By Proposition 8.10, the JH multiplicities of  $P_0$  in the objects  $x$  and  $\sigma_1^r x$  are equal, and both also have the same value for  $\overline{\text{hom}}(-, P_1)$ . Hence the proposition for  $x$  implies the proposition for  $\sigma_1^r x$ . By Proposition 8.6, any  $x$  supported either at  $[P_0, P_1]$  or  $[P_1, P_2]$  can be rewritten as  $x = \sigma_1^r y$  for some  $r \in \mathbf{Z}$  and  $y$  supported at  $[P_k, P_{k+1}]$  for  $k \neq 0, 1$ . Hence the proposition holds for  $x$ .  $\square$

The division of  $\mathbf{S}$  according to the HN support corresponds nicely to a geometric division of the circle  $\mathbf{P}^1(\mathbf{R})$ , which we now describe. The objects  $P_i$  divide  $\mathbf{P}^1(\mathbf{R})$  into a collection of closed arcs, namely  $\{[P_i, P_{i+1}] \mid i \in \mathbf{Z}\}$ . Together with  $P_\infty$ , these arcs cover all of  $\mathbf{P}^1(\mathbf{R})$ . We show that this decomposition of the circle is compatible with the automaton.

**Proposition 8.12.** *For each  $i \in \mathbf{Z}$ , the objects supported at state  $i$  are mapped to the arc  $[P_i, P_{i+1}]$ .*

*Proof.* Analogous to the proof of Proposition 7.13.  $\square$

**Proposition 8.13.** *The map  $h: \mathbf{S} \rightarrow \mathbf{P}^{\mathbf{S}}$  extends to a continuous map  $\mathbf{P}^1(\mathbf{R}) \rightarrow \mathbf{P}^{\mathbf{S}}$ , which is a homeomorphism onto its image.*

*Proof.* Analogous to the proof of Proposition 7.16.  $\square$

**8.4. Stability conditions and the mass map.** In this section we show that the mass map is a homeomorphism onto its image  $M$ . The basic technical tool used to do this is the automaton from Section 8.2.2.

We begin with the following analogue of Proposition 7.20.

**Proposition 8.14.** *Consider the following distinguished triangles:*

$$P_1[-1] \rightarrow P_0 \oplus P_0 \rightarrow P_{-1} \xrightarrow{+1} \text{ and } P_3[-1] \rightarrow P_2 \oplus P_2 \rightarrow P_1 \xrightarrow{+1}.$$

*If  $\tau$  is a standard stability condition of type I, then neither of the above triangles is collapsed by  $\tau$  in the sense of Definition 6.18. Furthermore, if  $\tau'$  is any stability condition such that neither of the above triangles is collapsed by  $\tau'$ , then  $\tau'$  is standard of type I up to the action of  $\mathbf{C}$ .*

*Proof.* It is immediate from Proposition 3.3 that neither of the above triangles is collapsed by any standard stability condition of type I. Now let  $\tau'$  be any stability condition. Recall that  $\tau'$  is in the braid group orbit of a standard stability condition of type I. Write  $\tau' = \beta\tau''$  for some braid  $\beta$  and some standard stability condition  $\tau''$ . As in the proof of Proposition 7.20, we may assume that  $\tau''$  is off-the-wall; otherwise the result follows by continuity.

If  $\beta$  is a power of  $\gamma$ , then  $\tau'$  is already standard of type I up to the action of  $\mathbf{C}$ . Otherwise, we show that one of the two triangles above is collapsed by  $\tau'$ . Equivalently, we show that  $\beta^{-1}$  applied to one of the two triangles above is collapsed by  $\tau''$ .

Consider a recognised expression of  $\beta^{-1}$  as described in Proposition 8.5. Since  $\beta$  is not a power of  $\gamma$ , this expression ends in  $\sigma_i$  for some  $i \in \mathbf{Z}$ . We observe using the automaton in Fig. 10, analogous to the argument in the proof of Proposition 7.20 that the first triangle is collapsed by  $\tau'$  if  $i \neq 0, 1$ . Otherwise, the second triangle is collapsed by  $\tau'$ . We omit the details.  $\square$

**Proposition 8.15.** *The map  $m: \text{Stab}(\mathcal{C})/\mathbf{C} \rightarrow \mathbf{P}^{\mathbf{S}}$  is a homeomorphism onto its image.*

*Proof.* The proof is analogous to the proof of Proposition 7.22, using the result of Proposition 8.14.  $\square$

**8.5. Closure of the stability manifold.** In this section we show that the closure  $\overline{M}$  of  $M$  in  $\mathbf{P}^{\mathbf{S}}$  is the disjoint union of  $M$  and  $P$ .

**Definition 8.16** (Gromov coordinates). Let  $\tau$  be a stability condition of type I. For each integer  $i \in \mathbf{Z}$ , let  $x_i(\tau)$  be the rational number defined as follows:

$$x_i(\tau) = \frac{m_\tau(P_{i-1}) + m_\tau(P_{i+1}) - 2m_\tau(P_i)}{2}.$$

We call the numbers  $(x_i(\tau))$  the *Gromov coordinates* of  $\tau$ .

Using the triangle inequality (Proposition 3.1) on the triangles (48), we deduce that all Gromov coordinates of  $\tau$  are non-negative. If  $\tau$  is off-the-wall, they are all positive.

**Proposition 8.17** (Linearity). *Let  $\tau$  be a stability condition of type I with Gromov coordinates  $(x_i)$ . For any object  $X \in \mathbf{S}$ , we have*

$$(55) \quad m_\tau(X) = \frac{1}{2} \sum_{i \in \mathbf{Z}} x_i \overline{\text{hom}}(P_i, X).$$

*Proof.* Recall that for any  $i \in \mathbf{Z}$ , we have

$$P_{2i} = \gamma^i P_0, \quad P_{2i+1} = \gamma^i P_1.$$

Therefore, we get

$$\overline{\text{hom}}(P_{2i}, X) = \overline{\text{hom}}(P_0, \gamma^{-i} X), \quad \overline{\text{hom}}(P_{2i+1}, X) = \overline{\text{hom}}(P_1, \gamma^{-i} X).$$

Proposition 8.11 allows us to compute  $\overline{\text{hom}}(P_0, \gamma^{-i} X)$  and  $\overline{\text{hom}}(P_1, \gamma^{-i} X)$  from the HN multiplicities of  $\gamma^{-i} X$ . In turn, the automaton in Fig. 10 allows us to compute the HN multiplicities of  $\gamma^{-i} X$  from the HN multiplicities of  $X$ . Suppose  $X$  is supported at the  $k$ th state of the automaton. Suppose the HN filtration of  $X$  consists of  $\alpha$  copies of  $P_k$  and  $\beta$  copies of  $P_{k+1}$ , up to shift. Then  $\gamma^{-i} X$  is supported on the  $(k - 2i)$ th state, and its HN filtration contains  $\alpha$  copies of  $P_{k-2i}$  and  $\beta$  copies of  $P_{k+1-2i}$ . In particular, for any  $i \in \mathbf{Z}$ , the JH multiplicities of  $P_1$  and  $P_0$  in  $\gamma^{-i} X$  are exactly

$$(\alpha|k - 2i| + \beta|k - 2i + 1|) \quad \text{and} \quad (\alpha|k - 2i - 1| + \beta|k - 2i|)$$

respectively. Using Proposition 8.11, we deduce that for any  $j \in \mathbf{Z}$ , we have

$$(56) \quad \overline{\text{hom}}(P_j, X) = 2(\alpha|k - j| + \beta|k - j + 1|).$$

Substitute

$$x_i = \frac{m_\tau(P_{i-1}) + m_\tau(P_{i+1}) - 2m_\tau(P_i)}{2}$$

in

$$\frac{1}{2} \sum_{i \in \mathbf{Z}} x_i \overline{\text{hom}}(P_i, X).$$

Then the coefficient of  $m_\tau(P_j)$  is exactly

$$\frac{\overline{\text{hom}}(P_{j-1}, X) + \overline{\text{hom}}(P_{j+1}, X) - 2\overline{\text{hom}}(P_j, X)}{4}.$$

Using (56), we see that the expression above equals  $\alpha$  if  $j = k$ , is  $\beta$  if  $j = k + 1$ , and is 0 otherwise. Therefore we see that

$$\frac{1}{2} \sum_{i \in \mathbf{Z}} x_i \overline{\text{hom}}(P_i, X) = \alpha \cdot m_\tau(P_k) + \beta \cdot m_\tau(P_{k+1}),$$

which is precisely the  $\tau$ -mass of  $X$ .  $\square$

Recall that  $\Lambda$  is the closure in  $\text{Stab}(\mathcal{C})/\mathbf{C}$  of the image of the standard off-the-wall type I stability conditions. We now study the topology of  $\Lambda$ , and compute the closure of  $m(\Lambda)$  in  $\mathbf{P}^{\mathbf{S}}$ .

**Proposition 8.18.** *Let  $\overline{\mathbf{H}}$  be the closed upper half plane. Consider the subset*

$$L = \{-1\} \cup \left\{ \frac{1-n}{n} \mid n \in \mathbf{Z} \setminus 0 \right\}.$$

*Then the map*

$$\tau \mapsto \frac{Z_\tau(P_1)}{Z_\tau(P_0)}$$

is a homeomorphism of  $\Lambda$  onto  $\overline{\mathbf{H}} \setminus L$ .

*Proof.* The map is clearly continuous. We first show that its image lies in  $\overline{\mathbf{H}} \setminus L$ . Recall that  $\Lambda$  is the closure of off-the-wall stability conditions of type I (up to the  $\mathbf{C}$ -action). These are stability conditions in which  $P_0$  and  $P_1$  are stable and satisfy the phase inequalities

$$\phi(P_0) < \phi(P_1) < \phi(P_0) + 1.$$

It follows that for any point  $\tau \in \Lambda$  the quantity  $\omega_\tau = Z_\tau(P_1)/Z_\tau(P_0)$  lies in the closed upper half plane. To see why  $\omega_\tau$  avoids  $L$ , note that the objects  $P_n$  are  $\tau$ -semistable for each  $n \in \mathbf{Z}$  as well as  $n = \infty$ . Therefore, we must have  $Z_\tau(P_n) \neq 0$  for  $n \in \mathbf{Z} \cup \{\infty\}$ . Since

$$[P_n] = |n|[P_1] + |n-1|[P_0] \text{ and } [P_\infty] = [P_0] + [P_1],$$

it is clear that  $\omega_\tau \notin L$ .

Now fix some  $\omega \in \overline{\mathbf{H}} \setminus L$ . We now prove that there is a unique  $\tau \in \Lambda$  with  $\omega = Z_\tau(P_1)/Z_\tau(P_0)$ . Consider the central charge  $Z: K(\mathcal{C}) \rightarrow \mathbf{C}$  defined as

$$Z(P_0) = 1, \quad Z(P_1) = \omega.$$

Recall from [7, Proposition 5.3] that a stability condition is uniquely specified by choosing:

- (a) the heart of a bounded t-structure, and
- (b) a central charge that sends the heart to the semi-closed upper half plane  $\mathbf{H} \cup \mathbf{R}_{>0}$  with the Harder–Narasimhan property.

The standard heart on  $\mathcal{C}$  is the extension closure of  $P_0$  and  $P_1$ . Being a finite-length category, any central charge as above with respect to the standard heart automatically enjoys the Harder–Narasimhan property. Thus if  $\omega$  lies in the semi-closed upper half plane, the central charge above uniquely specifies a type I standard stability condition.

We now treat the remaining cases; namely those where  $\omega$  lies on the negative real axis. In this case we can no longer use the standard heart for our reconstruction, because the central charge function does not land in the semi-closed upper half plane. We will show nevertheless that we can uniquely reconstruct points of  $\Lambda$  in these cases using tilts of the standard heart. These tilts will also be finite-length categories for which the Harder–Narasimhan property is automatic.

The set  $\mathbf{R} \setminus L$  is a disjoint union of open intervals of two kinds:

$$\mathbf{R} \setminus L = \bigcup_{n \geq 0} \left( -\frac{n}{n+1}, -\frac{n-1}{n} \right) \cup \bigcup_{n \geq 0} \left( -\frac{n+1}{n}, -\frac{n+2}{n+1} \right).$$

The tilt we choose will depend on the interval that contains  $\omega$ . Regard  $(0, \infty)$  as an interval of the first kind at the value  $n = 0$ ; we have already treated this case.

Suppose  $\omega$  lies in  $(-1/2, 0)$ , which is the interval of the first kind at the value  $n = 1$ . Consider the central charge  $Z' = Z \circ \sigma_1$ . Recall that  $\sigma_1^{-1}$  takes stability conditions of type I to those (possibly in the closure) of type II. Also note that  $\omega' = Z'(P_0)/Z'(P_1)$  lies in the interval  $(0, +\infty)$ . The previous argument applied to type II conditions shows that there is a unique  $\tau'$  of type II with central charge  $Z'$ . Then  $\tau = \sigma_1 \tau'$  gives the unique point of  $\Lambda$  with central charge  $Z$ .

Now suppose  $\omega = Z(P_1)/Z(P_0)$  lies in  $\left( -\frac{n}{n+1}, -\frac{n-1}{n} \right)$  for some  $n \geq 2$ . We construct the corresponding stability condition using the action of the element  $\gamma = \sigma_1 \sigma_0$ , which tilts the standard heart. Let  $m = \lfloor n/2 \rfloor$  and set  $Z' = Z \circ \gamma^m$ . Then  $\omega' = Z'(P_1)/Z'(P_0)$  lies in one of the two intervals  $(-1/2, 0)$  or  $(0, +\infty)$ , depending on the parity of  $n$ . In either case, the previous argument shows that there is a unique stability condition  $\tau'$  of type I with central

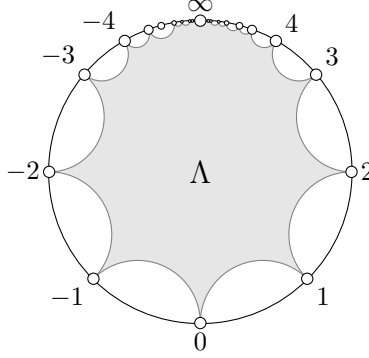


FIGURE 12. The set  $\Lambda$  of stability conditions of type I for the CY2 category associated to  $\hat{A}_1$ . The stability conditions are labelled by the ratio  $\frac{Z(P_0)}{Z(P_0)+Z(P_1)W}$ . The shaded infinite-sided polygon, except for its vertices, is homeomorphic to  $\overline{\mathbf{H}} \setminus L$ .

charge  $Z'$ . Since  $\gamma$  preserves  $\Lambda$ , the stability condition  $\tau = \gamma^m \tau'$  gives the unique point of  $\Lambda$  with central charge  $Z$ .

For  $\omega = Z(P_1)/Z(P_0)$  in the intervals of the second kind, we use a similar argument with  $Z' = Z \circ \gamma^{-m}$ . We omit the details.

We have shown that the map  $\tau \mapsto \omega_\tau$  is a bijection from  $\Lambda$  to  $\overline{\mathbf{H}} \setminus L$ . It is evident from the proof that the inverse map is continuous; therefore the map  $\tau \mapsto \omega_\tau$  is a homeomorphism from  $\Lambda$  to  $\overline{\mathbf{H}} \setminus L$ .  $\square$

It will be convenient in the remainder of this section to consider a change of coordinates from the previous proposition. Let  $\overline{\Delta}$  be a closed disk, considered as the one-point compactification of the closed upper half plane. Consider the map

$$\delta: \Lambda \rightarrow \overline{\Delta}$$

defined as

$$(57) \quad \delta_\tau = \frac{Z_\tau(P_0)}{Z_\tau(P_0) + Z_\tau(P_1)}.$$

By changing coordinates from Proposition 8.18, we see that  $\delta$  is injective and a homeomorphism onto its image, and also that the image of  $\delta$  is precisely

$$\overline{\Delta} \setminus (\mathbf{Z} \cup \{\infty\}).$$

Fig. 12 shows a sketch of  $\Lambda$  as a subset of the closed disk  $\overline{\Delta}$ .

We use the embedding  $\Lambda \xrightarrow{\delta} \overline{\Delta}$  to prove the following proposition.

**Proposition 8.19** (Closure of  $\Lambda$ ). *The closure of  $m(\Lambda)$  in  $\mathbf{P}(\mathbf{R}^{\mathbf{S}})$  is precisely*

$$\overline{m(\Lambda)} = m(\Lambda) \cup \{\overline{\text{hom}}(P_i) \mid i \in \mathbf{Z}\} \cup \{\overline{\text{hom}}(P_1) - \overline{\text{hom}}(P_0)\}.$$

*This closure is homeomorphic to  $\overline{\Delta}$ .*

*Proof.* Via the identification  $\Lambda \cong \overline{\Delta} \setminus (\mathbf{Z} \cup \{\infty\})$  via the map  $\delta$  from (57), the set  $\overline{\Delta} \setminus (\mathbf{Z} \cup \{\infty\})$  maps to  $\mathbf{P}(\mathbf{R}^{\mathbf{S}})$  by  $m \circ \delta^{-1}$ . We now show that this map extends continuously to  $\overline{\Delta}$ . By Proposition 8.17, we can express the mass functional  $m_\tau$  of  $\tau \in \Lambda$  as an infinite linear

combination of the Gromov coordinates of  $\tau$ . We now compute the limit of each Gromov coordinate of  $\tau$  as  $\delta_\tau$  approaches one of the points in  $\mathbf{Z} \cup \{\infty\}$ , and thereby check that the map  $m_\tau$  extends continuously.

Consider some  $\tau \in \Lambda$  with central charge  $Z_\tau$ ; let us compute the limit of the Gromov coordinates as  $\delta_\tau$  approaches some point of  $\mathbf{Z} \cup \{\infty\}$ . We know that up to a complex scalar, we have

$$Z_\tau(P_0) = 1, \quad Z_\tau(P_1) = \frac{1 - \delta_\tau}{\delta_\tau}.$$

We see that

$$Z_\tau(P_i) = \frac{|i|(1 - \delta_\tau) + |i - 1|\delta_\tau}{\delta_\tau}.$$

First suppose that  $\delta_\tau$  approaches some  $n \in \mathbf{Z}$ . In this case,

$$Z_\tau(P_i) \rightarrow \frac{|i|(1 - n) + |i - 1|n}{n}.$$

We can compute from Definition 8.16 that the limit of the  $j$ th Gromov coordinate is

$$x_j(\tau) \rightarrow \begin{cases} 0, & j \neq n, \\ 1, & j = n. \end{cases}$$

The sum in (55) obviously converges in the limit. Up to a simultaneous scalar, the limit of the mass functional is

$$m_\tau(-) \rightarrow \overline{\text{hom}}(P_n, -).$$

Now suppose that  $\delta_\tau$  approaches  $\infty$ . In this case,

$$Z_\tau(P_j) \rightarrow -|i| + |i - 1| = \begin{cases} -1, & i > 0, \\ 1, & i \leq 0. \end{cases}$$

We compute that

$$x_j(\tau) \rightarrow \begin{cases} 0, & i < 0 \text{ or } i > 1, \\ -1, & i = 0, \\ 1, & i = 1. \end{cases}$$

Again, the sum in (55) obviously converges in the limit. Up to a simultaneous scalar, the limit of the mass functional is

$$m_\tau(-) \rightarrow \overline{\text{hom}}(P_1, -) - \overline{\text{hom}}(P_0, -).$$

We have exhibited a factoring

$$\begin{array}{ccc} \Lambda & \xrightarrow{m} & \mathbf{P}(\mathbf{R}^S) \\ & \searrow \delta & \nearrow d \\ & \overline{\Delta} & \end{array}$$

and identified the additional points in the image of  $\overline{\Delta}$ . Since  $\overline{\Delta}$  is compact, no other points are in the closure of  $m(\Lambda)$ .  $\square$

*Remark 8.20.* The point  $\infty$  is already a limit point of the subset  $\mathbf{Z} \cup \{\infty\} \subset \overline{\Delta}$ . Therefore the additional point in the closure, namely the functional  $\overline{\text{hom}}(P_1) - \overline{\text{hom}}(P_0)$ , is already a limit point of the set  $\{\overline{\text{hom}}(P_i) \mid i \in \mathbf{Z}\}$ .

It is now easy to identify the closure of  $\text{Stab}(\mathcal{C})/\mathbf{C}$ , using the following contraction property, analogous to Proposition 7.28. Fix a metric on  $\mathbf{P}^1(\mathbf{R}) = S^1$ .



**Proposition 8.21.** *Let  $\epsilon > 0$  and let  $I \subset \mathbf{P}^1(\mathbf{R})$  be a compact subset. Then for all but finitely many elements  $g \in B_\Gamma$ , the image  $g(I)$  has diameter less than  $\epsilon$ .*

*Proof.* The action of  $B_\Gamma$  on  $\mathbf{P}^1(\mathbf{R})$  is via the map  $B_\Gamma \rightarrow \mathrm{PSL}_2(\mathbf{Z})$  of (49). The claim is true for  $\mathrm{PSL}_2(\mathbf{Z})$ , as shown in the proof of Proposition 7.28, and hence also for  $B_\Gamma$ .  $\square$

Let  $M$  and  $P$  be the images of  $\mathrm{Stab}(\mathcal{C})/\mathbf{C}$  and  $\mathbf{S}$  in  $\mathbf{P}^{\mathbf{S}}$ , respectively.

**Proposition 8.22** (Closure of  $M$ ). *The sets  $M$  and  $P$  are disjoint, and their union is the closure of  $M$ .*

*Proof.* The proof is analogous to the proof of Proposition 7.29.  $\square$

We conjecture that the union  $M \cup P$  is homeomorphic to a closed disk, as in the  $A_2$  case.

## APPENDIX A. RECTIFIABLE FILTRATIONS

In this section, we formulate sufficient conditions on a filtration to ensure that it can be re-arranged to the Harder–Narasimhan (HN) filtration.

Fix a triangulated category  $\mathcal{C}$  and stability condition on  $\mathcal{C}$ . For a semistable object  $x$ , let  $\phi(x)$  denote its phase. For any object  $x$ , denote by  $\lfloor x \rfloor$  (resp  $\lceil x \rceil$ ) the semistable factor of lowest (resp. highest) phase in its HN filtration. We retain the subscript notation for the various truncations induced by the stability condition from Section 2.1.

The first basic question is as follows. Consider a distinguished triangle

$$x \rightarrow y \rightarrow z \rightarrow x[1].$$

When can we easily obtain the HN filtration of  $y$  from that of  $x$  and  $z$ ? The following property of the map  $z \rightarrow x[1]$  leads to the answer.

**Definition A.1** (Rectifiable map, rectifiable triangle). We say that a map  $z \rightarrow x[1]$  is *rectifiable* if for every  $\alpha \in \mathbf{R}$ , the induced map  $z_{>\alpha} \rightarrow x_{\leq\alpha}[1]$  obtained as the composition

$$\begin{array}{ccc} z_{>\alpha} & \longrightarrow & z \\ & & \downarrow \\ & & x[1] \end{array} \longrightarrow x_{\leq\alpha}[1]$$

vanishes. Furthermore, we say that a distinguished triangle  $x \rightarrow y \rightarrow z \xrightarrow{+1}$  is a *rectifiable triangle* if the connecting map  $z \rightarrow x[1]$  is rectifiable.

*Remark A.2.* Definition A.1 uses the induced maps  $z_{>\alpha} \rightarrow x_{\leq\alpha}[1]$ . An alternative choice would have been to use the induced maps  $z_{\geq\alpha} \rightarrow x_{<\alpha}[1]$ . However, the two choices give equivalent definitions. Indeed, a given object has only finitely many HN pieces. Thus the maps  $z_{>\alpha} \rightarrow x_{\leq\alpha}[1]$  vanish for all  $\alpha$  if and only if the maps  $z_{\geq\alpha} \rightarrow x_{<\alpha}[1]$  vanish for all  $\alpha$ .

**Example A.3.** (1) The zero map  $z \rightarrow x[1]$  is rectifiable.

(2) If both  $z \rightarrow x_1[1]$  and  $z \rightarrow x_2[1]$  are rectifiable, then so is the direct sum  $z \rightarrow (x_1 \oplus x_2)[1]$ .

(3) Analogously, if  $z_1 \rightarrow x[1]$  and  $z_2 \rightarrow x[1]$  are rectifiable, then so is the direct sum  $(z_1 \oplus z_2) \rightarrow x[1]$ .

(4) Suppose  $\phi(\lfloor x \rfloor) \geq \phi(\lceil z \rceil)$ . Then any map  $z \rightarrow x[1]$  is rectifiable. Indeed, either  $z_{>\alpha}$  or  $x_{\leq\alpha}$  is zero.

**Proposition A.4.** *Suppose  $z \rightarrow x[1]$  is rectifiable. Then for any maps  $y \rightarrow z$  and  $x \rightarrow w$ , the induced map  $y \rightarrow w[1]$  is rectifiable. That is, rectifiable maps form a two-sided ideal.*

*Proof.* The map  $y_{>\alpha} \rightarrow w_{\leq\alpha}[1]$  factors as

$$y_{>\alpha} \rightarrow z_{>\alpha} \rightarrow x_{\leq\alpha}[1] \rightarrow w_{\leq\alpha}[1].$$

Since  $z \rightarrow x[1]$  is rectifiable, the map  $z_{>\alpha} \rightarrow x_{\leq\alpha}[1]$  vanishes. Hence, the map  $y_{>\alpha} \rightarrow w_{\leq\alpha}[1]$  vanishes too.  $\square$

We recall a useful lemma [3, Proposition 1.1.9].

**Lemma A.5.** *Let  $x \xrightarrow{u} y \xrightarrow{v} z \xrightarrow{+1}$  and  $x' \xrightarrow{u'} y' \xrightarrow{v'} z' \xrightarrow{+1}$  be distinguished triangles. Suppose that  $g: y \rightarrow y'$  is any morphism. Consider a diagram as follows.*

$$\begin{array}{ccccc} x & \xrightarrow{u} & y & \xrightarrow{v} & z \xrightarrow{+1} \\ \downarrow f & & \downarrow g & & \downarrow h \\ x' & \xrightarrow{u'} & y' & \xrightarrow{v'} & z' \xrightarrow{+1} \end{array}$$

The following statements are equivalent.

- (1) The composition  $v'gu$  is zero.
- (2) There is a morphism  $f: x \rightarrow x'$  that makes the first square commute.
- (3) There is a morphism  $h: z \rightarrow z'$  that makes the second square commute.
- (4) There is a morphism of triangles as in the diagram, where the middle map is  $g$ .

Furthermore, if  $\text{Hom}(x, z'[-1]) = 0$ , then the morphisms asserted above are unique.

We now examine how rectifiable maps behave under truncations. In this analysis, we repeatedly use the following distinguished triangle for an object  $x$  and an  $\alpha \in \mathbf{R}$ :

$$(58) \quad x_{>\alpha} \rightarrow x \rightarrow x_{\leq\alpha} \xrightarrow{+1}.$$

We call (58) the *truncation triangle* of  $x$  at  $\alpha$ . A morphism  $x \rightarrow y$  induces a map of distinguished triangles from the truncation triangle of  $x$  to the truncation triangle of  $y$ .

The following proposition says that a rectifiable map induces rectifiable maps on truncations.

**Proposition A.6.** *Suppose that  $x \rightarrow y \rightarrow z \xrightarrow{+1}$  is a distinguished triangle, in which the connecting map  $z \xrightarrow{e} x[1]$  is rectifiable.*

- (1) *There exist unique maps  $z_{\leq\alpha} \xrightarrow{e_{\alpha}^{\leq}} x_{\leq\alpha}[1]$  and  $z_{>\alpha} \xrightarrow{e_{\alpha}^{>}} x_{>\alpha}[1]$  fitting in a morphism of distinguished triangles as follows (see Warning A.7).*

$$\begin{array}{ccccc} z_{>\alpha} & \longrightarrow & z & \longrightarrow & z_{\leq\alpha} \xrightarrow{+1} \\ \downarrow e_{\alpha}^{>} & & \downarrow e & & \downarrow e_{\alpha}^{\leq} \\ x_{>\alpha}[1] & \longrightarrow & x[1] & \longrightarrow & x_{\leq\alpha}[1] \xrightarrow{+1} \end{array}$$

- (2) *The maps  $e_{\alpha}^{\leq}$  and  $e_{\alpha}^{>}$  are rectifiable.*
- (3) *We have distinguished triangles*

$$\begin{aligned} y_{>\alpha} &\rightarrow z_{>\alpha} \xrightarrow{e_{\alpha}^{>}} x_{>\alpha}[1] \xrightarrow{+1} \text{ and} \\ y_{\leq\alpha} &\rightarrow z_{\leq\alpha} \xrightarrow{e_{\alpha}^{\leq}} x_{\leq\alpha}[1] \xrightarrow{+1}. \end{aligned}$$

Analogous statements hold for the truncations  $(-)_{\geq\alpha}$  and  $(-)_{<\alpha}$ .

*Warning* A.7. The maps  $e_{\alpha}^{\leq}$  and  $e_{\alpha}^{>}$  asserted by the proposition are not the same as the maps induced by the truncation functors  $(-)^{\leq_{\alpha}}$  and  $(-)^{>_{\alpha}}$ . Indeed, the object  $x_{\leq_{\alpha}}[1]$  is not the same as the object  $(x[1])_{\leq_{\alpha}}$ , and similarly  $x_{>_{\alpha}}[1]$  is not the same as  $(x[1])_{>_{\alpha}}$ .

*Proof.* Let us prove each part in sequence.

- (1) This is a direct consequence of Lemma A.5. Existence follows from the rectifiability of  $e$ ; uniqueness follows from the fact that there are no morphisms from  $z_{>_{\alpha}}$  to  $x_{\leq_{\alpha}}$ .
- (2) Let us check that  $e_{\alpha}^{>}$  is rectifiable. We must check that for every  $\beta$ , the composite

$$(z_{>_{\alpha}})_{>_{\beta}} \rightarrow z_{>_{\alpha}} \xrightarrow{e_{\alpha}^{>}} x_{>_{\alpha}}[1] \rightarrow (x_{>_{\alpha}})_{\leq_{\beta}}[1]$$

is zero. The rightmost term is zero if  $\beta \leq \alpha$ , so there is nothing to prove. Suppose  $\beta > \alpha$ . Then the map above becomes

$$z_{>_{\beta}} \rightarrow z_{>_{\alpha}} \xrightarrow{e_{\alpha}^{>}} x_{>_{\alpha}}[1] \rightarrow x_{(\alpha, \beta]}[1].$$

Consider the diagram

$$\begin{array}{ccccc} z_{>_{\beta}} & \xrightarrow{a} & z_{>_{\alpha}} & \xrightarrow{b} & z \\ & & \downarrow e_{\alpha}^{>} & & \downarrow e \\ x_{\leq_{\alpha}} & \longrightarrow & x_{>_{\alpha}}[1] & \xrightarrow{c} & x[1] \\ & & \downarrow d & & \downarrow f \\ x_{\leq_{\alpha}} & \longrightarrow & x_{(\alpha, \beta]}[1] & \xrightarrow{g} & x_{\leq_{\beta}}[1]. \end{array}$$

Here the middle row is a rotation of the truncation triangle of  $x$  at  $\alpha$  and the bottom row is a rotation of the truncation triangle of  $x_{\leq_{\beta}}$  at  $\alpha$ . The maps from the middle row to the bottom row form a morphism of triangles induced by  $x \rightarrow x_{\leq_{\beta}}$ . The top right square commutes by part (1). We want to prove that the composition  $d \cdot e_{\alpha}^{>} \cdot a$  vanishes. Since  $e$  is rectifiable,  $f \cdot e \cdot b \cdot a$  vanishes, and hence  $g \cdot d \cdot e_{\alpha}^{>} \cdot a$  vanishes. From the bottom triangle, we see that  $d \cdot e_{\alpha}^{>} \cdot a$  is induced by a map  $z_{>_{\beta}} \rightarrow x_{\leq_{\alpha}}$ . Since  $\beta > \alpha$ , the only such map is the zero map for phase reasons. So  $d \cdot e_{\alpha}^{>} \cdot a$  vanishes.

By a similar argument, we see that  $e_{\alpha}^{\leq}$  is also rectifiable.

- (3) Consider the commuting square

$$\begin{array}{ccc} z_{>_{\alpha}} & \longrightarrow & z \\ \downarrow e_{\alpha}^{>} & & \downarrow e \\ x_{>_{\alpha}} & \longrightarrow & x[1]. \end{array}$$

By [3, Proposition 1.1.11], this square completes to a diagram as follows, where the rows and columns form distinguished triangles. Further, all squares commute except

for the one on the bottom right, which anti-commutes.

$$\begin{array}{ccccccc}
z_{>\alpha} & \longrightarrow & z & \dashrightarrow & z_{\leq\alpha} & \dashrightarrow & z_{>\alpha}[1] \\
\downarrow e_{\alpha}^{>} & & \downarrow e & & \downarrow e_{\alpha}^{\leq} & & \downarrow \\
x_{>\alpha}[1] & \longrightarrow & x[1] & \dashrightarrow & x_{\leq\alpha}[1] & \dashrightarrow & x_{>\alpha}[2] \\
\downarrow & & \downarrow & & \downarrow & & \downarrow \\
y'[1] & \dashrightarrow & y[1] & \dashrightarrow & y''[1] & \dashrightarrow & y'[2] \\
\downarrow & & \downarrow & & \downarrow & & \downarrow \\
z_{>\alpha}[1] & \dashrightarrow & z[1] & \dashrightarrow & z_{\leq\alpha}[1] & \dashrightarrow & z_{>\alpha}[2],
\end{array}$$

Observe that the vertical map  $z_{\leq\alpha} \rightarrow x_{\leq\alpha}[1]$  must be  $e_{\alpha}^{\leq}$  by part (1). The categories  $\mathcal{C}_{>\alpha}$  and  $\mathcal{C}_{\leq\alpha}$  are extension-closed. Thus  $y' \in \mathcal{C}_{>\alpha}$  and  $y'' \in \mathcal{C}_{\leq\alpha}$ . We have the triangle  $y' \rightarrow y \rightarrow y'' \xrightarrow{+1}$  in which  $y' \in \mathcal{C}_{>\alpha}$  and  $y'' \in \mathcal{C}_{\leq\alpha}$ . So this must be the truncation triangle of  $y$  at  $\alpha$ . That is,  $y' = y_{>\alpha}$  and  $y'' = y_{\leq\alpha}$ . This argument completes the proof.

The proof for  $(-)_{\geq\alpha}$  and  $(-)_{<\alpha}$  is analogous.  $\square$

For every  $\alpha \in \mathbf{R}$ , recall that  $\mathcal{C}_{\alpha}$  is the abelian category of semistable objects of phase  $\alpha$ . Also recall that we have the central charge  $Z$  and the mass  $m_q$  of a semi-stable object  $a$  of phase  $\alpha$  defined by  $m_q(a) = |Z(a)|q^{\alpha}$ .

**Proposition A.8.** *Let  $x \rightarrow y \rightarrow z \xrightarrow{e} x[1]$  be a distinguished triangle in which  $e$  is rectifiable. For every  $\alpha \in \mathbf{R}$  we have an exact sequence*

$$0 \rightarrow x_{\alpha} \rightarrow y_{\alpha} \rightarrow z_{\alpha} \rightarrow 0$$

in  $\mathcal{C}_{\alpha}$ . Consequently, we have  $m_q(y) = m_q(x) + m_q(z)$ .

*Proof.* Since  $e$  is rectifiable, we can apply Proposition A.6 twice, first for the  $(-)_{\leq\alpha}$  truncation and then to the  $(-)_{\geq\alpha}$  truncation, to obtain a distinguished triangle

$$x_{\alpha} \rightarrow y_{\alpha} \rightarrow z_{\alpha} \xrightarrow{+1}.$$

All three objects lie in  $\mathcal{C}_{\alpha}$ , which is an abelian subcategory of  $\mathcal{C}_{[\alpha, \alpha+1)}$ . Since the latter is the heart of a  $t$ -structure, the distinguished triangle becomes a short exact sequence.

Since  $Z$  is additive in exact triangles, we get

$$Z(y_{\alpha}) = Z(x_{\alpha}) + Z(z_{\alpha}).$$

But the three numbers all lie on the ray of argument  $\pi\alpha$ , so we get

$$|Z(y_{\alpha})| = |Z(x_{\alpha})| + |Z(z_{\alpha})|.$$

Multiplying throughout by  $q^{\alpha}$  gives  $m_q(y_{\alpha}) = m_q(x_{\alpha}) + m_q(z_{\alpha})$ . There are only finitely many numbers  $\alpha$  such that at least one of the objects  $x_{\alpha}$ ,  $z_{\alpha}$ , and  $y_{\alpha}$  is nonzero. Summing over all such  $\alpha$  yields  $m_q(y) = m_q(x) + m_q(z)$ .  $\square$

We now take up filtrations. For a filtration  $(x_0 \rightarrow \cdots \rightarrow x_n)$ , we fix objects  $x(i, j)$  that fit into fixed distinguished triangles

$$x_i \rightarrow x_j \rightarrow x(i, j) \xrightarrow{+1}.$$

*Remark A.9.* In the setup above, we observe the following consequences of the octahedral axiom. We understand  $x(-1, i)$  to mean the object  $x_i$ . Suppose that  $i < j < k < l$ .

- (1) We obtain distinguished triangles  $x(i, j) \rightarrow x(i, k) \rightarrow x(j, k) \xrightarrow{+1}$ .
- (2) The maps in the filtration and the distinguished triangles above induce morphisms of distinguished triangles

$$\begin{array}{ccccccc} x(i, j) & \longrightarrow & x(i, k) & \longrightarrow & x(j, k) & \xrightarrow{+1} & \longrightarrow \\ \downarrow \text{id} & & \downarrow & & \downarrow & & \\ x(i, j) & \longrightarrow & x(i, l) & \longrightarrow & x(j, l) & \xrightarrow{+1} & \longrightarrow, \end{array}$$

and

$$\begin{array}{ccccccc} x(i, j) & \longrightarrow & x(i, l) & \longrightarrow & x(j, l) & \xrightarrow{+1} & \longrightarrow \\ \downarrow & & \downarrow \text{id} & & \downarrow & & \\ x(i, k) & \longrightarrow & x(i, l) & \longrightarrow & x(k, l) & \xrightarrow{+1} & \longrightarrow. \end{array}$$

**Definition A.10** (Rectifiable filtration). We say that a filtration  $(x_0 \rightarrow \cdots \rightarrow x_n)$  is rectifiable if for every  $i \leq j \leq k$ , the connecting map  $x(j, k) \rightarrow x(i, j)[1]$  is rectifiable.

*Remark A.11.* In the setup above, the objects  $x(i, j)$  are unique up to (possibly non-unique) isomorphism. The rectifiability of the connecting map is independent of the particular choices.

*Remark A.12.* The definition of a rectifiable filtration is “translation invariant” in the following sense. Given a filtration  $x_0 \rightarrow x_1 \rightarrow \cdots \rightarrow x_n$ , set  $x'_i = x(0, i)$  and consider the induced filtration  $0 \rightarrow x'_1 \rightarrow \cdots \rightarrow x'_n$ . Then the first filtration is rectifiable if and only if the second one is. Indeed, the cones  $x(i, j)$  and the connecting maps between them in the first filtration are isomorphic to the corresponding objects in the second filtration.

**Example A.13.** (1) If  $(x_0 \rightarrow \cdots \rightarrow x_n)$  is rectifiable, then any filtration obtained from it by dropping intermediate terms is also rectifiable.  
 (2) If  $(0 = x_0 \rightarrow x_1 \rightarrow \cdots \rightarrow x_n = x)$  is the HN filtration of  $x$ , then it is rectifiable for phase reasons. Precisely, the connecting maps  $x(j, k) \rightarrow x(i, j)[1]$  are rectifiable due to Example A.3 (4).

The following lemma says that in Definition A.10, it suffices to take  $i = 0$  and  $k = n$ .

**Lemma A.14.** A filtration  $(x_0 \rightarrow \cdots \rightarrow x_n)$  is rectifiable if and only if for all  $j$ , the connecting map  $x(j, n) \rightarrow x(0, j)[1]$  is rectifiable.

*Proof.* The “only if” direction is obvious. We prove the “if” direction. Consider  $i \leq j \leq k$ . By Remark A.9, the connecting map  $x(j, k) \rightarrow x(i, j)[1]$  is the composite

$$x(j, k) \rightarrow x(j, n) \rightarrow x(0, j)[1] \rightarrow x(i, j)[1].$$

By Proposition A.4, if  $x(j, n) \rightarrow x(0, j)[1]$  is rectifiable, then so is the composite.  $\square$

The following proposition says that rectifiability can be checked inductively.

**Proposition A.15.** Let  $(x_0 \rightarrow \cdots \rightarrow x_n)$  be a filtration. Suppose for some  $i$ , both filtrations  $(x_0 \rightarrow \cdots \rightarrow x_i)$  and  $(x_i \rightarrow \cdots \rightarrow x_n)$  are rectifiable and the connecting map  $x(i, n) \rightarrow x(0, i)[1]$  is also rectifiable. Then  $(x_0 \rightarrow \cdots \rightarrow x_n)$  is rectifiable.

*Proof.* We induct on  $n$ . The base case  $n = 0$  is vacuous.

We now prove the induction step. If  $i = 0$  or  $i = n$ , then there is nothing to prove. So assume  $0 < i < n$ . By Lemma A.14, it suffices to show that for any  $j$ , the connecting map  $x(j, n) \rightarrow x(0, j)[1]$  is rectifiable. The case  $j = i$  follows from the hypotheses. We treat the case  $j > i$ ; the case  $j < i$  is similar. Fix an  $\alpha \in \mathbf{R}$ ; we need to show that the following map vanishes

$$x(j, n)_{>\alpha} \xrightarrow{g} x(0, j)_{\leq\alpha}[1].$$

By the hypotheses, the filtration  $(x_i \rightarrow \cdots \rightarrow x_n)$  is rectifiable. So the connecting map  $x(j, n) \rightarrow x(i, j)[1]$  is rectifiable. By Proposition A.6 applied to this map, we get the triangle

$$(59) \quad x(i, j)_{>\alpha} \rightarrow x(i, n)_{>\alpha} \rightarrow x(j, n)_{>\alpha} \xrightarrow{+1}.$$

The connecting map  $x(i, j) \rightarrow x(0, i)[1]$  is the composite

$$x(i, j) \rightarrow x(i, n) \rightarrow x(0, i)[1].$$

Since the second map is rectifiable, Proposition A.4 implies that the composite is also rectifiable. By Proposition A.6 applied to  $x(i, j) \rightarrow x(0, i)[1]$ , we get the triangle

$$(60) \quad x(0, i)_{\leq\alpha} \rightarrow x(0, j)_{\leq\alpha} \rightarrow x(i, j)_{\leq\alpha} \xrightarrow{+1}.$$

Consider the following diagram, in which the square commutes by Remark A.9.

$$\begin{array}{ccccc} x(i, n) & \longrightarrow & x(j, n) & \longrightarrow & x(i, j)[1] \\ \downarrow & & \downarrow & & \\ x(0, i)[1] & \longrightarrow & x(0, j)[1] & & \end{array}$$

We obtain the diagram below by pre-composing the first row with the natural maps  $(-)_{>\alpha} \rightarrow (-)$ , and post-composing the second row with the natural maps  $(-) \rightarrow (-)_{\leq\alpha}$ . Once again, the square commutes.

$$\begin{array}{ccccc} x(i, n)_{>\alpha} & \xrightarrow{u} & x(j, n)_{>\alpha} & \xrightarrow{v} & x(i, j)_{>\alpha}[1] \\ \downarrow b & & \downarrow g & & \\ x(0, i)_{\leq\alpha}[1] & \longrightarrow & x(0, j)_{\leq\alpha}[1] & & \end{array}$$

Note that  $b = 0$  because the original map  $x(i, n) \rightarrow x(0, i)[1]$  is rectifiable by the hypothesis. Thus the composition  $gu$  is zero, which means that  $g$  is a composition of  $v$  with a map from  $x(i, j)_{>\alpha}[1] \rightarrow x(0, j)_{\leq\alpha}[1]$ . Such a map must be zero due to phase reasons, and thus  $g = 0$  as desired. □

**Corollary A.16.** *A filtration  $0 = x_0 \rightarrow x_1 \rightarrow \cdots \rightarrow x_n$  is rectifiable if and only if for all  $i$ , the connecting map  $x(i-1, i) \rightarrow x_{i-1}[1]$  is rectifiable.*

*Proof.* Induct on  $n$ . To go from  $n-1$  to  $n$ , apply Proposition A.15 with  $i = n-1$ . □

*Remark A.17.* Using Proposition A.4 and Corollary A.16, one can show that the set of triangles  $x \rightarrow y \rightarrow z \xrightarrow{+1}$  where  $z \rightarrow x[1]$  is rectifiable form a extriangulated structure. Since we do not use this, we omit the details.

We now prove the analogues of Proposition A.6 and Proposition A.8 for rectifiable filtrations.

**Proposition A.18.** *Let  $(0 = x_0 \rightarrow x_1 \rightarrow \cdots \rightarrow x_n)$  be a rectifiable filtration. For any  $\alpha \in \mathbf{R}$ , we have exact triangles*

$$(x_i)_{>\alpha} \rightarrow (x_j)_{>\alpha} \rightarrow x(i, j)_{>\alpha} \xrightarrow{+1}.$$

Furthermore, the filtration

$$0 = (x_0)_{>\alpha} \rightarrow (x_1)_{>\alpha} \rightarrow \cdots \rightarrow (x_n)_{>\alpha}$$

is also rectifiable. The analogous statements hold for  $(-)_\geq \alpha$ ,  $(-)_< \alpha$ , and  $(-)_\leq \alpha$ .

*Proof.* Since  $x_0 = 0$ , we have  $x(0, i) = x_i$ . Since the original filtration is rectifiable, the connecting map  $x(i, j) \rightarrow x_i[1]$  is rectifiable. Proposition A.6 applied to this map gives the asserted triangle. By the same proposition, the connecting map  $x(i, j)_{>\alpha} \rightarrow (x_i)_{>\alpha}[1]$  is rectifiable. By Lemma A.14, the new filtration is rectifiable.

The proof for the other truncations is analogous.  $\square$

**Theorem A.19.** *Suppose the filtration  $(0 = x_0 \rightarrow x_1 \rightarrow \cdots \rightarrow x_n = x)$  is rectifiable. Let  $a_1, \dots, a_n$  be the factors of the filtration.*

(1) *For any  $\alpha \in \mathbf{R}$ , in  $\mathcal{C}_\alpha$  we have the short exact sequences*

$$0 \rightarrow (x_i)_\alpha \rightarrow (x_j)_\alpha \rightarrow x(i, j)_\alpha \rightarrow 0.$$

(2) *Set  $a_i = x(i-1, i)$ . In  $\mathcal{C}_\alpha$ , we have the filtration*

$$0 \subseteq (x_1)_\alpha \subseteq \cdots \subseteq (x_n)_\alpha$$

*whose factors are  $(a_1)_\alpha, \dots, (a_n)_\alpha$ .*

(3) *We have  $m_q(x) = \sum m_q(a_i)$ .*

*Proof.* For part (1) apply Proposition A.18, first for the  $(-)_\leq \alpha$  truncation of the original filtration, and then to the  $(-)_\geq \alpha$  truncation of the resulting filtration. Part (2) follows by taking  $(i, j)$  to be  $(i-1, i)$  in part (1). Part (3) then follows by the same argument as in the proof of Proposition A.8.  $\square$

In applications, it is useful to have a criterion for rectifiability only in terms of the factors. After proving two lemmas that allow us to swap the factors of a filtration, we establish such a criterion in Theorem A.22

**Lemma A.20.** *Consider a filtration  $x_1 \rightarrow x_2 \rightarrow x_3$  such that the connecting map  $x(2, 3) \rightarrow x(1, 2)[1]$  vanishes. Then there is an object  $x'_2$  and a filtration*

$$\begin{array}{ccccc} x_1 & \xrightarrow{\quad} & x'_2 & \xrightarrow{\quad} & x_3, \\ & \nwarrow \scriptstyle +1 & \swarrow & \nwarrow \scriptstyle +1 & \swarrow \\ & & x(2, 3) & & x(1, 2) \end{array}$$

such that the connecting map  $x(1, 2) \rightarrow x(2, 3)[1]$  also vanishes.

*Proof.* Consider the composition  $x_1 \rightarrow x_3$  in the original filtration, and let  $x(1, 3)$  be the object that completes it to a distinguished triangle. Recall that by the octahedral axiom, we have a distinguished triangle

$$x(1, 2) \rightarrow x(1, 3) \rightarrow x(2, 3) \xrightarrow{+1},$$

where the connecting map is the same as the one in the original filtration. Since that map is zero, we see that  $x(1, 3) \cong x(1, 2) \oplus x(2, 3)$ .

Now consider the map  $x_3 \rightarrow x(1, 2)$  obtained by mapping to  $x(1, 3)$  and projecting to  $x(1, 2)$ . Let  $x'_2$  be an object that completes it to a distinguished triangle

$$x'_2 \rightarrow x_3 \rightarrow x(1, 2) \xrightarrow{+1}.$$

Again, the octahedral axiom implies that we have a distinguished triangle

$$x_1 \rightarrow x'_2 \rightarrow x(2, 3) \xrightarrow{+1}.$$

The connecting map  $x(1, 2) \rightarrow x(2, 3)[1]$  is also zero by construction, as it is part of the split distinguished triangle

$$x(2, 3) \rightarrow x_3 \rightarrow x(1, 2) \xrightarrow{+1}.$$

□

**Lemma A.21.** *Suppose*

$$x_0 \rightarrow \cdots \rightarrow x_{i-1} \rightarrow x_i \rightarrow x_{i+1} \rightarrow \cdots \rightarrow x_n$$

*is rectifiable, and the map  $x(i, i+1) \rightarrow x(i-1, i)[1]$  is zero. Consider a filtration*

$$x_0 \rightarrow \cdots \rightarrow x_{i-1} \rightarrow x'_i \rightarrow x_{i+1} \rightarrow \cdots \rightarrow x_n = x,$$

*obtained by flipping  $x(i-1, i)$  and  $x(i, i+1)$  as in Lemma A.20. Then this new filtration is also rectifiable.*

*Proof.* By Proposition A.15, it suffices to show that

- (1) the filtration  $x_0 \rightarrow \cdots \rightarrow x_{i-1}$  is rectifiable;
- (2) the filtration  $x_{i-1} \rightarrow x'_i \rightarrow \cdots \rightarrow x_n$  is rectifiable; and
- (3) the connecting map  $x(i-1, n) \rightarrow x(0, i-1)[1]$  is rectifiable.

The first and third conditions follow directly from the rectifiability of the original filtration, so it remains to check the second. Once again, we apply Proposition A.15. It suffices to show that

- (1) the filtration  $x_{i-1} \rightarrow x'_i \rightarrow x_{i+1}$  is rectifiable;
- (2) the filtration  $x_{i+1} \rightarrow \cdots \rightarrow x_n$  is rectifiable; and
- (3) the connecting map  $x(i+1, n) \rightarrow x(i-1, i+1)[1]$  is rectifiable.

The second and third conditions follow directly from the rectifiability of the original filtration, so it remains to check the first. However, the connecting map in the first filtration is zero by Lemma A.20, and so it is rectifiable. The proof is complete. □

**Theorem A.22.** *Consider a filtration  $x_0 \rightarrow \cdots \rightarrow x_n$ , and let  $a_i = x(i-1, i)$ . Suppose for every  $i, j$  with  $i < j$ , at least one of the following conditions holds:*

- (1)  $\text{Hom}(a_j, a_i[1]) = \text{Hom}(a_i, a_j[1]) = 0$ , or
- (2)  $\phi(\lfloor a_i \rfloor) \geq \phi(\lceil a_j \rceil)$ .

*Then the filtration is rectifiable.*

*Proof.* We induct on  $n$ . The base cases  $n = 0, 1$  are trivial. Assume  $n \geq 2$ .

Suppose  $i$  is such that

$$\phi(\lceil a_i \rceil) < \phi(\lceil a_{i+1} \rceil).$$

Then we also have  $\phi(\lfloor a_i \rfloor) < \phi(\lceil a_{i+1} \rceil)$ . By the hypothesis, we must have  $\text{Hom}(a_{i+1}, a_i[1]) = \text{Hom}(a_i, a_{i+1}[1]) = 0$ . In particular, the connecting map  $a_{i+1} \rightarrow a_i[1]$  vanishes. Consider a filtration obtained by flipping  $a_i$  and  $a_{i+1}$  as in Lemma A.20. The sequence of factors of the new filtration is

$$a_1, \dots, a_{i-1}, a_{i+1}, a_i, a_{i+2}, \dots, a_n.$$



Observe that the new filtration also satisfies the hypotheses of the theorem. By Lemma A.21, it suffices to check that the new filtration is rectifiable.

By repeatedly swapping adjacent terms as above, we may assume that

$$\phi(\lceil a_1 \rceil) \geq \cdots \geq \phi(\lceil a_n \rceil).$$

By the inductive hypothesis, the filtration  $(x_1 \rightarrow \cdots \rightarrow x_n)$  is rectifiable. It remains to check that the connecting map  $x(1, n) \xrightarrow{e} x(0, 1)[1]$  is rectifiable.

Note that  $x(0, 1) = a_1$ . Partition  $\{2, \dots, n\}$  into two subsets  $\{2, \dots, i\}$  and  $\{i+1, \dots, n\}$  such that:

- (1) for all  $j \in \{2, \dots, i\}$  we have  $\phi(\lfloor a_1 \rfloor) < \phi(\lceil a_j \rceil)$ , and
- (2) for all  $j \in \{i+1, \dots, n\}$  we have  $\phi(\lfloor a_1 \rfloor) \geq \phi(\lceil a_j \rceil)$ .

If the first set is empty, take  $i = 1$ ; in this case,  $x(1, i) = 0$ . If the second set is empty, take  $i = n+1$ , and set  $x(i, n) = 0$ . Consider the diagram

$$\begin{array}{ccccc} x(1, i) & \longrightarrow & x(1, n) & \longrightarrow & x(i, n) \xrightarrow{+1} . \\ & & \downarrow e & & \\ & & a_1[1] & & \end{array}$$

The object  $x(1, i)$  has a filtration  $(0 = x(1, 1) \rightarrow x(1, 2) \rightarrow \cdots \rightarrow x(1, i))$  with factors  $a_2, \dots, a_i$ . By construction, for  $j \in \{2, \dots, i\}$ , we have

$$\phi(\lfloor a_1 \rfloor) < \phi(\lceil a_j \rceil)$$

and hence, by hypothesis, we must have

$$\text{Hom}(a_j, a_1[1]) = 0.$$

As a result, we get

$$\text{Hom}(x(1, i), a_1[1]) = 0.$$

Therefore, the map  $e$  factors through a map  $f$  as follows:

$$\begin{array}{ccccc} x(1, i) & \longrightarrow & x(1, n) & \longrightarrow & x(i, n) \xrightarrow{+1} . \\ & & \downarrow e & \swarrow f & \\ & & a_1[1] & & \end{array}$$

Now, the object  $x(i, n)$  has a filtration  $(0 = x(i, i) \rightarrow x(i, i+1) \rightarrow \cdots \rightarrow x(i, n))$  with factors  $a_{i+1}, \dots, a_n$ . For  $j \in \{i+1, \dots, n\}$ , we have

$$\phi(\lfloor a_1 \rfloor) \geq \phi(\lceil a_j \rceil),$$

and therefore

$$\phi(\lfloor a_1 \rfloor) \geq \phi(\lceil x(i, n) \rceil).$$

By Example A.3 (4), the map  $f$  is rectifiable for phase reasons. By Proposition A.4, the map  $e$  is also rectifiable.  $\square$

## APPENDIX B. SELF-EXTENSIONS OF A SPHERICAL OBJECT

The aim of this section is to record some properties of self-extensions of a spherical object. The results of this section play a significant role in the proof of Theorem 4.10, and may also be of independent interest. For  $d = 2$ , these appear in [22, § 2.3].

Fix  $d \geq 2$ , a field  $\mathbf{k}$ , a  $\mathbf{k}$ -linear triangulated category  $\mathcal{C}$ , and a  $d$ -spherical object  $a$  of  $\mathcal{C}$ . Fix a nonzero element  $\text{loop}_a \in \text{Hom}^d(a, a)$ . It will be convenient to set  $e = d - 1$ .

**Proposition B.1.** *Let  $a_n$  be an object of  $\mathcal{C}$  that admits a filtration*

$$(61) \quad \begin{array}{ccccccc} 0 = a_{-1} & \longrightarrow & a_0 & \longrightarrow & \cdots & \cdots & \cdots \longrightarrow a_n \\ & \nwarrow g_0 & \nearrow f_0 & & \nwarrow g_1 & \nearrow f_1 & & \nwarrow g_n & \nearrow f_n \\ & & a & & a[-e] & \cdots & \cdots & & a[-ne], \end{array}$$

where the connecting maps

$$f_{i-1} \circ g_i: a[-ie] \rightarrow a[-(i-1)e + 1]$$

are shifts of  $\text{loop}_a$ . Then we have the following.

- (1) The space  $\text{Hom}^\bullet(a, a_n)$  is two dimensional, with generators

$$i_n: a \rightarrow a_n \text{ and } l_n: a[-ne - d] \rightarrow a_n$$

in degrees 0 and  $ne + d$  respectively. The map  $a_0 \rightarrow a_n$  in (61) is a nonzero multiple of  $i_n$ .

- (2) The space  $\text{Hom}^\bullet(a_n, a)$  is two dimensional, with generators

$$t_n: a_n \rightarrow a[d] \text{ and } q_n: a_n \rightarrow a[-ne]$$

in degrees  $d$  and  $-ne$  respectively. The map  $f_n: a_n \rightarrow a[-ne]$  in (61) is a nonzero multiple of  $q_n$ .

- (3) The compositions

$$t_n \circ i_n: a \rightarrow a[d],$$

$$q_n \circ l_n: a[-ne - d] \rightarrow a[-ne]$$

are nonzero multiples of shifts of  $\text{loop}_a$ .

*Proof.* We prove the statements by induction on  $n$ . For  $n = 0$ , all three follow from the  $d$ -CY property of  $a$ . Let us assume them for  $(n - 1)$ , and prove them for  $n$ .

In the filtration (61), consider the last distinguished triangle

$$(62) \quad a_n \xrightarrow{f_n} a[-ne] \xrightarrow{g_n} a_{n-1}[1] \xrightarrow{+1} .$$

By assumption,  $f_{n-1} \circ g_n$  is nonzero, so  $g_n$  is also nonzero. Apply  $\text{Hom}(a, -)$  to (62) to get a long exact sequence

$$\cdots \rightarrow \text{Hom}^j(a, a_{n-1}) \rightarrow \text{Hom}^j(a, a_n) \rightarrow \text{Hom}^j(a, a[-ne]) \rightarrow \text{Hom}^{j+1}(a, a_{n-1}) \rightarrow \cdots .$$

By the inductive hypothesis,  $\text{Hom}^j(a, a_{n-1}) = 0$  unless  $j \in \{0, ne + 1\}$ . Since  $a$  is spherical,  $\text{Hom}^j(a, a[-ne]) = 0$  unless  $j \in \{ne, ne + d\}$ . So  $\text{Hom}^j(a, a_n) = 0$  for all  $j$  unless  $j \in \{0, ne, ne + 1, ne + d\}$ . We now examine the four remaining values of  $j$ .

For  $j = 0$ , the long exact sequence gives

$$0 \rightarrow \text{Hom}^0(a, a_{n-1}) \rightarrow \text{Hom}^0(a, a_n) \rightarrow 0.$$

From the inductive hypothesis, we conclude that  $\text{Hom}^0(a, a_n)$  is one-dimensional and is spanned by a copy of the map  $i_n: a_0 \rightarrow a_n$  from (61).

For  $j = ne$  and  $j = ne + 1$ , consider the piece of the exact sequence

$$0 \rightarrow \mathrm{Hom}^{ne}(a, a_n) \rightarrow \mathrm{Hom}^{ne}(a, a[-ne]) \xrightarrow{\bar{g}_n} \mathrm{Hom}^{ne+1}(a, a_{n-1}) \rightarrow \mathrm{Hom}^{ne+1}(a, a_n) \rightarrow 0.$$

Here  $\bar{g}_n$  is the map induced by  $g_n$  from (62). The source of  $\bar{g}_n$  is isomorphic to  $\mathbf{k}$ , spanned by the identity morphism. The identity morphism maps to  $g_n$  under  $\bar{g}_n$ . The target of  $\bar{g}_n$  is isomorphic to  $\mathbf{k}$  by the inductive hypothesis. It follows that  $\bar{g}_n$  is an isomorphism and  $\mathrm{Hom}^j(a, a_n) = 0$  for  $j = ne$  and  $ne + 1$ .

Finally, for  $j = ne + d$  the long exact sequence becomes

$$0 \rightarrow \mathrm{Hom}^{ne+d}(a, a_n) \rightarrow \mathrm{Hom}^{ne+d}(a, a[-ne]) \rightarrow 0.$$

Since  $a$  is  $d$ -spherical, it follows that  $\mathrm{Hom}^{ne+d}(a, a_n)$  is one-dimensional.

The induction step for part (1) is now complete. Parts (2) and (3) follow from the  $d$ -CY property of  $a$ .  $\square$

**Proposition B.2.** *For each  $n \geq 0$ , there exists an object  $a_n$ , unique up to isomorphism, which admits a filtration whose factors are  $a, a[-e], \dots, a[-ne]$  such that for all  $i$ , the connecting maps  $a[-ie] \rightarrow a[-ie + d]$  are shifts of  $\mathrm{loop}_a$ .*

*Proof.* We induct on  $n$ . For the base case  $n = 0$ , we have  $a_0 \cong a$ .

We assume the result for  $n$ , and prove it for  $n + 1$ . Observe that Proposition B.1 applies to  $a_n$ . Let  $a_{n+1}$  be the object that fits into a distinguished triangle

$$(63) \quad a[-ne - d] \xrightarrow{l_n} a_n \rightarrow a_{n+1} \xrightarrow{+1}.$$

Using the rotated triangle

$$a_n \rightarrow a_{n+1} \rightarrow a[-(n+1)e] \xrightarrow{+1}$$

and the assumed filtration of  $a_n$ , we obtain the desired filtration for  $a_{n+1}$ .

It remains to prove that the object  $a_{n+1}$  is unique up to isomorphism. Consider another object  $a'_{n+1}$  that also admits a filtration

$$(64) \quad 0 \rightarrow a'_0 \rightarrow \dots \rightarrow a'_n \rightarrow a'_{n+1}$$

satisfying the hypotheses. By the inductive hypothesis, we have an isomorphism  $a'_n \cong a_n$ . The last map in (64) then fits into a distinguished triangle

$$a[-ne - d] \xrightarrow{\iota'} a_n \rightarrow a'_{n+1} \xrightarrow{+1}.$$

By assumption, the composite of  $\iota'$  with the next map  $a'_n \rightarrow a[-(n-1)e]$  in (64) is the loop map; so  $\iota' \neq 0$ . By Proposition B.1,  $\mathrm{Hom}^{ne+d}(a, a_n)$  is one-dimensional, so  $\iota'$  is a nonzero scalar multiple of  $l_n$ . By comparing (63) and (64), we obtain an isomorphism  $a_{n+1} \cong a'_{n+1}$ .  $\square$

**Proposition B.3.** *For each  $n \geq 0$ , consider nonzero maps*

$$\begin{aligned} i_n &: a \rightarrow a_n, & l_n &: a[-ne - d] \rightarrow a_n \\ t_n &: a_n \rightarrow a[d], & q_n &: a_n \rightarrow a[-ne] \end{aligned}$$

*as in Proposition B.1. Then we have distinguished triangles*

$$\begin{aligned} a_{n-1}[-d] &\xrightarrow{t_{n-1}} a \xrightarrow{i_n} a_n \xrightarrow{+1} \quad \text{and} \\ a_n &\xrightarrow{q_n} a[-ne] \xrightarrow{l_{n-1}} a_{n-1}[1] \xrightarrow{+1}. \end{aligned}$$

*Proof.* Using Proposition B.2, the objects  $a_i$  fit into a filtration as in (61), where the connecting maps are shifts of  $\text{loop}_a$ . By the uniqueness statement in Proposition B.2, the cone of the map  $i_n: a = a_0 \rightarrow a_n$  is  $a_{n-1}[-e]$ . By rotating, we obtain a distinguished triangle

$$a_{n-1}[-d] \xrightarrow{\star} a \xrightarrow{i_n} a_n \xrightarrow{+1}.$$

By the second statement in Proposition B.1, the map labelled  $\star$  is a multiple of  $t_{n-1}$ . Let us show that  $\star$  cannot be the zero map. If  $\star$  were zero, then we would obtain a splitting  $a_n \rightarrow a$  of the map  $a \xrightarrow{i_n} a_n$ ; however by Proposition B.1 there are no such maps. Thus  $\star$  is a non-zero multiple of  $t_{n-1}$ , and we obtain the first distinguished triangle by rescaling.

Likewise, the cone of the map  $q_n: a_n \rightarrow a[-ne]$  is  $a_{n-1}[1]$ . By a similar argument to the above, we obtain the second distinguished triangle.  $\square$

**Proposition B.4.** *For each  $n \geq 0$ , the object  $a_n$  is indecomposable in  $\mathcal{C}$ .*

*Proof.* The statement is clear for  $n = 0$ : spherical objects are indecomposable because they have a one-dimensional space of endomorphisms of degree zero. It can be checked, e.g., via induction on  $0 \leq k \leq n$  and using the triangles

$$a_{k-1} \rightarrow a_k \rightarrow a[-ke] \xrightarrow{+1}$$

that  $\text{Hom}^0(a_k, a_n)$  is one-dimensional for each  $k \leq n$ . In particular,  $\text{Hom}^0(a_n, a_n)$  is one-dimensional for each  $n$ , and so the objects  $a_n$  are indecomposable.  $\square$

In the remainder of the section, we prove some homological properties of the objects  $a_n$ . These results are important ingredients in the proof of Theorem 4.10. Suppose we have a stability condition  $\tau$  on  $\mathcal{C}$  such that  $a$  is a  $\tau$ -semistable object in the  $[0, 1)$ -heart, with phase strictly between 0 and 1. Then the defining filtration of  $a_n$  from Proposition B.2 is also its  $\tau$ -Harder–Narasimhan filtration, because the factors are semistable and of decreasing phase.

Let  $I$  be the interval  $[0, e)$ . For  $m \in \mathbf{Z}$ , let by  $I - me$  be the translate  $[-me, -(m-1)e)$ . For an object  $x \in \mathcal{C}$ , we study the truncations  $x_{I-me}$ . For the object  $a_n$ , observe that

$$(a_n)_{I-me} = \begin{cases} a[-me] & \text{if } m \in \{0, \dots, n\} \\ 0 & \text{otherwise.} \end{cases}$$

The objects  $a_n$  satisfy the following “lifting property”.

**Lemma B.5.** *Let  $f: x \rightarrow a_n$  be a map such that the induced map*

$$x_{I-ne} \rightarrow (a_n)_{I-ne}$$

*vanishes. Then the map  $x_{\geq -(n-1)e} \rightarrow a_{n-1}$ , obtained by truncating  $f$ , extends to a map  $x \rightarrow a_{n-1}$ .*

*Proof.* We have the distinguished triangle

$$x_{\geq -(n-1)e} \rightarrow x \rightarrow x_{< -(n-1)e} \xrightarrow{+1}.$$

It suffices to prove that the composition  $x_{< -(n-1)e}[-1] \rightarrow x_{\geq -(n-1)e} \rightarrow a_{n-1}$  vanishes. We have the commutative diagram

$$\begin{array}{ccccc} x_{< -(n-1)e}[-1] & \longrightarrow & x_{\geq -(n-1)e} & \longrightarrow & x \\ \downarrow \bar{f} & & \downarrow & & \downarrow \\ a[-ne-1] & \xrightarrow{b} & a_{n-1} & \longrightarrow & a_n. \end{array}$$

We prove that the composite  $b \circ \bar{f}$  vanishes. The map

$$x_{I-ne}[-1] \rightarrow a[-ne-1]$$

induced by  $f$  vanishes by assumption. Therefore,  $\bar{f}$  factors through a map  $g$  as follows

$$\begin{array}{ccccc} x_{I-ne}[-1] & \longrightarrow & x_{<-(n-1)e}[-1] & \longrightarrow & x_{<-ne}[-1] \\ & & \downarrow \bar{f} & \swarrow g & \\ & & a[-ne-1] & & \end{array}$$

Now it suffices to prove that the composite

$$b \circ g: x_{<-ne}[-1] \rightarrow a_{n-1}$$

vanishes. The phases of the HN factors of the domain lie in the interval  $(-\infty, -ne-1)$ . The HN factors of the codomain are  $d$ -CY and their phases lie in the interval  $[-(n-1)e, +\infty)$ . Let  $s$  be a semistable HN factor of the domain and  $t$  a semistable HN factor of the codomain. Then  $\phi(t) > \phi(s) + d$ . By the  $d$ -CY property of  $t$ , we have

$$\dim \operatorname{Hom}(s, t) = \dim \operatorname{Hom}(t, s[d]),$$

and the right hand side must vanish since  $\phi(t) > \phi(s[d])$ . We conclude that

$$\operatorname{Hom}(x_{<-ne}[-1], a_{n-1}) = 0,$$

and hence the composite  $b \circ g$  vanishes.  $\square$

We can use the previous lemma to show that the map  $i_n: a \rightarrow a_n$  of degree zero is irreducible in the following way.

**Lemma B.6.** *Let  $x$  be an object that does not contain  $a$  as a direct summand and has no nonzero endomorphisms of negative degree. Then, for any  $n \geq 0$ , the map  $i_n: a \rightarrow a_n$  does not factor as  $a \rightarrow x \rightarrow a_n$ .*

For  $n > 0$ , the object  $a_n$  has endomorphisms of negative degree. Indeed, it can be checked that the composition of  $q_n: a_n \rightarrow a[-ne]$  with  $i_n[-ne]: a[-ne] \rightarrow a_n[-ne]$  is nonzero.

*Proof.* We continue to use truncations to the translates of the interval  $I$ .

Suppose we have maps  $s: a \rightarrow x$  and  $q: x \rightarrow a_n$  that compose to  $i_n: a \rightarrow a_n$ . By induction on  $n$ , we prove that either  $x$  has  $a$  as a direct summand or an endomorphism of negative degree. In the base case  $n = 0$ , we immediately obtain  $a$  as a direct summand. Let  $n > 0$  and assume that we know the result for  $n - 1$ . We study the following endomorphism of  $x$  of negative degree

$$(65) \quad x \xrightarrow{q} a_n \rightarrow a[-ne] \xrightarrow{s[-ne]} x[-ne].$$

We begin with an observation. Consider the distinguished triangle

$$a \xrightarrow{i_n} a_n \rightarrow a_{n-1}[-e] \xrightarrow{+1}$$

obtained by rotating the first triangle in Proposition B.3. Since all HN factors of  $a_{n-1}[-e]$  are of negative phase, we see that the truncation

$$(i_n)_{\geq 0}: a \rightarrow (a_n)_{\geq 0}$$

is an isomorphism. Truncating the composite  $a \xrightarrow{s} x \xrightarrow{q} a_n$ , we get

$$(66) \quad q_{\geq 0} \circ a_{\geq 0} = (i_n)_{\geq 0} \text{ is an isomorphism.}$$

Consider the map

$$(67) \quad q_{I-ne}: x_{I-ne} \rightarrow (a_n)_{I-ne}.$$

Suppose  $q_{I-ne} = 0$ . Then Lemma B.5 gives  $r: x \rightarrow a_{n-1}$  with  $r_{\geq -(n-1)e} = q_{\geq -(n-1)e}$ , and therefore,  $r_{\geq 0} = q_{\geq 0}$ . Consider the composite  $r \circ s: a \rightarrow a_{n-1}$ . From (66), we get

$$r_{\geq 0} \circ s_{\geq 0} = q_{\geq 0} \circ s_{\geq 0} \text{ is an isomorphism.}$$

In particular,  $r \circ s$  is nonzero, and therefore a multiple of  $i_{n-1}$ . In this case, we are done by the induction hypothesis applied to  $s: a \rightarrow x$  and  $r: x \rightarrow a_{n-1}$ .

Finally, suppose  $q_{I-ne}$  is nonzero. Since the composite  $a_{\geq 0} \xrightarrow{s} x_{\geq 0} \xrightarrow{q} (a_n)_{\geq 0}$  is an isomorphism, and the interval  $I = [0, e)$  is non-negative, the composite

$$a_I \xrightarrow{s} x_I \xrightarrow{q} (a_n)_I$$

is also an isomorphism. That is, the map  $s_I: a_I \rightarrow x_I$  has a left inverse. Now, consider the maps on the truncations induced by (65)

$$x_{I-ne} \xrightarrow{q} (a_n)_{I-ne} \xrightarrow{\sim} a[-ne] \xrightarrow{s} (x[-ne])_{I-ne}.$$

The first map is nonzero, the middle map is an isomorphism, and the last map has a left inverse. It follows that the composite is nonzero, and hence the endomorphism of  $x$  of negative degree in (65) is also nonzero. The induction step is complete.  $\square$

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